

Assessment of Wave Induced Responses of Articulated Ships

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Abstract

Accurate prediction of loads on mechanical couplings is crucial in assessment of loads on coupled structures, and in reliable prediction of the motions of coupled bodies. This thesis presents a numerical articulation method, which is capable of modeling various mechanical couplings. The proposed numerical method relies on contact elements. The first objective of the work is to prove the eligibility of contact elements in representation of mechanical couplings, and to highlight the shortcomings and positive aspects of contact elements for this representation. The second objective of the thesis is to study the influence of different hydrodynamic, kinematic, and coupling models on the motion behavior of articulated bodies and coupling forces. To accomplish these objectives extensive and systematic studies on a twofold pushing convoy are conducted. The convoy bodies are interconnected with hinge joints. The hinges are modeled by means of kinematic constraints, and with contact elements. Frequency and time domain boundary element methods with linearized motions of equation, as well as a Reynolds-averaged Navier-Stokes (RANS) code coupled with a nonlinear kinematic solver are used to compute the body motion responses and coupling forces. The simulation results are compared to model basin measurements. It is shown that the contact element model can simulate mechanical couplings efficiently and it provides a suitable method to idealize free and suppressed modes at articulation locations. Furthermore, it is shown, that RANS in association with the nonlinear equation of motion and kinematically constrained joints, as well as boundary element methods in adherence to the contact element model provide more accurate prediction of articulation forces, and motion responses of coupled bodies. However, frequency and time domain boundary element methods with kinematically constrained joints still deliver agreeable and reliable results. Moreover, a case study is presented, in which a river-sea pushing convoy is applied in time domain numerical simulations. The convoy bodies are interconnected with prismatic joints and are free to heave and pitch relative to each other. Prismatic joints are modeled by means of contact elements. The case study demonstrates a successful application of contact elements in simulation of complexly articulated multibody systems. Furthermore, the findings obtained in this work contribute sustainably to the numerical approaches in assessment of hydrodynamic and articulation loads on mechanically coupled floating bodies.

Kurzfassung

Im Folgenden wird ein Verfahren basierend auf Kontaktelementen zur Modellierung mechanischer Kopplungen vorgestellt. Dadurch wird ein Beitrag zur effizienten Berechnung der Belastungen auf den mechanischen Kopplungen eines Mehrkörpersystems geleistet. Zum einen befasst sich diese Arbeit mit der Validierung, sowie den Vor- und Nachteilen des Kontaktelementen-Ansatzes, zum anderen werden die Einflüsse verschiedener hydrodynamischen, kinematischen, und Kopplungsmodellen auf die Vorhersage der Kopplungskräfte und des Bewegungsverhaltens eines gekoppelten Mehrkörpersystems untersucht. Für diesen Zweck werden umfangreiche und systematische Simulationen mit einem seegehenden Schubverband durchgeführt. Hierbei sind die Körper des Verbandes über ein Drehgelenk miteinander gekoppelt. Dieses Drehgelenk des Schubverbandes wird anhand der kinematischen Bindungen auf Position- bzw. Beschleunigungsebenen sowie mit Hilfe von Kontaktelementen modelliert. Die angewandten Kopplungsmodelle werden mit verschiedenen hydrodynamischen und kinematischen Methoden gekoppelt. Hierbei werden die Frequenz- und Zeitbereich-Randelementverfahren mit linearisierten Bewegungsgleichungen, sowie ein Reynolds-Averaged-Navier-Stokes (RANS) Solver gekoppelt mit einem nichtlinearen Bewegungsgleichungslöser herangezogen. Die Simulationsergebnisse werden miteinander und mit den Modellversuchsergebnissen verglichen. Durch die durchgeführte Studie wurde der Ansatz zur Modellierung der mechanischen Kopplungen über die Kontaktelemente validiert. Es wurde festgestellt, dass das Bewegungsverhalten der gekoppelten Körper wie auch die Kopplungslasten sehr vergleichbar zu den Modellversuchen errechnet werden können. Dieses gelingt sowohl mit der RANS-Methode in Verbindung mit der nichtlinearen Bewegungsgleichung und den kinematischen Bindungen, als auch durch das Randelementverfahren mit dem Kopplungsmodell über die Kontaktelemente. Auch die Frequenz- und Zeitbereich-Randelementverfahren mit den kinematischen Bindungen liefern belastbare Ergebnisse. Zum Abschluss wird eine praktische Anwendung des Kontaktelementen-Ansatzes anhand eines Schubgelenkig gekoppelten seegehenden Schubverbandes präsentiert. Dadurch wird die Einsatzbarkeit des Kontaktelementen-Ansatzes zur Modellierung komplexer Kopplungsmechanismen unterstrichen.

Declaration

Part of the work described in this dissertation has previously been published in the following references:

- M. Ghesmi, A. von Graefe, V. Shigunov, B. Friedhoff, and O. el Moctar. Comparison and validation of numerical methods to assess hydrodynamic loads on mechanical coupling of multiple bodies. *Ship Technology Research*, pages 1–13, 06 2018
- M. Ghesmi and O. el Moctar. Application of contact elements to represent prismatic mechanical couplings. *MATEC Web of Conferences*, 272:01028, 01 2019

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Notations

$\hat{\square}$	complex amplitude
$\square_k$	related to k^{th} body or reference frame
${}^k\square$	components are with respect to k^{th} reference frame
$\square^t$	total
$\square^T$	transposed vector; transposed matrix
$\square_t$	time derivation, $\equiv \frac{\partial \square}{\partial t}$
$\square_{tt}$	second time derivative, $\equiv \frac{\partial^2 \square}{\partial t^2}$
$\square_z$	derivation with respect to z -coordinate, $\equiv \frac{\partial \square}{\partial z}$
$\dot{\square}$	first time derivative, $\equiv \frac{\partial \square}{\partial t}$
$\ddot{\square}$	second time derivative, $\equiv \frac{\partial^2 \square}{\partial t^2}$
α	stabilization damping coefficient
$\vec{\alpha}$	rotational motion
$\hat{\alpha}$	complex amplitude of rotational motion
$\tilde{\alpha}$	skew matrix of Euler angles
β	stabilization spring coefficient
Δ	Laplace operator, $\nabla \cdot \nabla$
Δx	base mesh size
ϵ	turbulence dissipation rate
ζ	free surface elevation
ζ^t	total free surface elevation
$\vec{\eta}$	perturbated motion
θ	pitch angle
λ	Lagrange-multiplier; wave length
μ	friction coefficient
μ_t	turbulent viscosity
$\vec{\xi}(\xi, \eta, \psi)$	potential source point position
ρ	fluid density
σ	potential source point strength
τ	time
$\boldsymbol{\tau}$	viscous stress matrix
ϕ	roll angle; velocity potential

Notations

ϕ_d	diffraction potential
ϕ^n	n^{th} -order perturbation velocity
ϕ^t	total velocity potential
ϕ_w	wave potential
φ^k	k^{th} canonical radiation potential
ψ	yaw angle
ω	wave angular frequency; turbulence frequency
ω_e	encounter wave angular frequency
$\vec{\omega}$	angular velocity
$\tilde{\omega}$	skew matrix of angular velocity
ω_ϕ	Natural roll frequency
∇	nabla operator; displaced volume
A	added mass matrix
B	breadth
B	potential damping matrix
C	Coriolis and centripetal matrix
C	volume fraction function
C_d	contact element damping coefficient
C_k	contact element stiffness coefficient
C_{ki}	contact element constant coefficients
d	water depth
$\vec{e}$	unit vector
E	identity matrix
E^{kin}	kinetic energy
E^{pot}	potential energy
f	total degrees of freedom
$\vec{f}$	perturbated force
$\vec{f}_{\text{hyd}}$	hydrodynamic force
$\vec{f}_1^0$	ship resistance force
$\vec{f}_3^0$	sinkage force
$\vec{f}_5^0$	steady trim moment
$\vec{f}^{\text{diff}}$	diffraction force
$\vec{f}^{\text{exc}}$	excitation force
$\vec{f}^{\text{rad}}$	radiation force
$\vec{F}$	force
Fn	Froude number
$\hat{\mathbf{F}}$	matrix containing complex force amplitudes
$\vec{F}_d$	damping force

$\vec{F}_f$	friction force
$\vec{F}_k$	axial compression force of contact element
$\vec{F}_t$	compressive normal force of contact element
$\vec{F}^{\text{exc}}$	excitation force
$\vec{F}^{\text{ext}}$	external force
$\vec{F}^r$	reaction force
g	gravitational acceleration; constraint condition
$G(x)$	Green function
$\mathbf{I}$	matrix of mass moment of inertia
h	wave amplitude
H	wave height
J_0	Bessel function of the first kind
$\mathbf{J}$	Jacobian matrix
k	wave number; turbulence kinetic energy
k_0	wave number related to shallow water
$\mathcal{K}_0$	inertial reference frame
$\mathcal{K}_i$	reference frame of i^{th} body
$\mathbf{K}(t)$	memory function
l_0	initial uncompressed length of contact element
L	structure length; Lagrange function
$\vec{L}$	angular momentum
m	mass
$\mathbf{M}$	mass matrix
$\vec{M}$	moment
$\hat{\mathbf{M}}$	matrix containing complex moment amplitudes
$\vec{n}$	normal vector
$\vec{n}_x, \vec{n}_y, \vec{n}_z$	normal vector components in x -, y - and z -directions
n_c	constraint number
$\mathcal{O}_i$	origin of coordinate system i
p	pressure
$\bar{p}$	time averages of fluctuations of pressure p'
p_a	atmospheric pressure
$\vec{P}$	linear momentum
$\vec{q}$	generalized coordinate
q_j	source strength of j^{th} Rankine source
r_k	refinement ratio
r_{xx}, r_{yy}, r_{zz}	radii of gyration about x -, y - and z -axes
$\vec{r}$	position vector with x, y, z as its components
$\delta\vec{r}$	virtual change of system coordinates
$\mathbf{R}$	restoring matrix

Notations

S	surface; wetted surface
t	time
T	draft; wave period
$\mathbf{T}$	transformation matrix
u	wave horizontal velocity; steady inflow speed
$\vec{u}$	translation vector
$\hat{u}$	complex amplitude of translational motion
$\vec{v}$	velocity vector
$\vec{v}_s$	velocity vector of control volume surface
$\overline{\vec{v}}$	time averages of fluctuations of velocity $\vec{v}'$
V	volume
w	wave vertical velocity
x	x component of position vector $\vec{r}$
y	y component of position vector $\vec{r}$
y^+	dimensionless wall distance
z	z component of position vector $\vec{r}$

Acronyms

API	Application Programming Interface
BEM	Boundary Element Method
CFD	Computational Fluid Dynamics
CFL	Courant-Fridrichs-Levy Number
CPU	Central Processing Unit
DST	Development Centre for Ship Technology and Transport Systems
FDM	Finite Difference Method
FEM	Finite Element Method
FPSO	Floating Production Storage and Offloading Unit
FVM	Finite Volume Method
GFM	Green Function Method
HRIC	High Resolution Interface Capturing Scheme
IMO	International Maritime Orgnization
JONSWAP	Joint North Sea Wave Observation Project
MLRW	Most Likely Response Wave
RANS	Reynolds-Averaged Navier–Stokes Equations
PIANC	Permanent International Association of Navigation Congresses
RAO	Response Amplitude Operator
RSM	Rankine Source Method
SIMPLE	Semi-implicit Method for Pressure Linked Equations
SST	Shear Stress Transport Model
TEU	Twenty-Foot Equivalent Unit
TLP	Tension Leg Platform
UNECE	United Nations Economic Commission for Europe
VLFS	Very Large Floating Structure
VOF	Volume-of-Fluid Method
<i>GM</i>	Metacentric Height
<i>LCG</i>	Longitudinal Center of Gravity
<i>VCG</i>	Vertical Center of Gravity

1 Introduction

In today's marine and offshore engineering, many hydrodynamic applications refer to interplay of multiple bodies. A loading tanker in tandem to an FPSO (Floating Production Storage and Offloading Unit), a supply vessel next to a TLP (Tension Leg Platform), and underway replenishment operation are some examples of multi-body systems. The proximity of two or several bodies may be of temporary nature such as in marine operations or permanent such as in multi-hull ships. Especially in the former case, the bodies are usually equipped with mechanical devices to smooth operations. Once a mechanical connection is introduced into a multibody system, numerical analyses and/or model basin experiments may, depending on the connection configuration, become complex. In an articulated multibody marine or offshore system, there exists in addition to the hydrodynamic multiple scattering effects between the structures, the interaction of the mechanical couplings with the body hydrodynamics. Additionally, the mechanical couplings may themselves be of diffracting or non-diffracting type, if submerged. Accurate assessment of coupling forces is of crucial importance for appropriate design of the coupling and the local structures of the related bodies. In many cases, it must be assured that the coupling machinery is capable of keeping the bodies constantly coupled. Provision or design of the appropriate machinery cannot be assured without substantial study of the interaction behavior of coupling and articulated bodies. Similarly, depending on the articulation type, global structure forces may be impacted in such a way that the maximum loads acting on structures may be shifted in position or amplified during articulation. Hence, a mathematical model is essential to simulate mechanically coupled multibody systems efficiently. Such a model helps to understand the interaction behavior of the bodies with each other and with the interconnections, to assess the hydrodynamic responses more realistically, is crucial to estimate the influence of the coupling on local and global loads of interconnected bodies, and is necessary to optimize the design of coupling mechanism.

In this work, an articulation model relied on contact elements is introduced. The model undergoes a validation process with available articulation formulations before its capability is demonstrated by reviewing a case study. In the following, the motivation on the application of contact elements to represent mechanical coupling is given. Then, the state-of-the-art numerical methods to model articulations are

overviewed. Afterwards, the objectives and outline of the thesis are given.

1.1 Background and Motivation

The insatiable continuous demand to have higher capacity ships has always been an advancing force in driving many technological fields in naval architecture. Realization of container mega boxers with TEU (Twenty-Foot Equivalent Unit) capacity of more than 20000 TEU is an example of such achievements. Not only the design and construction of higher capacity vessels but also their operation is a technical challenge. One operational hindrance is that not all terminals are able to accommodate such vessels; thus, redesign of available terminals or planning new dedicated terminals are demanded. In the latter case, due to the high drafts of large vessels some terminals are forced to set up their facilities out of reach of inland vessels. France Le Havre deep terminal and Germany's Jade-Weser port are two examples of such ports, where the inland vessels are territorially exposed to open sea conditions. Deep-sea terminals are connected to inland water infrastructure by so-called river-sea vessels, which are predominantly river vessels adapted for sea voyages. The adaption of river vessels to sail at sea requires special technical and safety measures. In the middle of the millennium's first decade various national regulations associated with river-sea vessels appeared. France and Belgium published their stability related regulations in 2007. France regulated the inland vessels traveling to Le Havre deep sea terminal and Belgium released prescriptive rules for the inland vessels traveling between Westerschelde and Belgian deep waters [121, 124, 125]. Following these rules PIANC (Permanent International Association of Navigation Congresses) initiated a safety committee for such vessels, and UNECE (United Nations Economic Commission for Europe) issued the Recommendations on Harmonized Europe-Wide Technical Requirements for Inland Navigation Vessels [3].

Germany initiated a federally funded project in 2012 to study the linking options of Germany's deep sea port to inland waterways by means of river-sea vessels [29]. Especially out of economic reasons, a river-sea convoy consisting of a pushing tug and a pushed barge was chosen as the favorite solution (see figure in page 18). A suitable seagoing pushing vessel propelled the barge at sea but it was relieved by an inland pusher in inland waters. In order to determine the articulation effects, none of the aforementioned regulations could be consulted because these regulations do not address river-sea convoys, and are safety-related than technically-oriented. One of the major challenges investigated within the research project was the determination of loads acting on the complex coupling mechanism of the convoy, which included hydraulics, springs, static and sliding frictional parts, mechanical

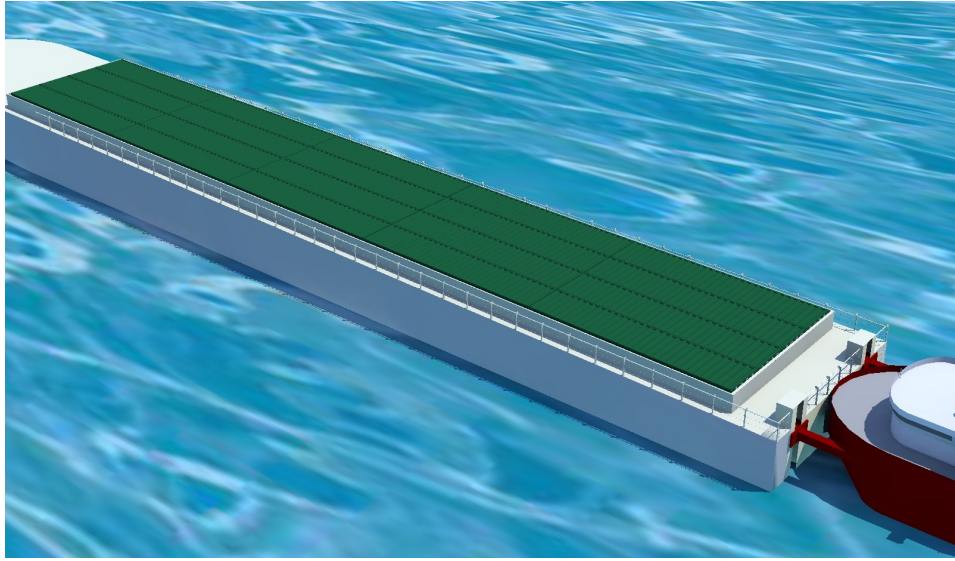


Figure 1.1: Mechanically coupled river-sea convoy.

end stops as well as pretension capability. The mechanical coupling mechanism was found to have a significant influence from hydrodynamic and structural points of view. It was proved that the coupling affected the motion responses of the bodies considerably. Furthermore, the coupling mechanism required a dedicated local structure design, and its reaction forces had influences on the global structural loads of the coupled bodies. The induced reactive joint forces varied depending on the pretension force exerted by the mechanism. Inaccuracy in proper design of the coupling mechanism, and in assessment of articulation effects on the local and global structure of the adjoined bodies could jeopardize the safety of the vessels, crew and cargoes. Such challenging articulation mechanism with prominent influential impact emphasized the necessity to have simplified and efficient models to predict reliably the articulation forces, while taking interaction effects between bodies and articulation into account.

This thesis lies on some outcomes of this research project, which intended to develop a numerical method to predict the forces on the mechanical coupling mechanism of the designed river-sea convoy reliably. The final project report was published in [55].

1.2 Overview of Existing Coupling Methods

There is no common categorization of mechanically coupled multibody systems; thus, different categorizations are cited in literatures based on articulation types, numerical methods, or theoretical models. In many engineering fields, articulating the bodies of a multibody system is equivalent to imposing dynamic constraints on the bodies because articulations impose constraints on the system dynamics. However, modeling articulation of multibody systems is not the only application of kinematic constraints. Therefore, the theories on the kinematic constraints subsume the theories on the articulations of multibody systems. Consequently, many approaches treat kinematic constraints rather than articulations of multibody systems. In the following, a concise overview of several methods applied in marine and offshore industry to treat articulated multibody systems is given concisely.

Penalty Methods

Penalty methods replace a constrained problem by a series of unconstrained problems, which involve penalty functions. The objective is to make the series of unconstrained problems converge ideally to the solution of the constrained problem. In case of an implicit constraint function, the deviation of the evaluated function from zero is the measure of violation, against which a penalty force must withstand. Penalty methods are computationally inexpensive but the simulation quality is dependent on the penalty force coefficients, which are not easy to determine. Additionally, depending on coefficients, the system may become numerically stiff, which may cause adverse numerical stability. Use of penalty methods in ship and offshore hydrodynamics is still rare. Two examples can be found in [47] and [35]. In the former publication, stiffness distribution on structure elasticity is investigated and in the latter, the dynamics of rigid and deformable multibodies are examined.

Lagrange Multiplier Method

Like penalty methods, Lagrange multiplier method uses forces to treat constraints. However, in penalty methods these internal forces are exerted after determination of the violation measurements, but in Lagrange multiplier method the internal forces must avoid occurrence of such violations in compliance with defined constraints. Application of Lagrange multiplier method is very common in marine and offshore industry and almost all widely used software packages in this field have some sort of application of this method implemented. Further on this method is given in section 2.3

Application of Generalized Coordinates

The penalty and Lagrange multiplier methods rely on determined internal forces to fulfill the system constraints. Another approach is to consider the

effect of constraints on the system, i.e. that the constraints reduce the degrees of freedom of a system. A system with n degrees of freedom and n_c constraints has $n - n_c$ independent coordinates. Independent coordinates are also called generalized, reduced or minimal coordinates. The position of a system must be definitely definable by generalized coordinates. Therefore, $n - n_c$ dynamic equations as per Lagrangian mechanics can be gained after substituting Lagrange function $L = E^{\text{kin}} - E^{\text{pot}}$, for kinetic and potential energy of the system, in Euler-Lagrange equations (Lagrange equation of the second kind). The prominent advantage of application of generalized coordinate is the reduction of coordinates, which is beneficial from numerical and computational points of view. The disadvantage of this method is investigation of constraints and coordinates to formulate generalized coordinates (parametrization). An example of application of generalized coordinates can be found in [142], in which constraints are defined to suppress degrees of freedom of a ship in different maneuvering scenarios.

Mode Expansion Method

Mode expansion method is probably the most widespread method in treating multibody systems in marine and offshore fields. Probably Newman [69, 70, 71, 72, 52, 74] has been the first, who showed the versatility of mode expansion method in association with boundary element fluid solvers. In literature, mode expansion method is also called generalized mode concept (not to be mixed with the aforementioned generalized coordinate method). In this concept, all degrees of freedom are described by mode shapes. Generalized modes can have any shapes and describe both rigid body and flexible modes. Therefore, this concept subsumes both seakeeping and interconnection theories in the sense that rigid or flexible bodies can be connected to other fixed or rigid bodies with rigid or flexible fixtures. The most prominent derivative theory from generalized mode concept is the hydroelasticity theory [10], in which, for example, dynamic responses, displacements, strains, stresses, internal forces and moments can be defined as generalized modes. The drawback of the generalized concept is that it relies on the mode superposition principle, which necessitates linearity of the analyzed system. There are numerous applications of this concept in marine and offshore fields. For example, Kim et al. [46] used the theory to analyze the bending moments and shear forces on the connection of two VLFSs (Very Large Floating Structures) in their side-by-side and fore-aft configurations for rigid and elastic modes in frequency domain. Fu et al. [30] analyzed the hydroelastic response of interconnected multimodule structures by comparing four model types consisting combination of rigid and flexible connectors and modules. Recently, Tuitman et al. [122] used this theory on two hinged barges in time domain.

1.3 Thesis Objectives and Outlines

Objective of the thesis is to develop a method to assess the articulation forces of mechanically coupled multibody systems reliably. Kinematic constraints constrain the kinematics of a point or a body. They are versatile in application, but they demand analytical prescription of constraints, and on which point/body they are to be imposed. Similarly, the mode expansion method is versatile, but it demands the system to be linear, and it requires exact definition of the modes at the outset. Therefore, constraints or modes need to be analytically formulated for a specific problem before being implemented programmatically. Considering the developments in the computer technology, the question arises if such analytical definitions can be circumvented numerically. Contact elements in conjunction with a time domain boundary element method are used to achieve this goal. Reliable prediction of motion behavior of articulated floating bodies and joint forces depends on the reliability of coupling models, hydrodynamic methods, as well as equation of motion. Therefore, the thesis objective has been extended to studying the effects of different coupling, hydrodynamic and equation of motion solvers on the accuracy of predicted motion responses and coupling forces of multibody systems.

Thus, the main goals of this work are:

- How do linear and nonlinear equations of motion affect the accuracy of predicted motion responses of articulated bodies?
- How do linear and nonlinear hydrodynamic models affect the accuracy of predicted loads on articulated multibodies and mechanical couplings?
- How effective and reliable are different coupling models in simulation of articulated multibody systems?
- Is the contact element model eligible to represent mechanical couplings?
- What are the advantages and shortcomings of the contact element model in representing mechanical couplings?
- How efficient is the contact element model in simulation of complex mechanical couplings?

In this context, *hydrodynamic coupling* refers to the condition, where the bodies of a multibody system are in close vicinity of each other without presence of any mechanical couplings. The term *articulation* is used generally throughout the context for mechanical couplings.

This thesis contains five chapters. **Chapter 2** deals with the numerical and theoretical aspects. In a more specific manner, in **section 2.1**, the theoretical back-

ground of boundary element methods and RANS methods used within this work are given. **Section 2.2** is dedicated to rigid body dynamics. It is also shown how rigid body dynamics are coupled with RANS and boundary element methods. In **section 2.3**, it is discussed how the formulations of hydrodynamic couplings are extended to treat mechanically coupled systems. Different types of articulations are reviewed and it is shown how kinematic constraints in conjunction with Lagrange equations of the first kind are used to constrain body dynamics. Furthermore, an overview of the theory of contact elements is given, and it is presented how they can be configured to represent various joint types.

In **chapter 3**, the contact element model in representation of mechanical couplings is validated. Additionally, the influence of different hydrodynamic methods, coupling models, as well as linear and nonlinear equations of motion on the motion behavior of articulated bodies and coupling force responses are studied. A seagoing pushing convoy consisting a barge and a tug are taken for these studies. In **section 3.1**, the model tests conducted for the convoy bodies in single, multibody, and articulated multibody configurations are described. Numerical simulations are conducted with linear boundary element frequency domain methods, a nonlinear boundary element time domain method, and with a RANS method. The success and accuracy of numerical simulations depend on numerical grids. Grid independence studies related to both RANS and boundary element simulations are presented in **section 3.2**. Comparison of the results of different hydrodynamic solvers with each other and with model basin measurements for bodies in their respective stand-alone conditions are presented in **section 3.3**. Results for hydrodynamically coupled arrangements are given in **section 3.4**, and the studies for mechanically coupled multibodies are shown in **section 3.5**. In such a manner, the complexity of simulations are increased in each section to facilitate analyses of additive parameters.

After validation of the contact element model in **chapter 3**, an application of this model is presented in **chapter 4**. The convoy mentioned in section 1.1 is taken for this purpose. In **section 4.1**, the coupling mechanism designed for the convoy, and the model test configurations are described. Related numerical simulations and simulation results are discussed in **section 4.2**. Finally, **chapter 5** summarizes the result of previous chapters.

2 Numerical Methods

2.1 Hydrodynamic Loads

Assessment of wave induced loads and motions of ships and offshore structures is important both in design and operation. This assessment is challenging because the flow around ships and offshore structures, and the resulting motions and loads caused by incident waves constitute often a three dimensional nonlinear problem. Early hydrodynamicists approached this complex problem, after studying the dominating hydrodynamic forces on the structure by categorizing the ship and offshore structures with respect to the ratio of structure length, L , to wave length, λ (L/λ) [23]. In this categorization, $L/\lambda=0.2$ acts as the limit for calculating the hydrodynamic forces. The structures such as jacket leg platforms with $L/\lambda \leq 0.2$, are called transparent structures, and those with $L/\lambda > 0.2$ are called compact structures [16]. Transparent structures hardly disturb the wave fields and undergo high relative velocities between the structures and waves, which lead to dominance of frictional and viscous forces. The hydrodynamic loads on transparent structures can be calculated by semi-empirical equations developed originally by Morison et al. [61] and extended by Clauss et al. [16]. On the contrary to transparent structures, the disturbance of the incoming wave field and the effect of diffracting waves by compact structures cannot be neglected. The hydrodynamicists approached this problem by applying linear wave theory, so that it can be assumed that the wave height, H , is small with respect to structure length L , $H/L \ll 1$. Hereby, the rate of change of the fluid velocity parallel to the structure wall is relatively small and the shear stress and the viscosity can be neglected. By neglecting the viscosity and assuming further that the fluid is incompressible, potential flow can be applied to solve the hydrodynamic load problem. In potential flow, there exists a velocity potential ϕ^t , such that the fluid velocity is defined: $\vec{v} \equiv \nabla \phi^t$. Then, it can be proved that the mass conservation of the fluid is governed by Laplace equation ($\Delta \phi^t = 0$), and the momentum conservation after assuming that the flow is irrotational can be integrated along the streamlines to obtain Bernoulli equation. Consequently, the pressure distribution can be computed from Bernoulli equation.

Furthermore, the assumption of $H/L \ll 1$ justifies the perturbation of the wave potential, so that the approximate perturbation solution is obtained by truncating

the perturbation series and keeping only the first two terms: the initial and first order potentials. The initial potential is independent of the wave amplitude and is the potential of basic steady flow, i.e. wave resistance problem, where the ship translates in otherwise calm water with constant velocity. The first order perturbation correction is the unsteady disturbance potential and is proportional to wave amplitude. Likewise, the assumption that the correlation between small wave amplitudes and small unsteady ship motions is linear justifies the linear decomposition of the unsteady problem into diffraction and radiation problems, as first introduced by Haskind [40]. In diffraction problem, the interaction of the motionless structure with incoming waves, and in radiation problem, the forced motion of the structure in absence of the incoming waves are studied. These assumptions have led to a series of methods for computation of hydrodynamic loads on ships and diffracting offshore structures. Initially, in addition to the above assumptions, further assumptions with respect to the structure geometry and forward speed were met to develop theories within the capabilities of modern mathematics. In thin ship theory [66], the beam of the ship compared to its length is assumed to be small. In flat ship theory [87], the draft is small compared to beam and length, and in slender ship or strip theories [97, 106] both draft and beam of the ship are assumed small compared to ship length. All mentioned methods, except the strip theory, could not attract attention from practical point of view. Strip theory is originated from aerodynamics and has been adapted by Korvin-Kroukovski et al. [49] for ship hydrodynamics. The most known strip theories are proposed by Salvesen, Tuck and Faltinsen [97], and Söding [106]. Strip methods solve a number of 2D flow problems for each “strip” and integrate the 2D results to a quasi 3D result for the whole ship to determine the motions. Strip methods are cheap, fast, and for most cases also quite accurate. However, they do not perform so well for high speed ships, blunt hull forms, ships with strong flare, and are invalid at low encounter frequencies which typically occurs in following seas [132]. Due to the advancements in computer technology, 3D methods have gained more attention from practical point of view. 3D methods, in contrary to above 2D methods, impose or assume basically no geometrical restrictions and are more accurate.

Most of the 3D methods belong to the family of boundary integral approaches. In boundary integral approach, Green’s theorem can be used to reduce the unknown three dimensional velocity potentials into two dimensional boundary integral equation(s). Depending on the Green function the body boundary and/or free surface are/is discretized with surface panels (hence the name Boundary Element Method, BEM, or panel methods). Panel methods solve the Laplace equation in the fluid domain by distributing sources and dipoles on these panels, each one associated with a source or dipole distribution of unknown strength.

One drawback of BEMs is that they deal with dense matrices. Therefore, the CPU time and computer memory usage arise dramatically with increasing number of unknowns. Another drawback is the difficulty in calculation of high order derivatives of velocity potentials, which makes BEMs inefficient in nonlinear handling of hydrodynamic load problem. Additionally, all BEMs linearize the free surface to solve seakeeping problem, and suffer from their incapability in representing complex free surface phenomena such as wave breaking and air trapping. Recent developments show that the dense-matrix problem of BEMs can be resolved to some extent by application of accelerating methods, e.g. the fast multipole method (FMM). von Graefe [127] showed nearly indifference of his applied FMM in low number of panels and its effectivity as the number of panels increases.

Although the problem of the calculation time in BEM seems to be mitigated, the problem in handling high derivatives in BEM is still an unsolved problem. As mentioned, the assumption of $H/L \ll 1$ justifies the perturbation of the wave potential. In the perturbed formulation of the total wave potential, higher order derivatives of the lower order potentials appear in the perturbation formulation of the higher order potentials. The higher order potentials cannot be computed accurately; for instance, the problem of second derivative of unsteady potential in the first order steady potential is addressed in [5, 4, 103, 130]. Different solutions have been introduced, with each solution rectifying the problem to some extent but causing other problems in other aspects, e.g. Beck and Magee [4] developed a fully nonlinear time domain approach, and Lin and Yue [57] a body nonlinear approach. Both approaches are proved to be numerically expensive. Second order derivatives on the body boundary condition can be eluded by developing steady potential in the body-fixed coordinate system. This approach improves the numerical accuracy in near field but causes some problems in far field and in multibody simulations [130, 103].

Despite their deficiencies, 3D BEMs have a long tradition in ship and offshore hydrodynamics - perhaps due to being the successors of strip methods - and are still the most common methods in calculation of hydrodynamic loads on ships and offshore structures. Two common advanced 3D BEMs are Neumann-Kelvin approaches (also called Green Function Methods, GFMs) and Rankine source methods (RSMs), with GFMs being the more widespread of the two. GFMs require only the body surface to be discretized with panels, and the free surface is handled analytically by linearizing the free surface condition about a constant steady streaming parallel flow. The assumption of steady parallel flow is not correct as shown by Ohkusu [84] because the added resistance, which is related to the second order of the wave amplitude cannot be calculated accurately. Alternative to GFM, RSM can be used, which has the disadvantage that both ship hull and the free surface in the near field around the ship must be discretized by panels. RSM

2 Numerical Methods

can capture steady flow almost completely but again, similar to GFM, linearize the free surface condition to compute unsteady distribution potential, ship motion problem.

On the contrary to BEMs, field methods directly discretize the Navier-Stokes equations numerically and do not impose any initial conditions, such as L/λ to categorize the structures based on the dominating hydrodynamic forces on them. This distinguishes the field methods from BEMs fundamentally. Field methods have enjoyed more attention in ship and offshore hydrodynamics in the last decade as a result of advancements in computer technology. They are still a core research topic and believed to supersede BEMs in near future. Some applications of field methods in maritime context can be found in [129, 65, 78]. Essentially, in contrast to BEMs, in which potential flow is prescribed, there is no necessity to restrict the governing flow of field methods to a specific flow model; however, it is possible to impose such restrictions, if required. This characteristics allows the field methods to handle any kinds of structure realistically. This comes at a price: generally field methods are computationally expensive because the whole computational domain is discretized in finite entities. The most common discretization methods are finite difference methods (FDM), finite volume methods (FVM), and finite element methods (FEM).

In FDMs, Navier-Stokes equations are handled in differential form. Although being simple to implement, FDMs have not found any significant practical footsteps in hydrodynamics, probably due to their difficulties in handling complex geometries. Application of FDMs in seakeeping problem can be found e.g. in [9, 22]. FEM is based on the use of shape functions on each element of the computational domain. Galerkin method is used to determine the coefficients of the shape function, requiring the integrals to be calculated on each element. The principal drawback of FEM is that the matrices of the linearized equations for unstructured grids are not as well structured as those for regular grids making it difficult to find efficient solution methods. An application of FEM in ship and offshore hydrodynamic context can be found e.g. in [141]. FVMs are by far the most widespread field method in ship and offshore hydrodynamics and can handle complex geometries efficiently. FVMs use the integral form of the Navier-Stokes equations and approximate the integral equations on each control volume. The volume integrals, which contain divergence terms, are converted to surface integrals, using the divergence theorem, and evaluated as fluxes at surfaces of each finite volume. The most used implementation of FVMs are second order accurate because of the difficulties in building higher order approximation schemes. There are an uncountable amount of application of FVM in ship and offshore hydrodynamics, e.g. [54, 96, 78, 79]. Further details about the numerical approaches in field methods can be extracted from textbooks like [25]. Within this thesis, Green function and Rankine source boundary methods, as well

as a finite volume method are employed. Therefore, the theoretical background of these methods are given in the next section concisely. For brevity, the $\cdot$ in dot products are omitted in the whole thesis, i.e. $\vec{a} \cdot \vec{b} \equiv \vec{a}\vec{b}$.

2.1.1 Governing Equations

Water can be regarded as incompressible to cope with velocities involved in ship flows. Although there exists several numerical models to solve ship flow problem, they are all derived from two principle equations, i.e. mass and momentum conservation. The mass and momentum conservation equations in integral form read respectively:

$$\frac{d}{dt} \int_V \rho dV + \int_S \rho(\vec{v} - \vec{v}_s) \vec{n} dS = 0 \quad (2.1)$$

$$\frac{d}{dt} \int_V \rho \vec{v} dV + \int_S \rho \vec{v}(\vec{v} - \vec{v}_s) \vec{n} dS = \int_S (\boldsymbol{\tau} - p \mathbf{E}) \vec{n} dS + \int_V \vec{F} dV \quad (2.2)$$

ρ denotes the fluid density, V the volume, S the bounding surface of control volume, and $\vec{n} dS$ the outward surface normal vector. $\vec{v}$ is the velocity vector of the fluid, $\vec{v}_s$ the velocity vector of control volume surface, and p the pressure. $\vec{F}$ denotes a volumetric body force, t the time, $\mathbf{E}$ the identity matrix, and $\boldsymbol{\tau}$ the viscous stress matrix with components proportional to deformation rates of the fluid. $\boldsymbol{\tau}$ for a Newtonian incompressible fluid reads:

$$\boldsymbol{\tau} = \mu [\nabla \vec{v} + (\nabla \vec{v})^T] \quad (2.3)$$

2.1.2 Reynolds-Averaged Navier-Stokes Equations

The Reynolds number is in the range of 10^9 for large scale and 10^7 for model scale marine applications. Considering that the size of turbulent vortices decreases with increasing Reynolds number, extremely fine resolution in space and time are required to resolve the turbulence. This makes FVM inapplicable for a wide range of engineering problems. As a solution, velocities and pressure may be divided into a time-averaged and a fluctuation part to bring the Navier-Stokes equations closer to a form where a numerical solution is possible.

$$\begin{aligned} \vec{v} &= \bar{\vec{v}} + \vec{v}' \\ p &= \bar{p} + p' \end{aligned} \quad (2.4)$$

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$\vec{\bar{v}}$ and $\bar{p}$ denote the time averages of the fluctuations of velocity $\vec{v}'$ and pressure p' , respectively. With the help of equation (2.4), equations (2.1) and (2.2) are recast into:

$$\frac{d}{dt} \int_V \rho dV + \int_S \rho(\vec{\bar{v}} - \vec{v}_s) \vec{n} dS = 0 \quad (2.5)$$

$$\frac{d}{dt} \int_V \rho \vec{\bar{v}} dV + \int_S \rho \vec{\bar{v}}(\vec{\bar{v}} - \vec{v}_s) \vec{n} dS = \int_S (\vec{\tau} - p\mathbf{E} - \rho \vec{\bar{v}}' \vec{v}') \vec{n} dS + \int_V \vec{F} dV \quad (2.6)$$

Averaging of the fluctuation part of convective term $\rho \vec{v}' \vec{v}'$ results in additional unknowns $\rho \vec{\bar{v}}' \vec{v}'$, Reynolds stress tensor. Traditionally, Boussinesq approach, which couples the Reynolds stress components to the gradient of the average velocities by turbulent viscosity μ_t , are used in practice.

$$\rho \vec{\bar{v}}' \vec{v}' = \mu_t [\nabla \vec{\bar{v}} + (\nabla \vec{\bar{v}})^T] - \frac{2}{3} \rho k \mathbf{E} \quad (2.7)$$

k is the turbulence average kinetic energy.

Many different turbulence models have been developed to solve equation (2.7). In [90], a literature review of turbulence models for ship flow applications is given. Turbulence models may be either algebraic (0-equation models) or based on differential equations (one-equation models, two-equation models). In $k-\epsilon$ turbulence model, a prevalent two-equation model, transport equations are solved for the turbulent kinetic energy, k , and the dissipation rate of velocity fluctuations, ϵ . $k-\epsilon$ model is robust, easy to implement and computationally relatively cheap. However, it is only valid for fully turbulent flows and performs poorly for complex flows involving severe pressure gradient, separation, and strong streamline curvature. $k-\epsilon$ models lack the sensitivity to adverse pressure gradients and is numerically stiff when equations are integrated through the viscous sublayer which are treated with damping functions that have stability issues [60]. Despite its popularity $k-\epsilon$ model is not convincing in benchmark tests for several ship geometries [7]. Another common two-equation turbulence model is $k-\omega$ model. It yields better agreement with experiments in many cases. Similar to $k-\epsilon$ models they solve a transport equation for kinetic energy, k , but instead of turbulence dissipation rate, ϵ , use the turbulence frequency, ω . This model allows for a more accurate near wall treatment with an automatic switch capability from a wall function to a low-Reynolds number formulation based on grid spacing. This model performs significantly better under adverse pressure gradient conditions, does not employ damping functions, and has straightforward Dirichlet boundary conditions, which lead to significant advantages in numerical stability. However, $k-\omega$ model underpredicts the amount of separation for severe adverse pressure gradient flows and

react relatively sensitive to grid quality [7].

In this work, neither of these two models, but Shear Stress Transport - $k-\omega$, SST - $k-\omega$, [59] is used. This model is a variant of the standard $k-\omega$ model, and uses respectively $k-\omega$ and $k-\epsilon$ models near and away from walls using a blending function. SST- $k-\omega$ model inherits the benefits of both underlying models. It behaves reliably in adverse pressure gradients and separating flows. Nevertheless, this model is dependent on wall distance and requires mesh resolution near the wall [7].

2.1.2.1 Discretization

The conservation equations have to be brought into discrete representation to allow for numerical solution. In finite volume methods, the whole domain is discretized into finite control volume of arbitrary shapes. The integral form of governing equations are used to approximate the integrals on each control volume. The volume integrals with divergence terms are converted to surface integrals and are evaluated as fluxes at surfaces of each finite volume. The control volumes contain the flow variables at their centers. For each control volume, an algebraic equation is set up using a finite approximation for volume and surface integrals and for spatial and temporal derivatives. Integrals are approximated using midpoint rule in combination with second order linear interpolation scheme. Gradients are approximated using second order central differencing scheme. However, in cases, where instability may occur, first order upwind difference scheme is used. The temporal derivatives are approximated by second order implicit three time level marching scheme. In general, all approximations are of second order accuracy.

2.1.2.2 Free Surface Treatment

An interface capturing method based on the Volume-of-Fluid approach is used to simulate free surface flows. Volume-of-Fluid method require a kind of a multi-phase model. Immiscible fluids are not considered separately, but are rather replaced by an effective fluid which is considered as a continuum in the whole solution domain. The physical properties of the effective fluid depend on physical properties of constituent fluids and a scalar indicator function, known as volume fraction C . These physical properties for a two phase water-air medium read:

$$\rho = C\rho_{\text{water}} + (1 - C)\rho_{\text{air}} \quad (2.8)$$

$$\mu = C\mu_{\text{water}} + (1 - C)\mu_{\text{air}} \quad (2.9)$$

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The volume fraction function is used to distinguish between two different fluids. A value of unity indicates usually the presence and the value of zero the absence of fluid. It can be shown from the continuity equation that the evolution of volume fraction coefficient is governed by an advection equation that requires strict conservation of mass in the numerical scheme [42]. Given a flow field $\vec{v}$, the advection equation for evolution of C reads:

$$\frac{d}{dt} \int_V C dV + \int_S C(\vec{v} - \vec{v}_s) \vec{n} dS = 0 \quad (2.10)$$

The approximation of the advection term needs special treatment, otherwise excessive diffusion of the advection equation occurs. Thus, the success of a Volume-of-Fluid method depends heavily on the scheme used for the advection of volume fraction function. A first order upwind scheme smears the interface and central differencing schemes may cause unphysical oscillation of the solution. Therefore, it has been necessary to develop schemes, which keep the free surface sharp, while also producing monotonic profiles for volume fraction function. Over the years, a multitude of different methods have been developed. These methods can roughly be divided into three categories: donor-acceptor formulation [42], higher order differencing schemes [20, 63], and line techniques [143, 92]. The vast majority of these schemes are based on higher order differencing schemes. Among these, High Resolution Interface Capturing scheme (HRIC) [62, 63, 20] is one of the most popular schemes widely used in CFD codes. This is the scheme used in the RANS simulations of this work. HRIC lies on a blending of bounded upwind and downwind schemes. The aim of this scheme is to combine the compressive properties of the downwind differencing scheme with the stability of the upwind scheme. Additionally, HRIC accounts for the local position and advance direction of the free surface to retain a sharp interface. More on HRIC can be found in [62, 63, 135].

2.1.3 Modeling of Waves

A dynamic boundary condition is defined for the RANS simulations to generate regular periodic sinusoidal linear Airy waves. Horizontal, $u(x, z, t)$, and vertical, $w(x, z, t)$, velocities, related pressure, $p(x, z, t)$, and surface elevation, $\zeta(x, t)$ of

Airy waves are described as following [138]:

$$\zeta(x, t) = h \cos(kx - \omega t) \quad (2.11)$$

$$u(x, z, t) = \omega h e^{kz} \cos(kx - \omega t) \quad (2.12)$$

$$w(x, z, t) = \omega h e^{kz} \sin(kx - \omega t) \quad (2.13)$$

$$p(x, z, t) = \frac{\omega^2 h e^{kz}}{k} \cos(kx - \omega t) - \frac{\omega^2 h^2 e^{2kz}}{2} \quad (2.14)$$

ζ is the unsteady free surface elevation, h is the wave amplitude, $k = 2\pi/\lambda$ is the wave number, and ω is the wave angular frequency. Considering a right-handed coordinate system, the waves propagate in the positive x-direction.

2.1.4 Boundary Element Methods

Neglecting the viscosity in equation (2.2) reduces the momentum conservation equation to Euler equation. A further simplification is the assumption of irrotational flow that results in potential flow. Potential flows are inviscid, irrotational, and allow to reformulate the vectorial velocity fields into the scalar velocity potential fields. For an incompressible potential flow, mass conservation is defined by Laplace equation.

$$\Delta \phi^t = 0 \quad (2.15)$$

ϕ^t denotes the total velocity potential. The Laplace equation ensures the divergence-freeness of the velocity field. Irrotationality of the potential field allows the integrability of the momentum conservation equation along the streamlines to yield Bernoulli equation.

$$p = -\rho(\phi_t^t + \frac{(\nabla \phi^t)^2}{2} + g\zeta^t - \frac{u^2}{2}) + p_a \quad (2.16)$$

The subscript t stand for time derivatives. Here, ρ is the fluid density, g gravitational acceleration, p atmospheric pressure which is assumed constant (equal to p_a) at the free surface and u speed of steady inflow. Therefore, equation (2.15), is sufficient to calculate the unknown velocities. The potential velocities should not only satisfy the Laplace-equation within the domain but also the boundary conditions.

The condition that no water penetrates through the free surface is defined by its elevation ($z = \zeta^t(x, y, t)$). Thus, a fluid particle with vertical coordinate z_F , which is at the free surface $z = \zeta^t$ remains at the free surface. Therefore, the vertical

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component of the fluid velocity is equal (with respect to a water particle) to time derivative of the free surface height. Considering the change of a point with the fluid (total derivative $\frac{D}{Dt} = \frac{\partial}{\partial t} + \nabla\phi^t\nabla$), the kinematic condition reads:

$$\frac{D\zeta^t}{Dt} = \frac{Dz_F}{Dt} = \phi_z^t \quad (2.17)$$

subscript z indicates derivation with respect to z -coordinate. The dynamic boundary condition states that the total pressure is equal to atmospheric pressure at free surface:

$$p - p_a = 0 \quad \text{at } z = \zeta^t \quad (2.18)$$

Combining equations (2.16) to (2.18) yields at $z = \zeta^t$:

$$\phi_{tt}^t + 2\nabla\phi^t\nabla\phi_t^t + \nabla\phi^t\nabla\left(\frac{1}{2}(\nabla\phi^t)^2\right) + g\phi_z^t = 0 \quad (2.19)$$

The subscript tt stands for second time derivatives. The elevation of free surface, which contributes in solving the velocity potentials via kinematic boundary condition can be calculated implicitly following equation (2.16), and under the assumption of atmospheric pressure on the free surface.

$$\zeta^t = -\frac{\phi_t^t + \frac{(\nabla\phi^t)^2 - u^2}{2}}{g} \quad \text{at } z = \zeta^t \quad (2.20)$$

Therefore, the fluid pressure and velocity fields are coupled indirectly with each other and no special coupling treatments like in RANS are necessary.

No water penetrates the ship's hull. Thus, the normal velocity should be equal to the velocity of the adjacent fluid on the wetted hull surface S .

$$(\nabla\phi^t - u)\vec{n} = 0 \quad (2.21)$$

$\vec{n}$ is the unit normal vector, and u corresponds steady flow velocity. In addition to the free surface and body boundary conditions, bottom and radiation conditions should be satisfied. Bottom condition signifies the condition of no flow through the bottom at $z = z_B$:

$$\phi_z^t = 0 \quad \text{at } z = z_B \quad (2.22)$$

Usually the bottom condition is satisfied numerically. The same numerical approach may be employed to model channel walls or bathymetry boundary regions. Depending on the numerical method analytical or numerical methods are required to satisfy the so-called radiation condition. Waves generated by the ship radiate away from the ship (radiation condition) and vanish at infinity.

Boundary integral methods have a long tradition in marine hydrodynamics. Hereof, Green theorem is used to reduce the three dimensional problem into a two dimensional one by reducing the unknown three dimensional velocity potentials into two dimensional boundary integral equations with Green functions and their normal derivatives as integrands. In boundary integral methods, velocity potentials are perturbed in terms of dependency on wave amplitude h .

$$\phi^t = \phi^0 + \phi^1 + \phi^2 + \dots \quad (2.23)$$

The superscript indicate the dependence order of the potential to h . In linearized wave theory, it is assumed that the amplitude of the oscillatory wave motion in comparison to the wave length is small. Then, the approximate perturbation solution is obtained by truncating the series and keeping only the first two terms. Each potential term must satisfy necessarily the Laplace equation.

$$\phi^t(\vec{r}, t) = \phi^0(\vec{r}) + \phi^1(\vec{r}, t) \quad (2.24)$$

$\phi^0(\vec{r})$ is independent of the wave amplitude and is the potential of basic steady state flow, i.e. the wave resistance problem. $\phi^1(\vec{r}, t)$ is the unsteady disturbance potential and is proportional to h .

The prediction of hydrodynamic forces on a ship or offshore structure presuppose the correct understanding of the steady and unsteady flow fields. Within perturbation theory the solution of steady flow problem is a prerequisite of unsteady flow problem. In many theoretical formulations, the steady velocity fields are completely neglected or represented by a parallel uniform flow to reflect the steady forward speed. In these approaches, Neumann-Kelvin approaches or Green function methods, the free surface condition is linearized about a constant steady flow. In case of zero speed, there exists analytical solutions for the free surface condition. The body boundary condition is solved numerically by discretizing the body surface with panels. In case of a surface-piercing vessel, the kernel Green function is augmented by a line integral at the intersection of the hull rim with the undisturbed free surface [12]. The main numerical effort in this approach is to evaluate the Green function accurately (especially for forward velocities). This approach is mathematically inconsistent in that no interaction with the stationary flow due to the forward speed is considered. However, neglecting this interaction

can only be justified if the body is slender [84]. For non-slender ships it is necessary to account for the steady disturbance and its interactions, particularly near the bow and stern. Furthermore, Ohkusu [84] proved that neglecting the steady flow and assuming the uniform and ambient flow is not a correct substitution for the effect of steady forward speed of a body on hydrodynamic forces in waves. Many scientists have attempted to incorporate the effect of non-uniformity of the steady flow around a bluff body into the linearized free surface condition (e.g. [146, 84]). Zhao et al. [146] tried to neglect the effect of the local steady flow around the body, i.e. to approximate the steady flow by the far-field steady flow over all of the free surface, but they did not achieve satisfactory results. Usually, the simple double body solution (as defined e.g. in [18]) corresponding to the zero Froude number limit of the steady problem is used. It allows to simulate the free surface as a horizontal solid wall. Double body method provides satisfactory results for zero or nearly zero forward speeds and not for moderate to high Froude numbers. Sutulo et al. [115] compared model test data and double body computations for a large tanker and a small tug in close vicinity. Large differences were found for the horizontal forces and yaw moments, which were ascribed to the simplifications of the double body approach. On the contrary to zero-speed Green function methods, which are still by far the most prevalent method in marine applications, non zero-speed Green function methods have never gained practical footsteps.

An alternative to Green function methods are Rankine source methods. Rankine source methods employ Rankine sources, which are simple Green functions ($\frac{1}{r}$), to form the kernel of domain boundary integrals. Apart from having a simple Green function, Rankine sources make it possible to accommodate a broad band of linear and nonlinear free surface solutions; hence, there is no need to substitute the steady flow with parallel flow as in Green function methods, because Rankine source methods treat free surfaces numerically by discretizing the free surfaces. Thus, Rankine source methods can model the real flow in all details except for effects of viscosity, flow separation, and wave breaking, i.e. the intrinsic characteristics inherited from simplifications imposed on the Navier-Stokes equation to obtain potential flow formulation. Rankine source methods were first used by [18, 31] to solve the wave resistance problem. Nakos [64] and Bertram [5] contributed substantially in successful application of Rankine sources in seakeeping problem in 3D space. Nakos used the solution of double body to approximate the steady state problem, whereas Bertram used the non-linear steady flow field previously developed by Jensen [32].

In this thesis, three different boundary element methods are used within this thesis: two zero-speed Green function methods, and one Rankine source method.

AQW The first employed zero-speed Green function method is called AQW hereinafter, and is implemented in *Ansys Aqwa* package [1]. It applies the following pulsating Green function for finite depth water.

$$G(\vec{r}, \vec{\xi}, \omega) = \frac{1}{r} + \frac{1}{r_2} + \int_0^\infty \frac{2(k + \frac{\omega^2}{g})e^{-kd} \cosh k(z+d) \cosh k(\zeta+d)}{k \sinh kd - \frac{\omega^2}{g} \cosh kd} k R J_0 dk \\ + i2\pi \frac{(k_0 + \frac{\omega^2}{g})e^{-k_0 d} \cosh k_0(z+d) \cosh k_0(\zeta+d)}{\sinh k_0 d + k_0 d \cosh k_0 d - \frac{\omega^2 d}{g} \sinh k_0 d} k_0 R J_0 \quad (2.25)$$

where J_0 is the Bessel function of the first kind and

$$R = \sqrt{(x - \xi)^2 + (y - \eta)^2} \quad (2.26)$$

$$r = \sqrt{R^2 + (z - \zeta)^2} \quad (2.27)$$

$$r_2 = \sqrt{R^2 + (z + \zeta - 2d)^2} \quad (2.28)$$

Where k and k_0 ($k = \frac{\omega^2}{g}$ and $k_0 \tanh k_0 d = \frac{\omega^2}{g}$) are wave numbers in deep and shallow waters, respectively, d water depth, and $\vec{\xi}(\xi, \eta, \zeta)$ is the source point position. This function fulfills the linearized free surface condition, the radiation condition as well as the Laplace equation. By using the Green theorem and employing source points of unknown strengths the unsteady velocity potential can be expressed as surface integrals of the form

$$\phi^1(\vec{r}) = \frac{1}{4\pi} \int_{S_0} \sigma(\vec{\xi}) G(\vec{r}, \vec{\xi}, \omega) dS \quad (2.29)$$

in which the source strength over the mean wetted hull surface can be determined by the hull surface boundary condition given by equation (2.21). In equation (2.29), σ denotes the unknown source strength of the source point ξ .

Since the direct evaluation of the frequency domain by the pulsating Green function in finite depth is time consuming, a Green function database to efficiently calculate the Green function and its first order derivatives is used by the package.

GLR The second method used in this work is an implementation of zero-speed pulsating Green function published in [120], and called **GLR** within this context.

The computing time involved in boundary integral equation methods depends linearly with the computing time required for evaluating the Green function and its gradients. In [120], a numerical evaluation of kernel Green function and its gradients is introduced. The method is especially well suited for deep sea and is based on five complementary expressions for the Green function and its gradient that are useful in neighboring regions of the quadrant, in which the Green function is defined. These expressions consist of asymptotic expansions, ascending series, two complementary Taylor series, and a numerical approximation based on a modified form of the Haskind integral representation. The series and the numerical approximation express the Green function and its gradient as sums of power series and terms involving functions of only one real variable. Reference is given to [120] for details on the kernel function and the associated numerical approximation of the function. This Green function method is implemented supplementarily in *GL Rankine* [110, 129, 128, 130, 127].

AQW and **GLR** have principle similarities and distinctions. One similar aspect of both methods is that the total velocity potentials in both **AQW** and **GLR** are calculated under consideration of steady inflow $-ux$ and under ignorance of steady perturbation potential ϕ^0 in frequency domain.

$$\phi^t(\vec{r}, t) = -ux + \text{Re}(\hat{\phi}^1 e^{i\omega t}) \quad (2.30)$$

Another similar capability of the solvers is their ability to remove irregular frequencies. Wave resonance may happen in surface piercing bodies due to unphysical standing waves within the bodies. The occurrence of irregular frequencies causes large errors in the solution over a substantial frequency band around these frequencies. Irregular frequencies are difficult to detect and are recommended to be removed. Zhu and Lee [148] studied different methods to remove the irregular wave frequency problem. Internal lid method is perhaps the most common method that assumes the existence of a fluid field within the body, which satisfies the free surface boundary condition. The interior free surface can be represented by a series of panels, which contributes in computation of equation (2.29).

Another similarity is that both zero-speed Green function methods are not applicable for forward speeds without further measurements. **GLR** simply replaces the wave frequency, ω , by the encounter frequency, ω_e , and boundary conditions are adapted for the reference frame moving with the mean steady forward speed. The encounter frequency approach used in **GLR** is efficient for very small forward speeds, but its accuracy in response predictions deteriorates with increasing forward speeds (see section 3.3). **AQW**, on the other hand, not only replaces ω with

ω_e but also uses the encounter frequency approach published in [97] to correct the added mass and damping terms, too. As it will be shown later, the correction method employed in AQW is more effective than GLR's correction method.

Among distinguished differences of GLR is its application of patch method [107], in which the boundary conditions are fulfilled on the average over each panel and not at discrete collocation points. Additionally, the panels are dissingularized in GLR by locating the source points ξ_j at some distance above the panel centroid and outside the fluid domain. AQW uses Hess and Smith panels on the discretized hull surface. These panels fulfill the boundary conditions at discrete collocation points, where the potentials are singular. Furthermore, the flow potentials are considered constant over the panel in AQW.

RSM The third boundary integral method used within this work is a Rankine source method, and is notated RSM herein. It is common in Rankine source methods to incorporate the steady forward velocity separately in ϕ^t formulation.

$$\phi^t(\vec{r}, t) = -ux + \phi^0(\vec{r}) + \text{Re}(\hat{\phi}^1 e^{i\omega t}) \quad (2.31)$$

Rankine sources are dissingularized and positioned outside the fluid domain or inside the body at small normal distance from center points of each panel. A Rankine source induces a potential

$$G(\vec{r}, \vec{\xi}_j) = \frac{1}{|\vec{r} - \vec{\xi}_j|} \quad (2.32)$$

and does not satisfy the free surface as well as radiation conditions analytically. The free surface problem is solved by discretizing the free surface and allocating Green function sources (equation (2.32)) to each panel. In [21], a modified kinematic free surface condition is applied together with staggered grid approach, in which the source points are shifted one grid downstreams, to enforce the open boundary and radiation conditions. Similarly, ad hoc solutions are usually applied to avert splashing and wave breaking, lest the numerical collapse occurs. Splashes are normally a trivial problem if free surface grids are not finer than required to resolve the interesting waves. Söding [108] showed that it is sufficient in his code to use a grid spacing of $0.75u^2/g$ in front of the body, and $0.6u^2/g$ beside and behind the body to resolve the waves accurately. Breaking waves, on the other hand, often require special treatments, e.g. application of a modified free surface condition as in [100]. Analogously, the steady potential is discretized but can be treated linearly or nonlinearly. Double body approximation of steady velocity po-

tential may be applied for slow forward velocities if no free surface discretization of steady potential is to be applied.

As long as the effect of steady velocity field is not neglected completely in boundary integral context, the terms representing the effect of oscillatory motion of the body within the steady velocity field, so called m -terms, occur in the linearized body boundary conditions. They are also corollaries of transferring the boundary condition from instantaneous position of the body to its mean position to develop a consistent perturbation solution in inertial coordinate system [68]. m -terms, as named first by Ogilvie and Tuck [82], cause special computational difficulties. Zhao et al. [144] showed that the direct evaluation of m -terms by a finite difference method introduces a significant amount of error. Also Bertram et al. [5] cited that numerical representation of m -terms had posed a considerable problem in development of their Rankine source panel code. Several different approaches to treat m -terms have been introduced. Bertram et al. [5] applied a simple slender-body theory to approximate the m -terms by representing the steady flow by a source distribution along the centerline of the ship. Nakos [64] used the Stokes theorem for the accurate evaluation of the second derivatives of the steady potential by converting it into the first derivative on the body surface and a vanishing waterline integral. Zhao et al. [145] introduced two numerical methods: a direct pressure integration and a conservation of momentum approach. Hachmann [36] reformulated the free surface boundary and developed the steady potential around the body-fixed coordinate system instead of inertial boundary condition. Hereto, the m -terms are not contained in the body boundary condition but in the linearized free surface boundary condition, where the influence of m -terms seems to be not significant.

Söding [108] developed the Hachmann approach further to contrive a three dimensional Rankine source code, *GL Rankine*. This method has been developed to regard especially the cases of high forward speed up to Froude number of 0.45. Recently, von Graefe [127] reengineered as well as extended the code to treat ship to ship interaction and shallow water problems. The numerical background of *GL Rankine* is well documented in [21] and [127]. In this work, the Rankine source method within *GL Rankine* is used. The steady perturbation potential is approximated in this work by double body approach. In double body approach, the free surface boundary condition can be simplified

$$\phi_z^0 = 0 \quad \text{at } z = 0 \quad (2.33)$$

so that no free surface panels are required any more and the Rankine sources are mirrored about $z = 0$. Enforcing a symmetric flow about this plane fulfills the kinematic but violates the dynamic free surface condition.

In all boundary element methods, the assumption of small unsteady motion of the ship justifies the linear decomposition of the unsteady problem into diffraction and radiation problems. Thereof, the unsteady potential can be decomposed into incident wave, diffraction and radiation potentials. Additionally, this assumption allows the decomposition of the radiation potential into six conical potentials φ^k for each body b to represent rigid motions, with each satisfying an inhomogeneous Neumann boundary condition on the instantaneous position of the body surface.

$$\phi^1(\vec{r}, t) = \left(\hat{\phi}^w + \hat{\phi}^d + \sum_{k=1}^{6b} \hat{\varphi}^k \right) e^{i\omega t} \quad (2.34)$$

The linearized potential of the incident wave in deep water with respect to inertial system reads [137]

$$\phi^w(x, z, t) = \text{Re} \left(\frac{ih\omega}{k} e^{kz} e^{i(\omega t - kx)} \right) \quad (2.35)$$

When computing the complex amplitude of the diffraction potential, $\hat{\phi}^d$, the body motion amplitudes are set to zero and equation (2.36) is inserted into the boundary conditions. Similarly, equation (2.37) is inserted into the boundary conditions to derive the equation for computation of complex amplitude of conical radiation potential $\hat{\varphi}^k$.

$$\hat{\phi}^1 = \hat{\phi}^w + \hat{\phi}^d = \hat{\phi}^w + \sum_{j=1}^n \hat{q}_j G(\vec{r}, \vec{\xi}_j) \quad (2.36)$$

$$\hat{\phi}^1 = \hat{\varphi}^k = \sum_{j=1}^n \hat{q}_j G(\vec{r}, \vec{\xi}_j) \quad (2.37)$$

2.2 Rigid Body Dynamics

Following the theoretical formulation of RANS and boundary element methods for computation of fluid loads on structures, rigid body dynamics will be described in this section. Consequently, it will be shown how rigid body and hydrodynamic solvers interact with each other.

2.2.1 Dynamics of Single Bodies

The bodies are assumed rigid within this work. It means that the position of mass particles within each respective body do not change relative to each other. The theory of rigid body dynamics of marine structures follows Newton-Euler principle. Two coordinate systems, one inertial, $\mathcal{K}_0$, and one body-fixed, $\mathcal{K}_1$, are defined. As per Newton's second law of motion, the force exerted on a body in inertial coordinate system is in direct relationship with the body's mass and linear acceleration.

$$\vec{F} = m\dot{\vec{v}} \quad \text{with } \vec{v} = \dot{\vec{r}} \quad (2.38)$$

Euler showed that equation (2.38) is equivalent to his first law of motion. The Euler first and second law of motion are the fundamental equations of Newton-Euler dynamics:

$$\vec{F} = \frac{d\vec{P}}{dt} = \frac{d(m\vec{r})}{dt} \quad \text{linear momentum law} \quad (2.39)$$

$$\vec{M} = \frac{d\vec{L}}{dt} = \frac{d(\mathbf{I}\vec{\omega})}{dt} \quad \text{angular momentum law} \quad (2.40)$$

where $\mathbf{I}$, $\vec{\omega}$, $\vec{P}$, $\vec{L}$, $\vec{F}$ and $\vec{M}$ are the matrix of mass moment of inertia, angular velocity vector, linear and angular momenta, force and moment, respectively. Equations (2.39) and (2.40) are valid both in inertial and body-fixed coordinate systems. Equation (2.40) under coincidence of the origin of the body-fixed coordinate system with the body's center of gravity reads [140]:

$$\vec{M} = {}^0\mathbf{I}\dot{\vec{\omega}} + \tilde{\vec{\omega}} {}^0\mathbf{I}\vec{\omega} \quad (2.41)$$

Mass moment of inertia in inertial coordinate system ${}^0\mathbf{I}$ is dependent on body position, whereas it is time invariant in body-fixed coordinate system, ${}^1\mathbf{I}$. Therefore, it is preferred to compute mass moment of inertia in body-fixed coordinate system, and then transform it to inertial coordinate system. Mass moment of inertia can be transformed from body-fixed to inertial coordinate system by means of transformation matrix $\mathbf{T}$. $\mathbf{T}$ consists of three successive Euler angles ϕ, θ, ψ , which represent rotation about x, y and z (roll, pitch, yaw) of the body-fixed coordinate system, respectively; hence:

$${}^0\mathbf{I} = \mathbf{T} {}^1\mathbf{I} \mathbf{T}^T \quad (2.42)$$

with

$$\begin{aligned}
 \mathbf{T}(\psi, \theta, \phi) &= \mathbf{T}_1(\psi) \mathbf{T}_2(\theta) \mathbf{T}_3(\phi) \\
 &= \begin{bmatrix} \cos \psi & -\sin \psi & 0 \\ \sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \phi & -\sin \phi \\ 0 & \sin \phi & \cos \phi \end{bmatrix} \\
 &= \begin{bmatrix} \cos \psi \cos \theta & \cos \psi \sin \theta \sin \phi - \sin \psi \cos \phi & \cos \psi \sin \theta \cos \phi + \sin \psi \sin \phi \\ \sin \psi \cos \theta & \sin \psi \sin \theta \sin \phi + \cos \psi \cos \phi & \sin \psi \sin \theta \cos \phi - \cos \psi \sin \phi \\ -\sin \theta & \cos \theta \sin \phi & \cos \theta \cos \phi \end{bmatrix}
 \end{aligned} \tag{2.43}$$

Using equations (2.42) and (2.43), equation (2.41) for inertial coordinate system reads:

$$\vec{M} = \mathbf{T}^1 \mathbf{I} \mathbf{T}^T \dot{\vec{\omega}} + \tilde{\omega} \mathbf{T}^1 \mathbf{I} \mathbf{T}^T \vec{\omega} \tag{2.44}$$

Consequently, the equation of motion in its matrix form can be obtained from equations (2.39) and (2.44).

$$\begin{bmatrix} m\mathbf{E} & 0 \\ 0 & \mathbf{T}^1 \mathbf{I} \mathbf{T}^T \end{bmatrix} \begin{bmatrix} \ddot{\vec{r}} \\ \dot{\vec{\omega}} \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & \tilde{\omega} \mathbf{T}^1 \mathbf{I} \mathbf{T}^T \end{bmatrix} \begin{bmatrix} \dot{\vec{r}} \\ \vec{\omega} \end{bmatrix} = \begin{bmatrix} \vec{F} \\ \vec{M} \end{bmatrix} \tag{2.45}$$

$\mathbf{E}$ is the identity matrix and $\vec{r}$ is an arbitrary body distance vector with respect to $\mathcal{K}_0$.

For convenience, the notation $\vec{q}$ is used to denote generalized coordinates¹ of the body reference. Generalized coordinates are very common in multibody, kinematically constrained and fluid-structure-interaction simulations. This is why, many codes consider the generalized coordinates to ease their extensibility. In Cartesian coordinate system $\vec{q}$ has six elements (for each body).

$$\vec{q} = [x \ y \ z \ \phi \ \theta \ \psi]^T \tag{2.46}$$

¹not to be mistaken with generalized coordinates or generalized modes mentioned in section 1.2.

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First and second time derivatives of $\vec{r}$ are related to $\vec{q}$ as following:

$$\begin{aligned}\dot{\vec{r}} &= \frac{d\vec{r}}{dt} = \frac{\partial \vec{r}}{\partial \vec{q}} \dot{\vec{q}} \\ \ddot{\vec{r}} &= \frac{d\dot{\vec{r}}}{dt} = \frac{\partial \dot{\vec{r}}}{\partial \vec{q}} \dot{\vec{q}} + \frac{\partial \dot{\vec{r}}}{\partial \dot{\vec{q}}} \ddot{\vec{q}}\end{aligned}\quad (2.47)$$

Therefore, equation (2.45) for generalized coordinates is recast in equation (2.48) [21].

$$\begin{bmatrix} m\mathbf{E} & 0 \\ 0 & \mathbf{T}^1 \mathbf{I} \mathbf{T}^T \mathbf{J}_\alpha \end{bmatrix} \ddot{\vec{q}} + \begin{bmatrix} 0 & 0 \\ 0 & \mathbf{T}^1 \mathbf{I} \mathbf{T}^T \mathbf{J}_\omega + \dot{\mathbf{T}} \mathbf{T}^1 \mathbf{I} \mathbf{T}^T \mathbf{J}_\alpha \end{bmatrix} \dot{\vec{q}} = \begin{bmatrix} \vec{F} \\ \vec{M} \end{bmatrix} \quad (2.48)$$

$\mathbf{J}_\alpha$ and $\mathbf{J}_\omega$ are Jacobins related to rotation and angular velocity, respectively.

$$\mathbf{J}_\alpha = \begin{bmatrix} 0 & -\sin \psi & \cos \psi \cos \theta \\ 0 & \cos \psi & \sin \psi \cos \theta \\ 1 & 0 & -\sin \theta \end{bmatrix} \quad (2.49)$$

$$\mathbf{J}_\omega = \begin{bmatrix} 0 & -\dot{\phi} \cos \psi \sin \theta & -\dot{\phi} \cos \psi \sin \theta - \dot{\theta} \cos \psi \\ 0 & -\dot{\phi} \sin \psi \sin \theta & \dot{\phi} \sin \psi \cos \theta - \dot{\theta} \sin \psi \\ 0 & -\dot{\phi} \cos \theta & 0 \end{bmatrix} \quad (2.50)$$

Equations (2.45) and (2.48) can be simplified in the form:

$$\mathbf{M} \ddot{\vec{r}} + \mathbf{C} \dot{\vec{r}} = \vec{F} \quad (2.51)$$

$\mathbf{C}$ is the body's Coriolis and centripetal matrix. Note that contents of $\mathbf{M}$ and $\mathbf{C}$ matrices are different depending on whether $\vec{r}$ is referred to generalized coordinate.

The RANS code used within this work uses the generalized form of motion equation (equation (2.48)). Temporal discretization of equation (2.51) is usually done by second order Crank-Nicholson method. In [48], higher order explicit Adams-Bashforth and higher-order implicit Adams-Moulton methods up to sixth order are studied. It is reported that these methods are less practicable due to their adverse effects on the convergence behavior of numerical simulations. Equation (2.51) is numerically stiff and demands relatively small time steps [131]. In order to solve equation (2.51), $\mathbf{I}$ -matrix is found at first. $\mathbf{C}$ -matrix is usually treated explicitly in solving the discretized form of this equation [48]. The hydrodynamic forces are

gained from the coupled RANS solver (section 2.1.2), and are treated as source terms, $\vec{F}$. Since the elements of both $\mathbf{C}$ and $\vec{F}$ are functions of the solution of the motion equation, iterative procedures are required to solve Coriolis, centripetal, and hydrodynamic forces as well as body motions.

Application of linear wave theory and assumption of linear correlation between amplitudes and ship motions justify the linearization of equation (2.51). The linearized equation of motion reads:

$$\mathbf{M}\ddot{\vec{\eta}} = \vec{f} \quad (2.52)$$

where $\ddot{\vec{\eta}}$ and $\vec{f}$ in equation (2.52) are the perturbed equivalents of $\ddot{\vec{r}}$ and $\vec{F}$ in equation (2.51). $\vec{f}$ consists of hydrostatic force, $\mathbf{R}\vec{\eta}$, hydrodynamic force, $\vec{f}^{\text{hyd}}$, and other excitation forces, $\vec{f}^{\text{exc}}$, such as wave, rudder, mooring and contact element forces. Cummins [17] showed that the hydrodynamic forces of potential radiation flow are related to frequency-dependent added mass $\mathbf{A}(\omega)$ and potential damping, $\mathbf{B}(\omega)$, so that:

$$\vec{f}^{\text{hyd}} = -\mathbf{A}(\infty)\ddot{\vec{\eta}}(t) - \int_0^t \mathbf{K}(t - \tau)\dot{\vec{\eta}}(\tau)d\tau \quad (2.53)$$

$\mathbf{A}(\infty)$ is the added mass in infinite frequency, t and τ denote time and $\mathbf{K}(t)$ is the memory function. Using equation (2.53) the linearized motion equation (equation (2.52)), can be written in the famous integro-differential form of the equation of motion in time domain [27].

$$(\mathbf{M} + \mathbf{A}(\infty))\ddot{\vec{\eta}}(t) + \int_0^t \mathbf{K}(t - \tau)\dot{\vec{\eta}}(\tau)d\tau + \mathbf{R}\vec{\eta}(t) = \vec{f}^{\text{exc}} \quad (2.54)$$

Time domain simulation of the dynamic response of marine structures by direct evaluation of convolution intergals may be computationally demanding depending on the time step, simulation length and degrees of freedom of the model [45]. This is why significant efforts have been dedicated to finding alternative representation of equation (2.54) or its convolution term. Generally, these representations can be divided into three different categories [117]. The first representation is replacing the frequency dependent added mass and damping by constant coefficients. The second approach is the approximation of the equation of motion by a state-space model, as originally proposed by Cummins in [17]. A recent approach of this kind can be found in [86]. The third and probably the most common intensively researched approach is approximation of the convolution integral by a state-space formulation. [117] and most recently [41] are two examples of this approach.

If the excitation forces are harmonic, i.e. periodic with constant amplitude,

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$\vec{f}^{\text{exc}} = \hat{f}e^{i\omega t}$, and if it is still assumed that the ship motion responses are in linear correlation with the excitation forces, i.e. $\hat{\eta} \propto \hat{f}$ and $\vec{\eta} = \hat{\eta}e^{i\omega t}$, the frequency domain form of equations (2.52) and (2.54) can be used to determine the ship motion responses.

$$-\omega^2 \mathbf{M} \hat{\eta} = \hat{f} \quad (2.55)$$

$$\left((-\omega^2 (\mathbf{M} + \mathbf{A}(\omega)) + i\omega \mathbf{B}(\omega) + \mathbf{R}) \right) \hat{\eta} = \hat{f}^{\text{exc}} \quad (2.56)$$

Similar to equation (2.24), pressure is divided into a stationary, p^0 , and an instantaneous, p^1 , parts. From the steady pressure, ship resistance force ($\vec{f}_1^0$), sinkage force ($\vec{f}_3^0$) and steady trim moment ($\vec{f}_5^0$) are calculated.

$$\vec{f}_1^0 = \int_S p^0 \vec{n}_x dS \quad (2.57)$$

$$\vec{f}_3^0 = \int_S p^0 \vec{n}_z dS \quad (2.58)$$

$$\vec{f}_5^0 = \int_S p^0 (\vec{r} \times \vec{n}_y) dS \quad (2.59)$$

$\vec{n}_x$, $\vec{n}_y$ and $\vec{n}_z$ are the components of hull normal vector $\vec{n}$ in x , y and z directions, respectively. The unsteady components of pressure derived from perturbation of Bernoulli equation are used to solve the radiation and diffraction forces and moments in the ship motion problem.

2.2.2 Dynamics of Multibody Systems

The equation of motion related to single bodies cannot be applied to multibodies without further measurements. In a multibody system consisting of two diffracting structures, the incoming wave, after its scattering by one of the structures, becomes the incident wave for the second structure. This generates a multiple scattering effect between both structures. When these waves coincide in-phase, there will be an escalation of the magnitude of the wave excitation forces on the floating bodies compared to that of a single stand-alone floating body. On the opposite, when these waves coincide out-of-phase, the magnitude of the wave excitation forces on the floating structure attenuates. Similarly, the motion of a structure creates radiated waves, which interfere with the second structure and are altered. The importance and significance of hydrodynamic interaction in multibody systems depend on the geometrical parameters of the involved bodies relative to the others, hydrodynamics of each respective body, separation gap between the bodies, and the environmental conditions. Therefore, the response amplitude operators for each of

the hydrodynamic interacting structures in multibody conditions are different from those of stand-alone conditions. Theoretically, the hydrodynamics of multibodies is a straightforward extension of single body hydrodynamics. In frequency domain, equation (2.56) can be rearranged as a 6×6 equation system for the unknown ship motions,

$$\underbrace{\begin{pmatrix} \hat{\mathbf{F}}^{\text{rad},\hat{u}} & \hat{\mathbf{F}}^{\text{rad},\hat{\alpha}} \\ \hat{\mathbf{M}}^{\text{rad},\hat{u}} & \hat{\mathbf{M}}^{\text{rad},\hat{\alpha}} \end{pmatrix}}_{\hat{\mathbf{B}}_{11}} + \omega_e^2 \mathbf{M} \underbrace{\begin{pmatrix} \hat{u} \\ \hat{\alpha} \end{pmatrix}}_{\hat{\eta}_1} = \underbrace{\begin{pmatrix} -\hat{\mathbf{F}}^{\text{w}} \\ -\hat{\mathbf{M}}^{\text{w}} \end{pmatrix}}_{\hat{f}_1^{\text{exc}}} \quad (2.60)$$

$\hat{\mathbf{F}}^{\text{rad},\hat{u}}$, $\hat{\mathbf{F}}^{\text{rad},\hat{\alpha}}$, $\hat{\mathbf{M}}^{\text{rad},\hat{u}}$, $\hat{\mathbf{M}}^{\text{rad},\hat{\alpha}}$ are the matrices containing the complex amplitudes of radiation forces and moments due to unit ship translational and rotational motions. $\mathbf{M}$ is the mass matrix containing masses and moments of inertia of the body. $\hat{\mathbf{F}}^{\text{w}}$ and $\hat{\mathbf{M}}^{\text{w}}$ are the vectors containing the complex amplitudes of forces and moments due to the incident wave and diffraction. For the interaction problem of two ships, equation (2.60) becomes [127]:

$$\begin{bmatrix} \hat{\mathbf{B}}_{11} & \hat{\mathbf{B}}_{12} \\ \hat{\mathbf{B}}_{21} & \hat{\mathbf{B}}_{22} \end{bmatrix} \begin{bmatrix} \hat{\eta}_1 \\ \hat{\eta}_2 \end{bmatrix} = \begin{bmatrix} \hat{f}_1^{\text{exc}} \\ \hat{f}_2^{\text{exc}} \end{bmatrix} \quad (2.61)$$

Here $\hat{\mathbf{B}}_{11}$ and $\hat{\mathbf{B}}_{22}$ are the same matrices as in the single body cases for the first and second ships, respectively. $\hat{\mathbf{B}}_{12}$ contains forces and moments due to radiated waves of the first ship acting on the second ship. $\hat{\mathbf{B}}_{21}$ contains these corresponding forces of the second ship acting on the first ship. $\hat{\eta}_1$ and $\hat{\eta}_2$ are the complex motion amplitudes of the first and second ship, respectively. $\hat{f}_1^{\text{exc}}$ and $\hat{f}_2^{\text{exc}}$ are the vectors containing incident wave and diffraction forces and moments. For more than two bodies, the whole equation of motion is assembled accordingly. Equation (2.61) can be simplified in the following form:

$$\hat{\mathbf{B}}\hat{\eta} = \hat{f}^{\text{exc}} \quad (2.62)$$

In analogy, the linearized equation of motion in time domain reads (cf. equation (2.52)):

$$\mathbf{B}\ddot{\eta} = \vec{f}^{\text{exc}} \quad (2.63)$$

$\mathbf{B}$ is the matrix containing the mass, moment of inertia, and radiation forces and moments of each ship as well as the interacting radiation forces and moments of the ships. $\ddot{\eta}$ and $\vec{f}^{\text{exc}}$ are the vectors containing ship acceleration and excitation forces, respectively.

2 Numerical Methods

The implicit coupled fluid and body motion solvers used for RANS simulations in this work allow to consider the interaction effects between bodies implicitly. Considering that the bodies' Coriolis and centripetal matrices are treated explicitly, the equation system to solve nonlinear motions of a two-ship system reads [105]:

$$\begin{bmatrix} \mathbf{M}_1 & 0 \\ 0 & \mathbf{M}_2 \end{bmatrix} \begin{bmatrix} \ddot{\vec{r}}_1 \\ \ddot{\vec{r}}_2 \end{bmatrix} = \begin{bmatrix} \vec{F}_1 \\ \vec{F}_2 \end{bmatrix} \quad (2.64)$$

Equation (2.64) is recast in short form into:

$$\mathbf{M}\ddot{\vec{r}} = \vec{F} \quad (2.65)$$

$\mathbf{M}$ is the matrix containing the mass and moments of inertia of bodies. $\ddot{\vec{r}}$ is the vector comprising unknown generalized body accelerations. $\vec{F}$ is the force vector containing the hydrodynamic forces and moments, which are obtained from the first term on the right hand side of equation (2.2) (diffusive term), the body weights, explicit centrifugal and Coriolis moments, as well as other external forces and moments.

On the contrary to the straightforward theoretical extension, the numerical handling of motion equations of multibody systems may be intricate. In the following, a concise description of the problems, with which the numerical approaches cope, will be given.

2.2.2.1 Numerical Approaches in Multibody Dynamics

Probably, [83] is one of the first publications on multibody hydrodynamics, in which two cylinders in a catamaran configuration is studied two-dimensionally. One of the first three dimensional techniques has been published in [24], which envisaged the direct numerical solution of wave effects on three-dimensional floating bodies. Since the boundary element methods have a long history in marine and offshore hydrodynamics, many methods developed in the last decades rely on boundary element methods to treat hydrodynamics of floating multibodies. For instance, over the years, zero-speed Green function methods, Rankine source methods, Cummins-based time domain methods, Green function methods with translating-pulsating sources in time domain, etc, while treating the forces linearly, weakly non-linearly, or nonlinearly have emerged. In [74], a roadmap of the tasks done by researchers at Massachusetts Institute of Technology to realize multibody analyses in their zero-speed Green function code in frequency domain is given. In the context of Rankine source methods in frequency domain, the result

of a recent research in published in [127]. In time domain, most methods are based on Cummin's equation such as in the recent publications [147, 136]. Currently, almost all widely-used codes assess the multibody seakeeping problem of floating structures by considering one diffraction and six radiation equations for each body (see equation (2.34)). Furthermore, radiation equations of each body is usually solved while considering other bodies fixed [127].

One drawback in all boundary element methods is the absence of flow viscosity. In seakeeping problems, the lack of flow viscosity causes overprediction of roll motion of bodies - regardless of the bodies being isolated or in a multibody system. In multibody systems, another problem, namely gap resonance, is also ascribed to the lack of viscosity in ideal flows. Gap resonance has gained especial attention in the last decades due to increasing demands in side-by-side offshore operations. Depending on the interacting body geometries, the imprisoned wave at the gap between two or more bodies may be excited to resonance with different mode shapes. Three reference basic mode shapes, i.e. piston, longitudinal, and transverse sloshing modes, are normally referred to. In reality, the resonance in the wave height can largely be damped by viscous effects, i.e. skin friction, and especially the flow separation [51]. Therefore, the lack of viscosity in potential flows leads to overprediction of hydrodynamic forces and body motions. Lids of different formulations have been introduced to remedy this problem in boundary element methods. In [43], a flux condition is imposed by applying a rigid lid along the gap. Newman [75] modified the lid approach to make wave-like motions available on the lid by rendering the lid flexible via a set of polynomials. In [13], a damping force, as a representation of energy dissipation, is introduced directly into the free surface boundary conditions.

Multibody dynamics in relation with finite volume methods is not an unexplored field. [56, 119, 38] are some examples of application of RANS in multibody dynamics in different engineering fields. However, comparing to the application of boundary element methods, finite volume methods still play a minor role in ship and offshore industry. Ship hydrodynamics of single bodies with finite volume methods, especially RANS, has been a focal research topic in the last decade in order to make RANS eligible for daily engineering tasks. Despite such dazzling advances for single ship hydrodynamics, few attempts have been done in multibody hydrodynamic field. One of the first publications on application of RANS in marine multibody hydrodynamics is probably [142]. [94, 95, 85, 58] belong to the recent publications of this kind in marine and offshore field. One advantage of finite volume fluid solvers is their ability to couple to nonlinear kinematic solvers. Additionally, viscous finite volume fluid solvers can simulate the gap resonance phenomena. For instance, in [28, 51] a domain composition is done, and the flow inside the moon pool area of a catamaran is treated with a viscous Navier-Stokes

solver, whereas the flow outside of the moonpool is regarded ideally.

2.2.3 Coupling Hydro- and Rigid Body Dynamics

Merely, the wetted surface hull surface of a body up to mean water level is discretized in frequency domain Green function seakeeping methods. The body does not change its position. It eliminates the necessity to make any changes to body position or its grid panels. Likewise, the mean water level is taken as the upper limit for discretization of the wetted hull surface in time domain boundary element methods, in which Froude-Krylov and restoring forces are considered linearly. Contrarily, grid adaption is required if nonlinear Froude-Krylov and restoring forces for instantaneous free surface is considered in time domain. In such cases, the body panels adjacent to instantaneous free surface are stretched, squeezed, or otherwise geometrically deformed to track the free surface shape. Alternatively, the body surface above the mean water level is also discretized and the panels intersected by free surface are cut geometrically through.

In finite volume methods, control volumes need to be adapted. Grid adaption is not so straightforward or negligible in finite volume methods because the whole domain is discretized. Therefore, special treatments are required to couple the body motions with the domain grid to resolve bilateral effect of flow forces and body kinematics. There are several different coupling methods but grid motion, sliding interfaces, mesh morphing, and Chimera grids are among the common methods used in marine applications.

In grid motion method, the entire domain grid follows the body motion. This approach is simple and robust, but is not appropriate for multibody analyses. Furthermore, in case of large body motions large angle between domain and free surface occurs. If the free surface is to be captured by Volume-of-Fluid methods, a large zone should be refined to capture free surface efficiently. Application of grid motion in marine application can be found in [54, 88]. Grid motion method is not used within this work because this method was not found suitable for multibody simulations.

Sliding interface techniques define an interface between a rotating and a non-rotating part. These techniques suit best for simulation of propeller rotation. As of late, Piel [88] used this technique to examine the roll motion of a vessel of parametric geometry to derive the roll damping coefficients regressively. Although sliding interfaces are suitable for simulations of some motion degrees of freedom, their application in multibody seakeeping simulations, in which both rotatory and translatory motions might be free, can be complicated. Sliding interface methods

are not employed within this work.

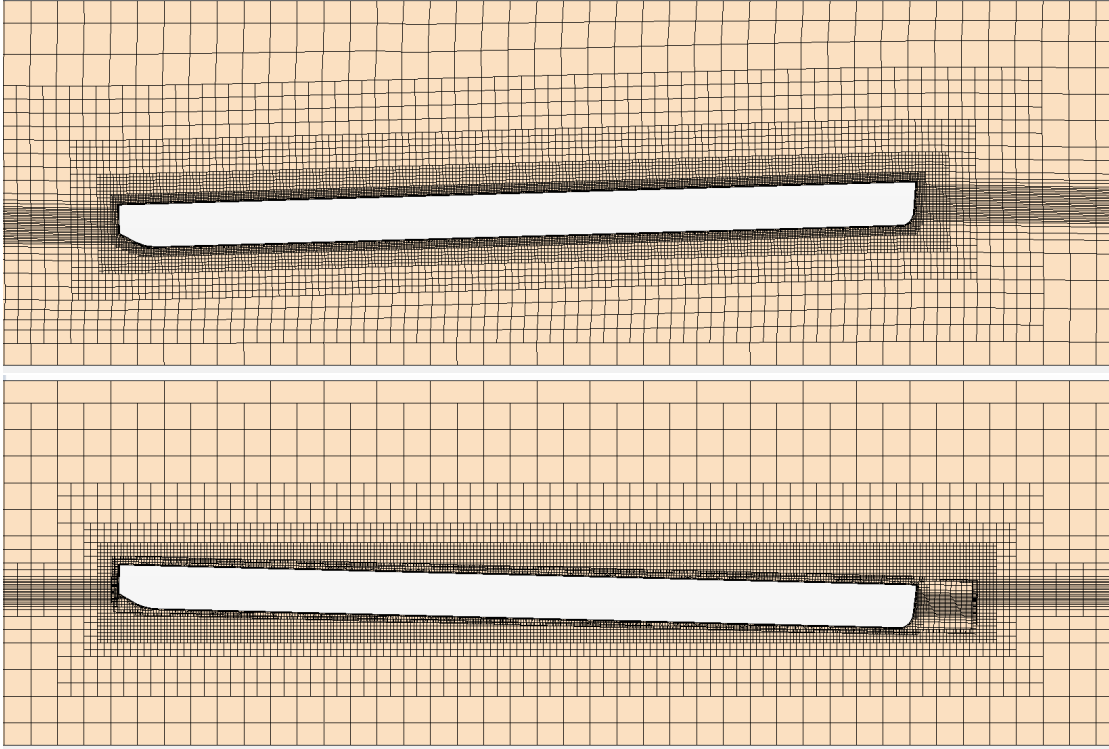


Figure 2.1: Mesh morphing (top) vs. Chimera grid (bottom) techniques.

Mesh morphing methods move the body with its surrounding grid while keeping the distant part of the domain mesh fixed, steadily displaced, or depending on the problem otherwise conditioned. There are several different techniques to propagate the body motion throughout the domain from simple efficient [77] to award-winning [8] techniques. Mesh morphing method is used in this work for articulated multibody analyses, for which a part of the grid around the bodies is moved while keeping the distant part fixed. However, the grid in the constantly adapting near-field may deform too much, and attain unacceptable geometrical properties such as high skewness and non-orthogonality. Therefore, especial care has been taken to generate a suitable mesh that allows consistently stable simulation over a relatively long simulation time.

Chimera grid methods, also known as overlapping or overset grid methods, provide a flexible and efficient spatial discretization method. Chimera grid methods consist of a set of body conforming grids, which overlap the physical domain grid(s). Body conforming grids are related to the bodies and follow their motions while overlapping one or more background grids, with which they exchange flow infor-

mation. Chimera grids suit well for simulation of multiple bodies. However, prices have to be paid to take these advantages. The programming and coupling of the grids can be complicated, and additional storage and computational effort are normally required. The solution procedure is usually sequential and applied on one grid after another. The interpolation process performed between the grids at the overlapping parts may introduce additional errors, and cause convergence problems if the solution exhibits high gradients across the interfaces. Furthermore, it is not easy to ensure overall mass and momentum conservation. Chimera grids have been applied successfully in the last decade in simulations of multibody dynamics in aerospace field. Although this technique has been applied successfully in marine hydrodynamics of single bodies, e.g. in [48, 11] and multibodies, e.g. in [142], fewer evidences of its application in articulated multibody hydrodynamics are available. Chimera grid method is another grid motion technique used intensively within this work. This method is used in all unarticulated multibody analyses, as well as in all single bodies, where high motion amplitudes are expected. Figure 2.1 illustrates applications of mesh morphing and overlapping grid techniques.

2.3 Articulated Multibody Systems

There are many types of mechanical coupling systems with different degrees of complexity. It is a common simulation practice to decompose the coupling mechanisms into principle joints, for which extensive researches are done. Among the most common principle joints are: rigid, hinge, prismatic, cylindrical and screw joints [140, 101]. In this work, totally, three different articulation types are studied: rigid, hinge, and prismatic joints. Rigid joints do not allow any translation and rotation at/about articulation points. Hinge or revolute joints allow only relative rotation between two bodies about an axis called revolute joint axis. Translational motion of connected bodies, as well as the rotation about axes other than the revolute joint axis are not allowed by hinge joints. Another principle articulation type is prismatic or translation joint. Prismatic joints allow relative translation between two bodies along the joint axis. The use of prismatic joint eliminates the freedom of a body to translate relative to the other body in any other direction, except along the joint axis. It also eliminates the freedom of the body to rotate with respect to the other body. Within this work, hinge joints are used for validation studies in chapter 3. However, rigid joints are taken additionally to support the validation procedure, especially in proving the correct implementation of the corresponding formulations. Prismatic and hinge joints are used later in chapter 4 to simulate the coupling mechanism of a case study. Two methods are used in this work to treat articulations. The first method follows the classical kinematic

constraint approach and the second method employs contact elements. In the following, the mathematical formulations of these methods are outlined briefly.

2.3.1 Kinematically Constrained Multibody Systems

The number of degrees of freedom of each free (unconstrained) floating body is equal to six, $f = 6$. Therefore, the total degrees of freedom of a multibody system with n bodies increases to $f = 6n$. Exits n_c constraints, f decreases, so that $f = 6n - n_c$ (Kutzbach criteria). Constraints are said to be holonomic, if they can be written implicitly with respect to the position vector, $\vec{r}$, in the following form [140]:

$$g(\vec{r}_1, \dots, \vec{r}_n, t) = 0 \quad (2.66)$$

For n_c implicit holonomic constraints, equation (2.66) reads:

$$\begin{bmatrix} g_1(\vec{r}_1, \dots, \vec{r}_n, t) \\ \vdots \\ g_{n_c}(\vec{r}_1, \dots, \vec{r}_n, t) \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix} \quad (2.67)$$

$$\vec{g}(\vec{r}, t) = \vec{0}$$

Two procedures can be followed to formulate the dynamic equation of constrained multibody systems: embedding techniques and augmented formulation. In embedding techniques, the system of dynamic equations are formulated in terms of degrees of freedom. This technique leads to a minimum set of dynamic equations that does not contain any constraint forces directly. In the augmented formulation, the dynamic equations are formulated in terms of a set of redundant coordinates. The resultant is a system of differential and algebraic equations, which are expressed in terms of dependent coordinates, as well as constraint forces [140, 101]. Since, direct assessment of the constraint forces is desirable for validation, the augmentation formulation is used in this work.

In the augmented formulation, the total force $\vec{F}$ in linear momentum equation (equation (2.39)) comprises the total excitation force, $\vec{F}^{\text{exc}}$, comprising e.g. gravity and hydrodynamic forces, and total reaction force, $\vec{F}^r$. According to D'Alembert-Lagrange principle, the reaction forces have the components merely normal to implicit constraints [101]. The normal vector at position $\vec{r}_i$ is given by $\partial \vec{g}_i / \partial \vec{r}$, an element of multibody system Jacobian. $\partial \vec{g}_i / \partial \vec{r}$ can be gained by application of the virtual displacement concept. This concept refers to a change in system configuration as the result of any arbitrary infinitesimal change of the position

2 Numerical Methods

coordinate, $\vec{r}$. Therefore, the constraint equations (2.66) and (2.67) yield [140]:

$$\vec{g}(\vec{r}, t) = \mathbf{J}(\vec{r}, t) \delta \vec{r} = \vec{0} \quad (2.68)$$

where $\mathbf{J}$ is $n_c \times n$ system Jacobian:

$$\mathbf{J}(\vec{r}, t) = \begin{bmatrix} \frac{\partial g_1(\vec{r}, t)}{\partial \vec{r}_1} & \dots & \frac{\partial g_1(\vec{r}, t)}{\partial \vec{r}_n} \\ \vdots & & \vdots \\ \frac{\partial g_{n_c}(\vec{r}, t)}{\partial \vec{r}_1} & \dots & \frac{\partial g_{n_c}(\vec{r}, t)}{\partial \vec{r}_n} \end{bmatrix} = \quad (2.69)$$

As we are dealing with the coordinates, whose number is not equal to the degrees of freedom of the constrained system. The resultant equation system is not mathematically closed. Therefore, a set of redundant coordinates, $\vec{\lambda}$, are introduced, which help to compute the constraint reaction forces directly:

$$\vec{F}_i^r = \begin{bmatrix} \frac{\partial g_1}{\partial \vec{r}_i^T} & \dots & \frac{\partial g_{n_c}}{\partial \vec{r}_i^T} \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \vdots \\ \lambda_{n_c} \end{bmatrix} = \mathbf{J}_i^T \vec{\lambda} \quad i = 1, \dots, n \quad (2.70)$$

The linear λ -factors are independent from each other and are called Lagrange-multipliers. Lagrange multipliers do not have any specific physical meaning. However, the related articulation reaction forces are derived immediately from reaction equation (equation (2.70)).

Explicit reaction equation (equation (2.70)), and implicit holonomic articulation equations (equation (2.67)), together with equations (2.39) and (2.40), while considering $\vec{F} = \vec{F}^{\text{exc}} + \vec{F}^r$, formulate the equation of motion of a holonomically constrained multibody system. These equations form the Lagrange equations of the first kind, which are a differential algebraic equation system [140].

$$\begin{aligned} \mathbf{M} \ddot{\vec{r}} &= \vec{F}^{\text{exc}} + \mathbf{J}^T \vec{\lambda} \\ \vec{0} &= \vec{g} \end{aligned} \quad (2.71)$$

This algebraic equation system can be solved after its transformation into an ordinary differential equation [140]:

$$\begin{bmatrix} \mathbf{M} & -\mathbf{J}^T \\ \mathbf{J} & 0 \end{bmatrix} \begin{bmatrix} \ddot{\vec{r}} \\ \vec{\lambda} \end{bmatrix} = \begin{bmatrix} \vec{F}^{\text{exc}} \\ \mathbf{J} \dot{\vec{r}} \end{bmatrix} \quad (2.72)$$

Numerical discretization of equation (2.72) violates the constraint condition and requires numerical stabilization [101]. One of the most prevalent stabilization

method is introduction of fictive spring and damping forces as per Baumgarte [2]. The motion of equation of a holonomically constrained multibody system in compliance with the augmentation formulation, and without consideration of the Coriolis force reads:

$$\begin{bmatrix} \mathbf{M} & -\mathbf{J}^T \\ \mathbf{J} & 0 \end{bmatrix} \begin{bmatrix} \ddot{\vec{r}} \\ \ddot{\vec{\lambda}} \end{bmatrix} = \begin{bmatrix} \vec{F}^{\text{exc}} \\ \dot{\mathbf{J}}\dot{\vec{r}} + \ddot{\vec{g}} + 2\alpha\dot{\vec{g}} + \beta^2\vec{g} \end{bmatrix} \quad (2.73)$$

Here, $\mathbf{M}$ and $\vec{F}^{\text{exc}}$ are respectively the mass matrix and external force vector containing components from all bodies of the multibody system. α and β are stabilization damping and spring coefficients, respectively.

Linearized equations of motions (equations (2.54) and (2.56)) require the constraints to be imposed on the body positions. Therefore, equation (2.66) is used to formulate the constraints. On the contrary, the nonlinear equation of motion equation (2.51) requires the body accelerations to be constrained. Implicit holonomic constraints on body acceleration, $\ddot{\vec{g}}$, is defined as

$$\ddot{\vec{g}}(\vec{r}, \vec{v}, t) = \vec{0} \quad (2.74)$$

where $\ddot{\vec{g}}$ can be obtained from second total time derivatives of position constraints $\vec{g}$.

$$\ddot{\vec{g}} = \mathbf{J}\ddot{\vec{r}} + \dot{\mathbf{J}}\dot{\vec{r}} \quad (2.75)$$

In the following, more specific description on the articulation of bodies is given. It will be shown how two bodies are so articulated at a point to constrain their relative translations, i.e. a ball joint configuration. This configuration is extended afterwards to formulate a hinge joint. Then, it is demonstrated how the formulation of a hinge joint can be extended to make both vessels connect rigidly. Additionally, an alternative formulation for ball, hinge, and rigid joints is given. Subsequently, it is shown how contact elements are employed to represent hinge and prismatic joints. A short description on the theory of contact elements is also given.

2.3.1.1 Implicit Constraint Equations for a Ball Joint

Hinge and rigid joints implemented within this work are of holonomic scleronomic and two-sided type. They can be seen as a further development of a ball joint. A ball joint allows free rotation of attached bodies at the articulation point but constrains their translation at that point. In the following, the formulation of the ball joint for a multibody system consisting of two bodies is presented.

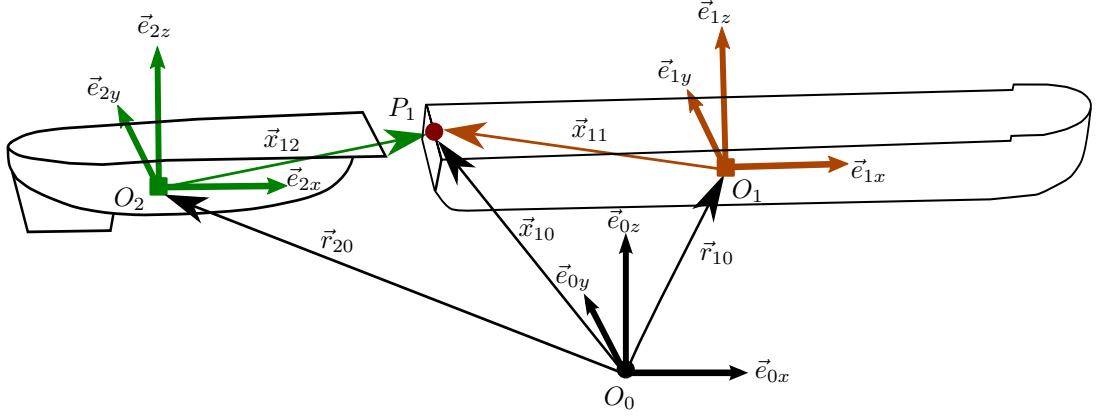


Figure 2.2: Coupling two bodies at one articulation point.

Both bodies are articulated at one point, P_1 , (figure 2.2). Considering that point P_1 is related to body 1, the position vector $\vec{x}_{10}$, with respect to the origin of the inertial coordinate system O_0 , can be expressed by its position vector $\vec{x}_{11}$, with respect to the body's origin O_1 :

$$\vec{x}_{10} = \vec{r}_{10} + \vec{x}_{11} \quad (2.76)$$

After transforming $\vec{x}_{11}$ by means of transformation matrix ${}^{01}\mathbf{T}$ from body-fixed coordinate system to the inertial system via

$${}^0\vec{x}_{11} = {}^{01}\mathbf{T}^1\vec{x}_{11} \quad (2.77)$$

equation (2.76) yields

$${}^0\vec{x}_{10} = {}^0\vec{r}_{10} + {}^{01}\mathbf{T}^1\vec{x}_{11} \quad (2.78)$$

The upper prefix indicates the reference frame, a vector is referenced to. Regarding that P_1 acts as an articulation point of both bodies, it can be considered also as a point pertained to body 2. Analogous to equation (2.78) ${}^0\vec{x}_{10}$ can be written for the second body:

$${}^0\vec{x}_{10} = {}^0\vec{r}_{10} + {}^{01}\mathbf{T}^1\vec{x}_{11} = {}^0\vec{r}_{20} + {}^{02}\mathbf{T}^2\vec{x}_{12} \quad (2.79)$$

In frequency domain, small ship motion, and linear correlation between ship responses and excitation forces are assumed (cf. equation (2.55)). Therefore, equation (2.79) can be cast into:

$${}^0\vec{x}_{10} = {}^0\vec{r}_{10} + {}^{01}\mathbf{S}^1\vec{x}_{11} = {}^0\vec{r}_{20} + {}^{02}\mathbf{S}^2\vec{x}_{12} \quad (2.80)$$

$\mathbf{S}$ is the linearized $\mathbf{T}$ transformation matrix [27].

$$\mathbf{S} = \begin{bmatrix} 1 & -\psi & \theta \\ \psi & 1 & -\phi \\ -\theta & \phi & 1 \end{bmatrix} = \tilde{\boldsymbol{\alpha}} + \mathbf{E} \quad \text{mit} \quad \tilde{\boldsymbol{\alpha}} = \begin{bmatrix} 0 & -\psi & \theta \\ \psi & 0 & -\phi \\ -\theta & \phi & 0 \end{bmatrix} \quad (2.81)$$

$\tilde{\boldsymbol{\alpha}}$ is the skew matrix of Euler angles and $\mathbf{E}$ is the identity matrix. Therefore:

$${}^0\vec{r}_{10} + {}^{01}\tilde{\boldsymbol{\alpha}} {}^1\vec{x}_{11} + \mathbf{E} {}^1\vec{x}_{11} = {}^0\vec{r}_{20} + {}^{02}\tilde{\boldsymbol{\alpha}} {}^2\vec{x}_{12} + \mathbf{E} {}^2\vec{x}_{12} \quad (2.82)$$

The vectorial form of equation (2.82) using $\vec{\alpha}_{i0}$, the rotation vector from i^{th} body-fixed reference frame to inertial reference frame, reads:

$${}^0\vec{r}_{10} + \vec{\alpha}_{10} \times {}^1\vec{x}_{11} + {}^1\vec{x}_{11} = {}^0\vec{r}_{20} + \vec{\alpha}_{20} \times {}^2\vec{x}_{12} + {}^2\vec{x}_{12} \quad (2.83)$$

If body 1 and body 2 undergo a translation with the vectors $\vec{u}_1$ and $\vec{u}_2$, respectively, and by application of equation (2.76) on ${}^0\vec{r}_{10}$ and ${}^0\vec{r}_{20}$, equation (2.83) yields:

$$\vec{u}_1 + \vec{\alpha}_{10} \times {}^1\vec{x}_{11} = \vec{u}_2 + \vec{\alpha}_{20} \times {}^2\vec{x}_{12} \quad (2.84)$$

Therefore, with the constraint

$$g(\vec{r}) = \vec{u}_1 + \vec{\alpha}_{10} \times {}^1\vec{x}_{11} - \vec{u}_2 - \vec{\alpha}_{20} \times {}^2\vec{x}_{12} \quad (2.85)$$

the translation of bodies at articulation point P_1 can be constrained. In the following, equation (2.85) is expanded and applied to formulate constraints for hinge and rigid joints.

2.3.1.2 Implicit Constraint Equations for Hinge and Rigid Joints

Coupling both vessels at one common point (c.f. section 2.3.1.1) makes both vessels pivot about it. By introduction of an additional pivot point, both vessels can be so configured to revolute about the axis, which connects both pivot points with each other (figure 2.3). In order to achieve a hinged constrained motion by means of both articulation points, five constraint equations as per equation (2.85) are required. Assuming that both vessels are to be free in pitch and undergo small motions, with reference to figure 2.3, these five constraint equations read:

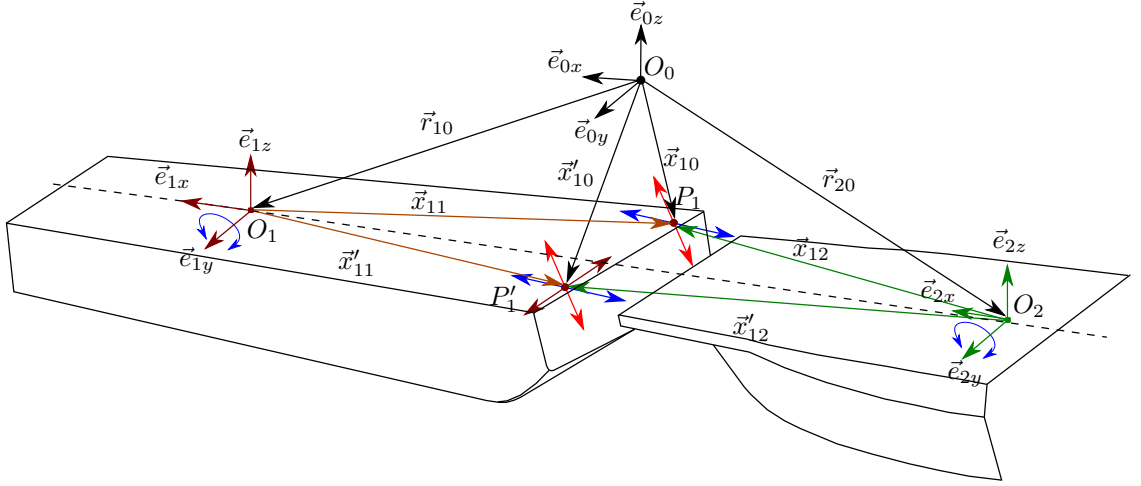


Figure 2.3: Application of two articulation pivot points makes it possible to revolute both bodies about the connecting axis of the points. Double ended arrows indicate the directions, the related points are constrained at.

$$\begin{aligned}
 g_1(\vec{r}) &= (+u_{1x} + \theta^1 x_{11z} - \psi^1 x_{11y}) \vec{e}_{1x} + (-u_{2x} - \theta^2 x_{12z} + \psi^2 x_{12y}) \vec{e}_{2x} = 0 \\
 g_2(\vec{r}) &= (+u_{1x} + \theta^1 x'_{11z} - \psi^1 x'_{11y}) \vec{e}_{1x} + (-u_{2x} - \theta^2 x'_{12z} + \psi^2 x'_{12y}) \vec{e}_{2x} = 0 \\
 g_3(\vec{r}) &= (+u_{1y} - \phi^1 x_{11z} + \psi^1 x_{11x}) \vec{e}_{1y} + (-u_{2y} + \phi^2 x_{12z} - \psi^2 x_{12x}) \vec{e}_{2y} = 0 \\
 g_4(\vec{r}) &= (+u_{1z} + \phi^1 x_{11y} - \theta^1 x_{11x}) \vec{e}_{1z} + (-u_{2z} - \phi^2 x_{12y} + \theta^2 x_{12x}) \vec{e}_{2z} = 0 \\
 g_5(\vec{r}) &= (+u_{1z} + \phi^1 x'_{11y} - \theta^1 x'_{11x}) \vec{e}_{1z} + (-u_{2z} - \phi^2 x'_{12y} + \theta^2 x'_{12x}) \vec{e}_{2z} = 0
 \end{aligned} \tag{2.86}$$

The indices x, y, z indicate the x, y, z - components of the vectors. Now, by adding another constraint equation, i.e. equal pitch rotation of both vessels, the hinge constraint configuration can be transposed to represent a rigid connection:

$$g_6(\vec{\alpha}) = \theta_1 - \theta_2 = 0 \tag{2.87}$$

The mathematical model for the hinge articulation with five constraints, $n_c = 5$, has seven degrees of freedom and the rigid body model with $n_c = 6$ has six degrees of freedom. Hence, the matrices of equation of motion are extended by five or six indices, respectively.

Alternatively, in lieu of using two points to define a hinge or a rigid joint, only one

point can be used if a new reference frame with the articulation point positioned at the origin of this new reference frame is defined, (figure 2.4). Therefore, not only the translation but also the rotation at the origin of articulation reference frame can be treated directly. In such a way, it suffices to constrain both vessels rigidly or hingedly via one point at the common center line of both vessels.

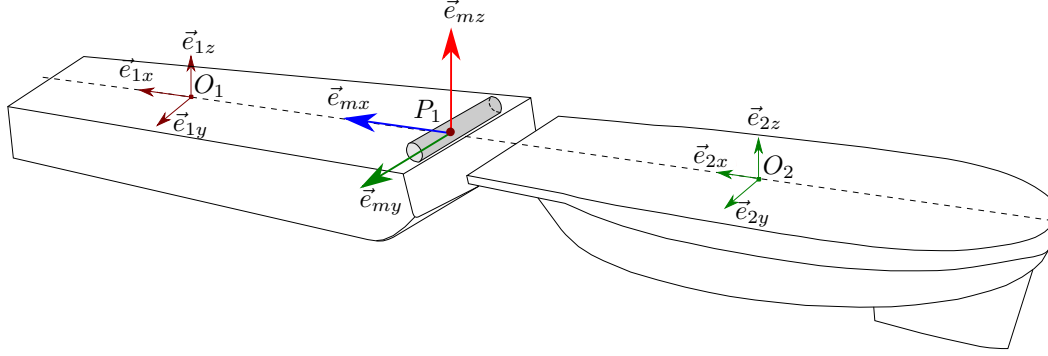


Figure 2.4: Implementation of hinge joint at one articulation point by mean of local reference frame.

$$\begin{aligned}
 g'_1(\vec{r}) &= \left(+u_{1x} + \theta^1 x_{11z} - \psi^1 x_{11y} - u_{2x} - \theta^2 x_{12z} + \psi^2 x_{12y} \right) \vec{e}_{mx} = 0 \\
 g'_2(\vec{r}) &= \left(+u_{1y} - \phi^1 x_{11z} + \psi^1 x_{11x} - u_{2y} + \phi^2 x_{12z} - \psi^2 x_{12x} \right) \vec{e}_{my} = 0 \\
 g'_3(\vec{r}) &= \left(+u_{1z} + \phi^1 x_{11y} - \theta^1 x_{11x} - u_{2z} - \phi^2 x_{12y} + \theta^2 x_{12x} \right) \vec{e}_{mz} = 0 \\
 g'_4(\vec{\alpha}) &= (+\phi_1 - \phi_2) \vec{e}_{mx} = 0 \\
 g'_5(\vec{\alpha}) &= (+\psi_1 - \psi_2) \vec{e}_{mz} = 0 \\
 g'_6(\vec{\alpha}) &= (+\theta_1 - \theta_2) \vec{e}_{my} = 0
 \end{aligned} \tag{2.88}$$

The constraint equations $g'_1(\vec{r})$ to $g'_3(\vec{r})$ constrain the translation motion and $g'_4(\vec{r})$ to $g'_6(\vec{r})$ the rotation at the specified articulation reference frame. All six constraint equations are required if a rigid joint is to be defined; hence, the system degrees of freedom is reduced to six. In case of a hinge joint, only $g'_1(\vec{r})$ to $g'_5(\vec{r})$ are required to allow relative pitch motion of bodies about $\vec{e}_{my}$ revolute axis - it leads to seven degrees of freedom for the multibody system. The bodies are articulated in a ball joint at P_1 if $g'_1(\vec{r})$ to $g'_3(\vec{r})$ are applied. This configuration results in nine degrees of freedom of the system.

Relationship between body, inertial, and articulation local coordinate systems is built through transformation matrices. Recalling that ${}^{0i}\mathbf{S}$ is the linearized transformation matrix from body-fixed coordinate system to inertial system, and ${}^{im}\mathbf{S}$ from articulation local coordinate system to body-fixed coordinate system, the

linearized transformation matrix from articulation local coordinate system to the inertial coordinate system, ${}^{0m}\mathbf{S}$, reads [27]:

$${}^{0m}\mathbf{S} = {}^{0i}\mathbf{S} {}^{im}\mathbf{S} \quad (2.89)$$

2.3.2 Contact Elements

Contact elements are used in this work to represent hinge and prismatic joints. Contact elements defined herein are damper-spring massless elements. In the following a short overview on the theory of contact elements is given. Further details on the theory of contact elements can be found in [1].

A contact element acts only in compression between a point associated to one structure, and a contact plane associated to the other body. A point and a normal vector define the contact plane. The compressive normal force, $\vec{F}_t$, is formed from specified relation between distance, and the specified internal damping. Denotes l_0 the initial uncompressed length of the contact element, the magnitude of the axial compression force, $\vec{F}_k$, on the contact element may be defined as a polynomial function of the compression Δl :

$$F_k = \begin{cases} \sum_{i=1}^n C_{ki} \Delta l^i & \text{if } \Delta l > 0 \\ 0 & \text{otherwise} \end{cases} \quad (2.90)$$

where C_{ki} are contact element constants. The nonlinearity of the contact force is helpful to represent physical characteristics of a contact element, that a contact element is hardly compressed anymore after some compression threshold is reached. Contact damping force, $\vec{F}_d$, is applied in the direction of the element compression force, $\vec{F}_k$, and is related to the element compression velocity. Its magnitude is:

$$F_d = C_d C_k \frac{d\Delta l}{dt} \quad (2.91)$$

where C_d and C_k are damping and stiffness coefficients, respectively. The magnitude of friction force, F_f , between both related bodies with the coefficient μ is

$$F_f = \mu F_k \quad (2.92)$$

Friction force, $\vec{F}_f$, acts in opposite direction to the velocity. Numerical instability due to instantaneous change of friction force can be avoided by a smoothing function, like $\tanh$ function. The smoothing functions are in relation to the rate of

change of relative body motions. Subsequently, the total contact force reads:

$$\vec{F}_t = \vec{F}_k + \vec{F}_d + \vec{F}_f \quad (2.93)$$

Contact elements are used within this work to represent mechanical couplings because they provide the ability to circumvent kinematic constraint formulations. Kinematic constraints need to be formulated for an articulation at the outset prior to implementation. On the contrary, the formulation of contact elements need to be formulated and implemented once. Then, the representation of different joint types can be realized by appropriate configurations of contact elements. Consequently, the work load in providing analytical formulations in kinematic constraints is shifted to configuring the contact elements during simulation setups. It can be disputed that different joint types can be formulated, and implemented once for repeated uses, as it is the case for some software packages. However, it does not change the principle essentially, that if a joint is not predefined, formulation needs to be set at first and then implemented.

Contact element forces can be incorporated as external forces in the equation of motion. Contact elements are used in this work in adherence to time domain boundary element method to solve seakeeping behavior of multibody systems. With reference to equation (2.54), the motion equation of a system in time domain after integration of contact element forces read:

$$(\mathbf{M} + \mathbf{A}(\infty)) \ddot{\vec{\eta}}(t) + \int_0^t \mathbf{K}(t - \tau) \dot{\vec{\eta}}(\tau) d\tau + \mathbf{R}\vec{\eta}(t) = \vec{f}^{\text{exc}} + \sum \vec{F}_t \quad (2.94)$$

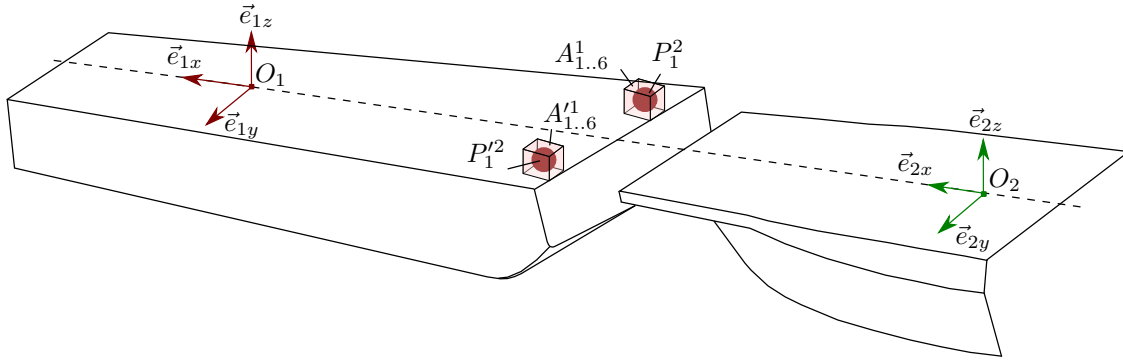


Figure 2.5: Two ball joints are defined to model a hinge joint. Each two ball joint comprises six contact elements, $A_{1..6}$, which form a bounding box about a point. The contact elements are referenced to the barge and the confined point is attached to the tug. The upper suffixes 1 and 2 indicate the reference body.

A contact element comprises a contact plane related to a structure, and a contact point pertained to the other. A plane exerts reaction force as soon as the point contracts it, while the both, the plane and the point, conform to the motion of their respectively related bodies. Composition of several contact elements allows representation of joints. For instance a ball joint can be modeled similar to a mechanical ball joint, i.e. a ball stud in a confining encompassing jacket. The jacket can be simulated as a box by planes of contact elements, and the bud stud by a common contact element point shared by all planes (figure 2.5). The confining box comply with the motions of one body while the confined point complies with the motions of the other body. The gap between planes and points is kept to a minimum to imprison the point in its dedicated extents. Due to the barrier characteristic of the plane, the point is not allowed to flee out of its extents. As it is obvious, more work is required to set up a simulation and to arrange the contact elements but it circumvent the necessity to define kinematic constraints.

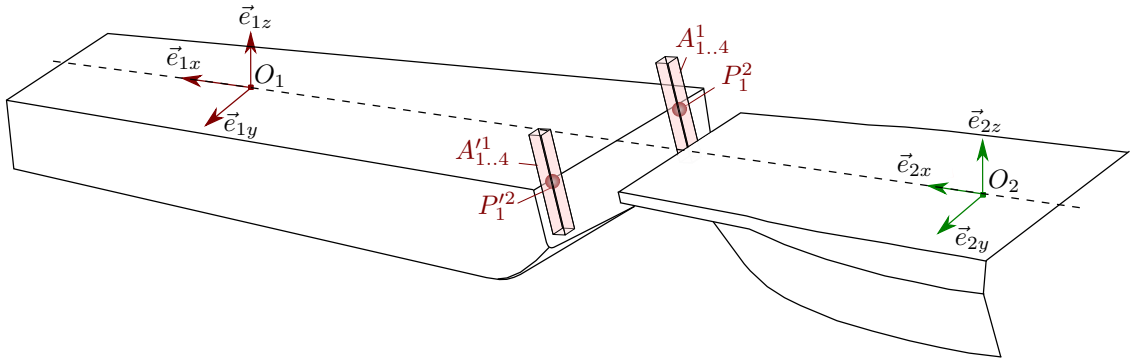


Figure 2.6: A point and four contact elements construct a prismatic joint.

Similar to the preceding section, two ball joints can be employed to construct a hinge joint. With reference to figure 2.5, planes $A_{1..6}^1$ and $A_{1..6}^{1'}$ confine P_1^2 and $P_1'^2$, respectively. The planes follow the motion of body 1, and the points are attached to body 2. The upper index of a plane or a point indicates the body, to which the plane or the point refer. It is obvious from figure 2.5 that such a configuration realizes automatically the interaction of couplings and body motions, in a sense that, the tug interacts through the contact element points, via the contact element planes, with the barge.

A prismatic joint can easily be represented by configuring the contact planes to build a duct. Since the point is free to move along the duct, the configuration is capable of modeling free translation along a prismatic joint axis. Like the aforementioned box the side planes confine the point. Figure 2.6 illustrates how contact elements can be arranged to model prismatic joints.

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In the previous chapter, the theoretical background of multibody hydrodynamics was discussed. The formulations of kinematic constraints to articulate two bodies by means of ball, hinge, and rigid joints were given. It was also elaborated how contact elements can be utilized to represent ball and prismatic joints. In this chapter, validation of contact elements to represent articulations is studied. At first, simulations for single bodies are given. Afterwards, hydrodynamically coupled multibody simulations are presented. Finally, simulations are performed for articulated multibody systems. Hinge joints are chosen as the validation articulation type. Different hinge articulation formulations described in section 2.3 are applied here. Rigid joints are also used, but auxiliarily, to validate formulations. Results are compared with code-to-code results as well as with model test measurements. In this work, two model test campaigns are referred to. The model test campaigns were of complementary nature to each other, albeit accentuated different objectives. The first model tests were conducted for validation purposes and are described in the following. The forthcoming simulations in sections 3.3 to 3.5 are related to these model tests. The second model basin tests were related to a case study and are presented in section 4.1.2.

3.1 Model Tests - The First Campaign

In the first model test campaign, the focus was set on measuring the motion responses of the convoy vessels, and articulation forces. The tests were conducted at the *Development Centre for Ship Technology and Transport Systems* (DST) in a shallow water tank of 200 m of length, 10 m of width, and 1.2 m of depth. The models were driven with a low speed carriage capable of having the maximal speed of 6.5 m/s. The shallow water basin is equipped with hydraulic wave makers, which can be adjusted continuously between flap and piston mode. The adjustable water depth and the design of wave makers allowed simulation of various conditions. The generated waves were absorbed with fixed iron grid beach at the basin end. The water depth was adjusted for 17 m full scale to be in compliance with the depth of convoy sailing zone. The model tests were conducted for 1:20 model scale

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with bare hulls without propulsion or steering devices. The line drawings of the ships modeled in the first model test campaign are shown in figures 3.1 to 3.3. The centers of gravity of the ships associated to the model test loading condition is also shown on these figures. The main particulars as well as the respective model test loading condition is stated in table 3.1. Right-handed coordinate systems were used and body-fixed coordinate systems were defined at the centers of gravity of each respective body, see figures 3.1 and 3.2. The free surface at rest lies at $z = 0$.

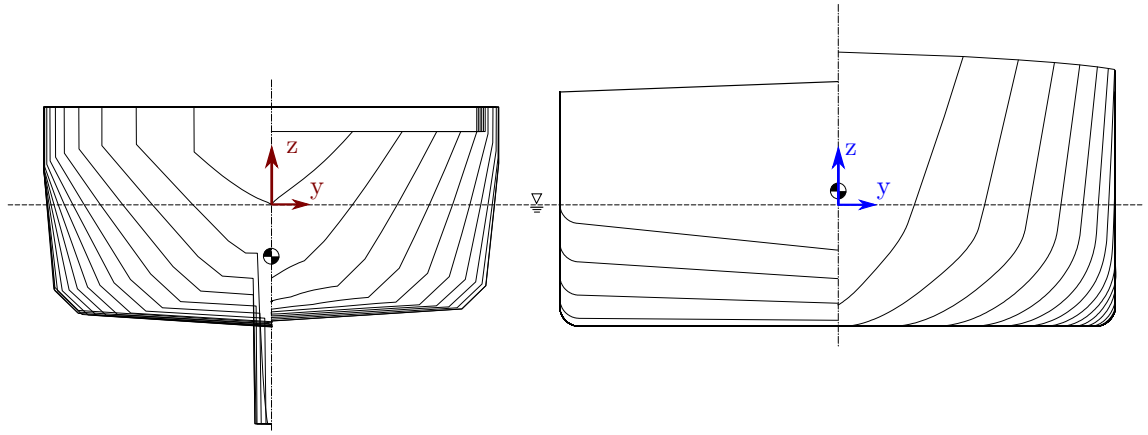


Figure 3.1: Body plan of ships used in model tests; tug (left), barge(right).

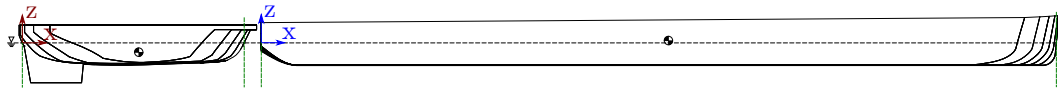


Figure 3.2: Sheer plan of ships used in model tests; tug (left), barge(right)

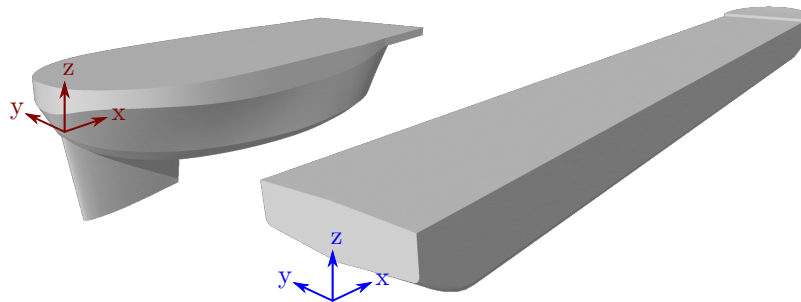


Figure 3.3: Perspective view of ships used in model tests; tug (left), barge(right)

3.1 Model Tests - The First Campaign

Table 3.1: Main particulars of convoy bodies. Units are in [m] if not otherwise stated.

Full scale										
	L	B	T	∇ [m ³]	LCG	VCG	GM	r_{xx}	r_{yy}	r_{zz}
Barge	90.0	11.4	2.5	2420	44.07	0.36	2.82	2.78	23.08	23.08
Tug	25.0	9.3	2.5	365	12.68	-1.07	2.80	2.18	5.38	5.38

Model scale (model scale factor: 20)										
	L	B	T	∇ [m ³]	LCG	VCG	GM	r_{xx}	r_{yy}	r_{zz}
Barge	4.5	0.57	0.12	0.30	2.20	0.02	0.14	0.14	1.15	1.15
Tug	1.25	0.46	0.12	0.05	0.63	-0.05	0.14	0.11	0.27	0.27

$L, B, T, \nabla, LCG, VCG, GM, r_{xx}, r_{yy}, r_{zz}$ stand for respective ship length, breadth, draft, displaced volume, longitudinal and vertical centers of gravity, metacentric height, and radii of gyration about x, y and z -axes, respectively.

At first, the tests were performed for each single convoy body. The models in single body test runs were towed with the carriage while being free to heave and to pitch (figure 3.4).

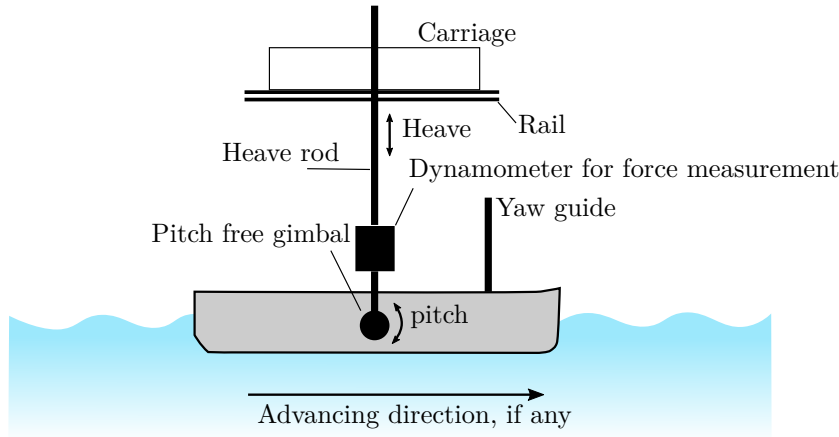


Figure 3.4: Test setup for single body runs.

Depending on the setup, the connecting forces were measured with single component load cells in longitudinal and transverse directions. Motions were measured with laser distance meters relative to the carriage or gyro- and accelerometers. A vertical bar connected to the carriage was installed at some position in the forward part of the barge and after part of the tug to suppress yaw motions. Following the single body test runs, a test run series for both bodies without having any mechanical connections was performed. Essentially, similar arrangement like single body

3 Validation

test arrangements was used. Figure 3.5 illustrates the unarticulated multibody test arrangement. Table 3.2 abridges the test matrices used in single body and unarticulated multibody runs.

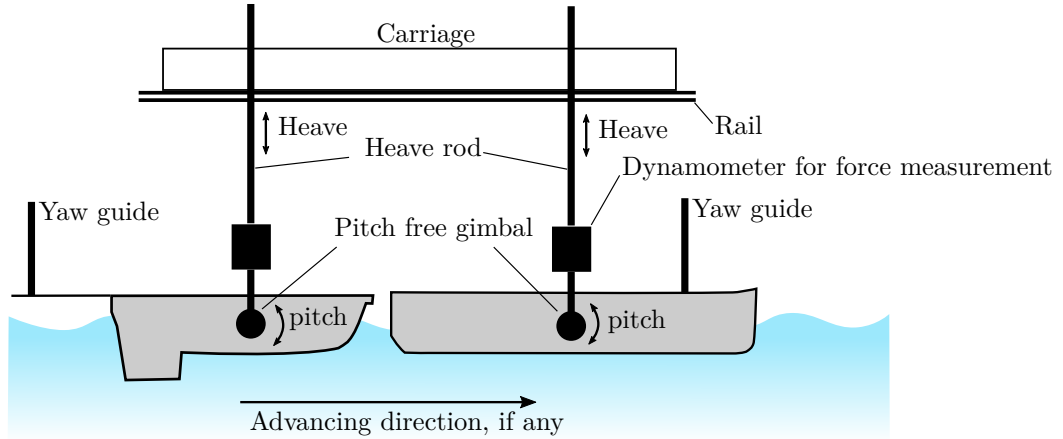


Figure 3.5: Test setup for unarticulated multibody runs.

Following single and unarticulated multibody runs, mechanical articulations were adopted in the model tests with hinge and rigid mechanical couplings. Due to rigid connection of the barge to the carriage, surge and sway motions could be suppressed. Likewise, yaw guides were used to suppress yaw motions. The tug was connected merely to the barge via mechanical couplings, and followed the motions imposed by the articulation. Initially, one test run was done with the barge being free to heave and pitch while connected hingedly to the tug, but the arrangement was extended to allow the barge free to roll - that subsequently allowed the engaged tug to roll. Following the hingedly articulated multibody tests, the arrangement was modified to connect the bodies rigidly. Likewise, the barge was set free to heave, roll, and pitch, with the tug obeying the articulation condition. Hinges were modeled by means of two ball-socket joints on the port and starboard sides of the barge transom, and rigid articulation by means of a rigid connection bar, which ran on the centerline of both ships. The articulation forces and moments were measured with two three-component force gauges connected to the hinge or rigid joints at the positions where the ball-socket joints were installed. Figures 3.6 and 3.7 depict the articulated multibody test arrangements schematically, and table 3.3 gives an abridged overview of the respective test matrices. The wave height were maximally 2 m in full scale for most frequencies. For the higher frequencies, amplitudes were reduced by half to avoid wave breaking. Figures 3.8 and 3.9 show an snapshot of each test run series.

3.1 Model Tests - The First Campaign

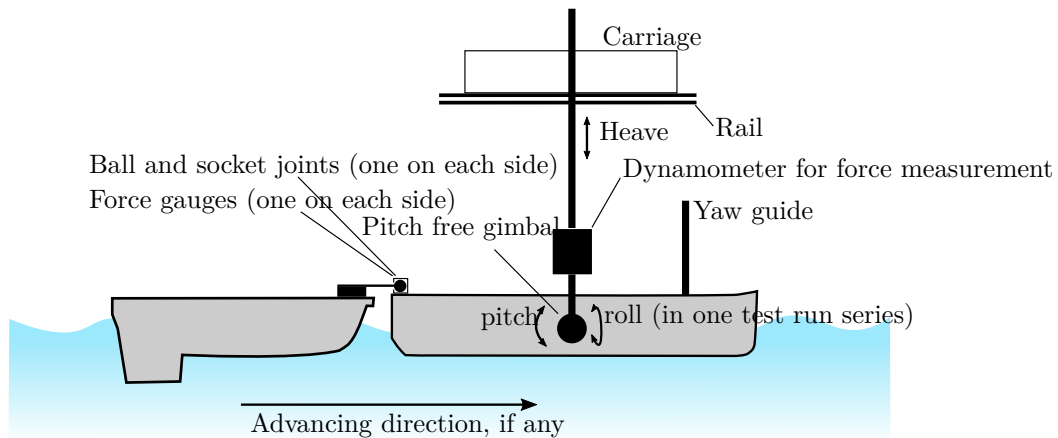


Figure 3.6: Test setup for the runs, for which the bodies are articulated hingedly.

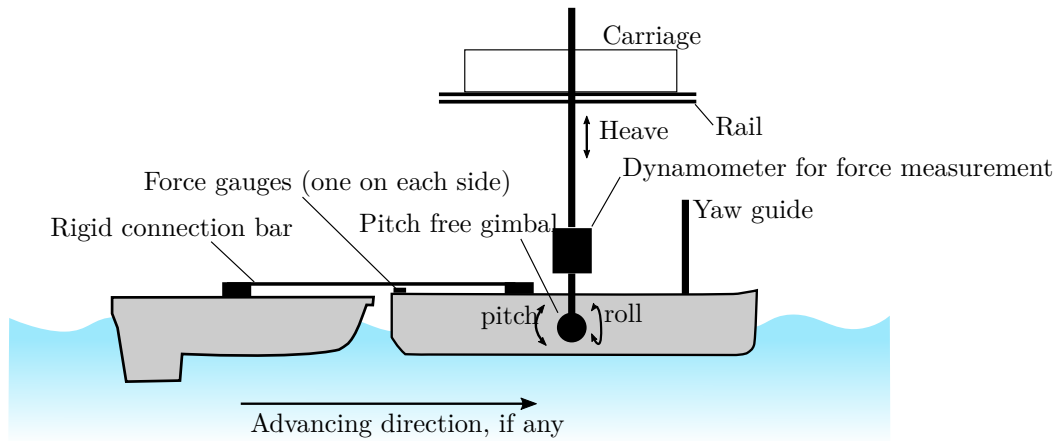


Figure 3.7: Test setup for the runs, for which the bodies are articulated rigidly.

Table 3.2: Overview of model test configurations for single body and unarticulated multibody cases.

Test run series for	single tug	single barge	unarticulated multibody
body(bodies) free to	heave, pitch	heave, pitch	heave, pitch
wave direction (180°: head waves)	180°	180°	180°
convoy speed [knots] (full scale)	0.0, 6.0, 6.5, 7.0	0, 6.0, 8.0	0, 6.0, 8.0

Table 3.3: Overview of model test configurations for articulated cases.

articulation type	rigid	hinge	hinge
barge is free to	heave, roll, pitch	heave, pitch	heave, roll, pitch
wave direction (180°: head waves)	0°, 60°, 180°	180°	0°, 60°, 120°, 180°
convoy speed [knots] (full scale)	0.0	0, 6.0, 8.0	0.0

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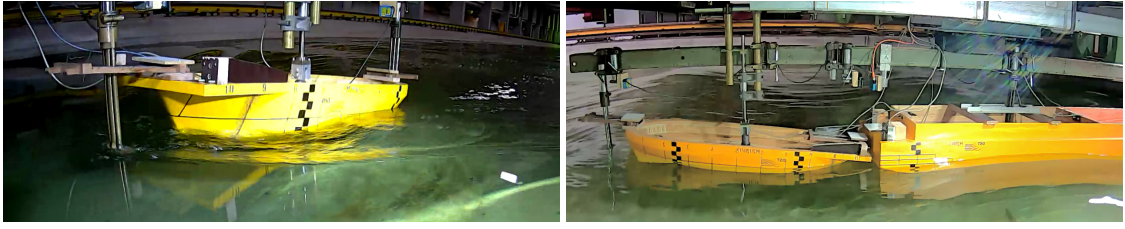


Figure 3.8: Single tug (left), unarticulated convoy bodies(right) (courtesy of DST).

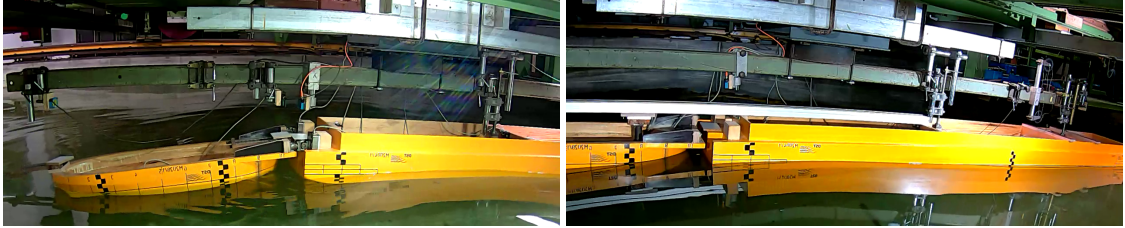


Figure 3.9: Hingedly (left) and rigidly (right) articulated convoy bodies (courtesy of DST).

3.2 Discretization Studies

A consistent numerical analysis provides solutions which approach the actual results, as the grid resolution gets close to zero. One significant issue in numerical computations is what level of grid resolution is appropriate. This is a function of flow conditions, type of analysis, geometry, numerical methods, and computational resources. In the following, numerical grid studies for boundary element and RANS methods used in this work are given. Heave and pitch responses, as well as articulation forces are the responses studied for various grid sizes and time steps. All discretization simulations are done for full scale ships. Right-handed coordinate systems are applied. Body-fixed coordinate systems are defined at the centers of gravity of respective bodies. The free surface levels are positioned at $z=0$ m.

3.2.1 Discretization Studies for Boundary Element Method Solvers

For discretization studies, the simulations are done, at first, in frequency domain, and with the program *Ansys Aqwa*. Similar grid study results were gained from the frequency domain package *GL Rankine*. This is why, only those studies performed in *Ansys Aqwa* are presented here. For the frequency domain simulations, the

bodies are not connected with each other and undergo zero forward speed. For investigating the influence of panel size on the frequency domain AQW results, three different grids, i.e. a coarse, a medium, and a fine grid are tested. The grids are refined systematically with refining factor of $\sqrt{2.25}$. The coarse grid comprises totally 2429 panels, of which 1807 are diffracting. The mid and fine grids contain totally 4990 and 9588 panels, with 3752 and 6967 diffracting elements, respectively. Figure 3.10 illustrates the grids used in frequency domain AQW simulations. Figure 3.11 shows the resultant heave and pitch motion responses per wave amplitude for these three body grids against wave frequency. It is obvious that the results are nearly identical. Hence, it is concluded that the panel size and number of grids have negligible influence on the heave and pitch motion response amplitudes in frequency domain simulations.

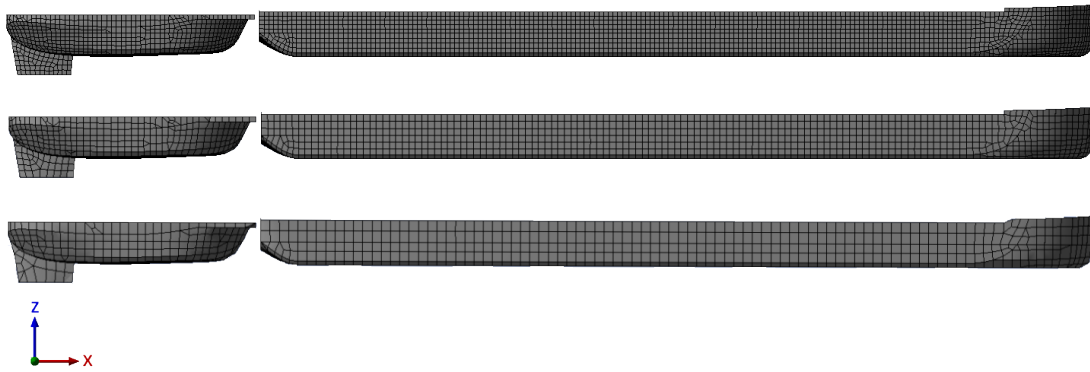


Figure 3.10: Fine (top), mid (middle) and coarse (bottom) body grids used for grid independence studies in boundary element AQW simulations.

In time domain AQW simulations, the same three body meshes used in AQW frequency domain simulations are employed to study the influence of grid size and panel number on the results. Again, simulations are performed for regular head waves at zero body forward speed. The ships are articulated by means of a hinge, which is modeled by kinematic constraints. The hydrostatic and Froude-Krylov forces are computed nonlinearly in each time step by accounting for instantaneous wetted surface. Diffraction forces are assumed linear and are transposed from prerequisite frequency domain computations. Additionally, the frequency dependent damping and added mass coefficients are gained from frequency domain AQW simulations. Figure 3.12 plots the heave and pitch motion responses of bodies per wave amplitude for the studied body grids against wave frequency. Figure 3.12 reveals how negligible the heave and pitch motion response amplitudes are influenced by the grid size and number of body panels. Similar conclusion that the number of body panels influences the motion response amplitudes marginally is

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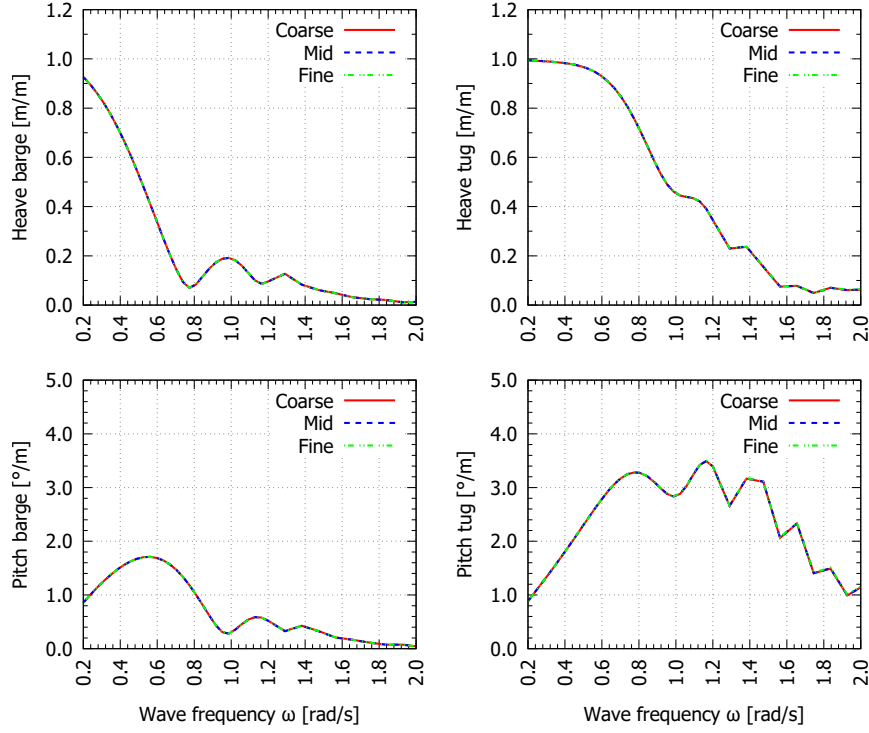


Figure 3.11: Heave (top) and pitch (bottom) motion responses per wave amplitude for head waves at zero body forward speed obtained from frequency domain AQW simulations with fine, mid and coarse body grids.

also reported in [93]. This marginal influence is also evident in the hinge force response amplitudes. Figure 3.13 shows representatively the times series of computed vertical hinge forces for different body grids for regular head waves of 8.6 s of period, ≈ 94 m of length and 1.16 m of amplitude for zero body forward speed.

In order to investigate the influence of time domain discretization, especially the influence of different time steps on the responses, AQW simulations are done with three different time steps, i.e. $\Delta t = 0.01, 0.005$ and 0.001 seconds. Again, the simulations are done for regular head waves at zero body forward speeds. The tug and the barge are interconnected with a kinematically constrained hinge. Figure 3.14 plots the response amplitude operator of heave and pitch motions for the studied time steps against wave frequency. It is obvious that the response amplitudes of these motions do not change significantly for different time steps. Similar conclusion is also reported by Riesner et al. [93], who studied the influence of different time steps on the motion responses of a large container ship. They recommend to keep $\Delta t < 0.07$ s if large roll motions for large container ships are expected; otherwise, larger time steps of up to 0.2 s may be used. Similar to

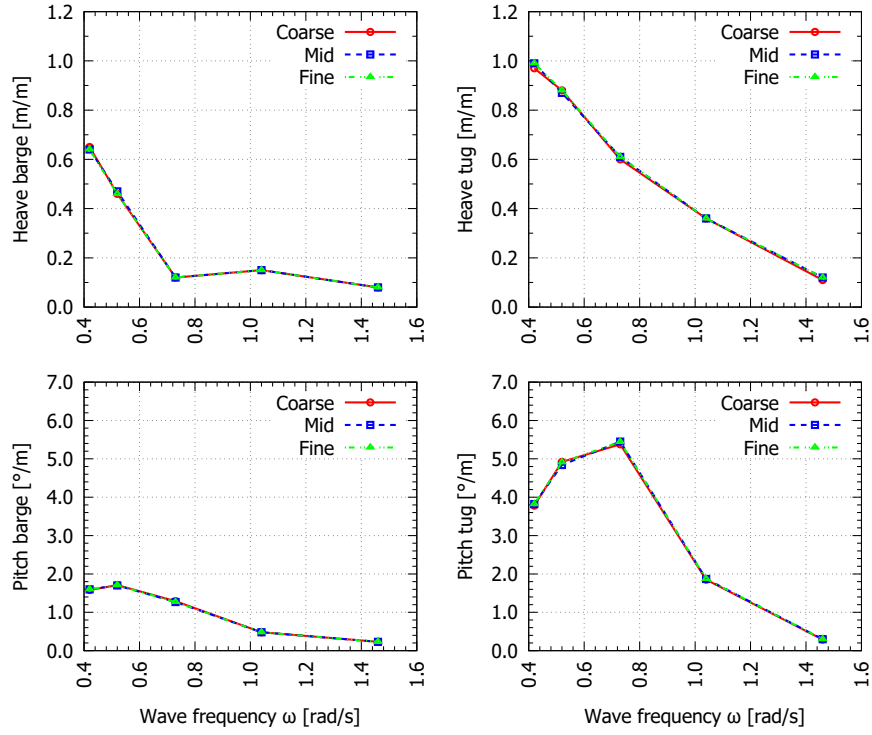


Figure 3.12: Heave (top) and pitch (bottom) motion responses per wave amplitude for head waves at zero body forward speed obtained from time domain AQW simulations with fine, mid and coarse body grids.

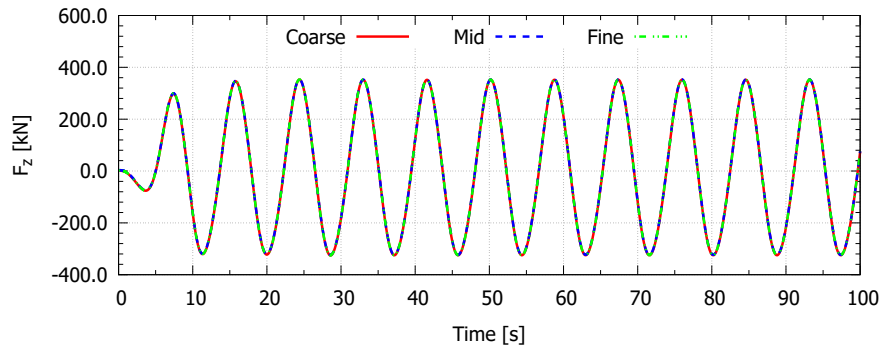


Figure 3.13: Influence of various panel sizes on vertical, F_z , hinge forces. The AQW simulations are conducted for head waves of 8.6 s of period, ≈ 94 m of length and 1.16 m of amplitude at zero body forward speed.

motion response amplitudes, the amplitudes of hinge force responses are hardly influenced by different time steps. Figure 3.15 shows the influence of studied time steps on the time series of computed vertical hinge forces for regular head waves

3 Validation

of 8.6 s of period, ≈ 94 m of length and 1.16 m of amplitude for zero body forward speed.

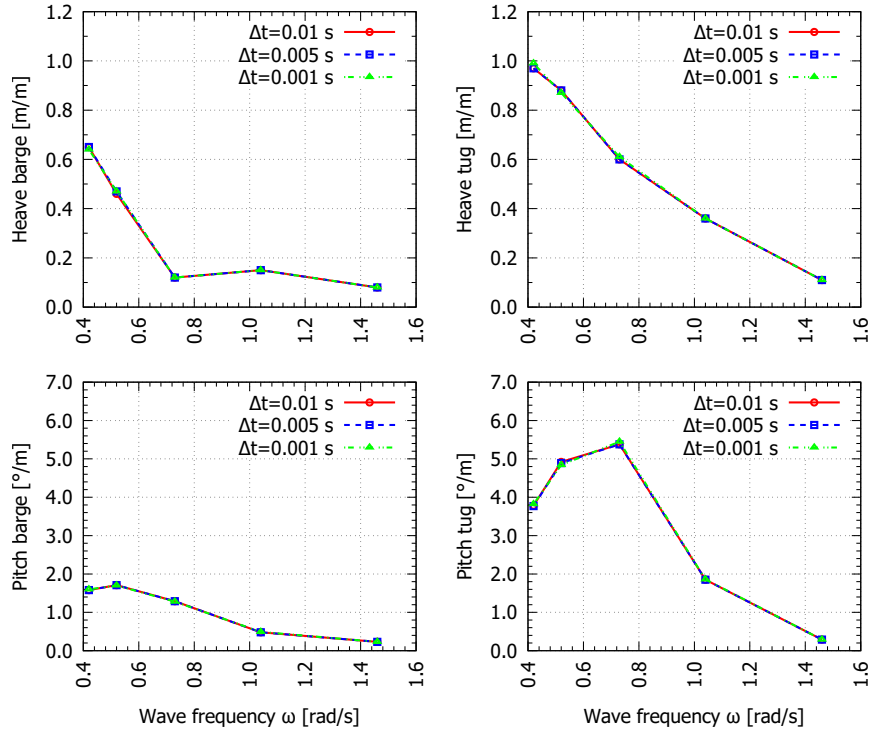


Figure 3.14: Heave (top) and pitch (bottom) motion responses per wave amplitude for head waves at zero body forward speed obtained from time domain AQW simulations for different time steps.

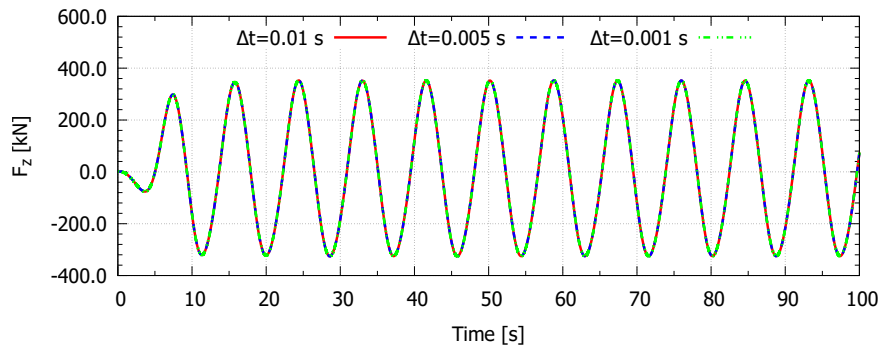


Figure 3.15: Influence of various time steps on vertical hinge forces, F_z . The AQW simulations are conducted for head waves of 8.6 s of period, ≈ 94 m of length and 1.16 m of amplitude at zero body forward speed.

3.2.2 Discretization Studies for Reynolds-Averaged Navier-Stokes Equation Solver

Grid quality and size play a decisive role in RANS simulations because they directly affect the convergence behavior and computation time. On the contrary to stationary flow problems, in which poor quality grids can still be counterweighted by small underrelaxation computations to converge computations, instationary flow problems require good quality grids with relatively high underrelaxation factors to attain convergence of the solutions in an acceptable time span. Unsteady simulation types are used within this work to simulate ship motion with RANS. Therefore, grids of reliable quality are required to ensure that the fluid solver converges robustly throughout the simulation time. Regarding grid independence studies, reference is given to [91, 112, 113]. In ITTC [91], recommendations are provided on grid independence studies in marine field, and in [112, 113], comprehensive details on verification and validation of numerical methods in field methods are given. In field methods grid independence study usually involves consideration of simulations on three or more successively finer grids, so that:

$$r_k = \frac{\Delta x_2}{\Delta x_1} = \frac{\Delta x_3}{\Delta x_2} = \frac{\Delta x_i}{\Delta x_{i-1}} \quad (3.1)$$

Δx_i is the grid size of i^{th} grid and r_k is the refinement ratio. ITTC [91] recommends $\sqrt{2} < r_k < 2$. This method results in an error band on the computational solution, which indicates the possible difference between discrete and continuum values. Additionally, it must be guaranteed in a grid refinement study that fluid characteristics are transmitted throughout the domain with the same velocity; otherwise, the comparability of data is violated.

Starccm+ package is employed to perform grid independence computations. Computations are done for full scale ships articulated with kinematically constrained hinge couplings. The bodies are exposed to regular head waves and undergo zero forward speed. Computations are performed with unstructured grid consisting of hexahedral control volumes. With reference to convoy length, L , the inlet is located about $1.0 L$ ahead of the barge's forward perpendicular; the outlet about $4.0 L$ away from the tug's aft perpendicular. No symmetry plan is used. The centerlines of the ships and computational domain are aligned. The side boundaries are positioned $1.5 L$ away from centerline. The bottom and top boundaries are located 17 m and 25 m below and above water surface, respectively. The free surface zone is refined to resolve the free surface phenomena efficiently. The cells behind the tug are stretched smoothly to damp the waves on the ways towards the outlet. Additionally, an augmented wave damping measure as per [15] is applied.

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To account for the body boundary layers, the area in immediate vicinity of ship hulls are resolved with prism layers and wall functions are applied. Second order spacial and temporal discretization schemes are used. Three different grids with the refinement factor of $\sqrt{2.25}$ are taken. The coarse, mid and fine grids comprise 1.9×10^6 , 2.8×10^6 and 4.6×10^6 control volumes, respectively. The influence of turbulence model and wall functions is minimized by keeping the dimensionless wall distance, y^+ , on hull's wetted surface constant at about 70. Figure 3.16 shows representatively the side view of mid grid.

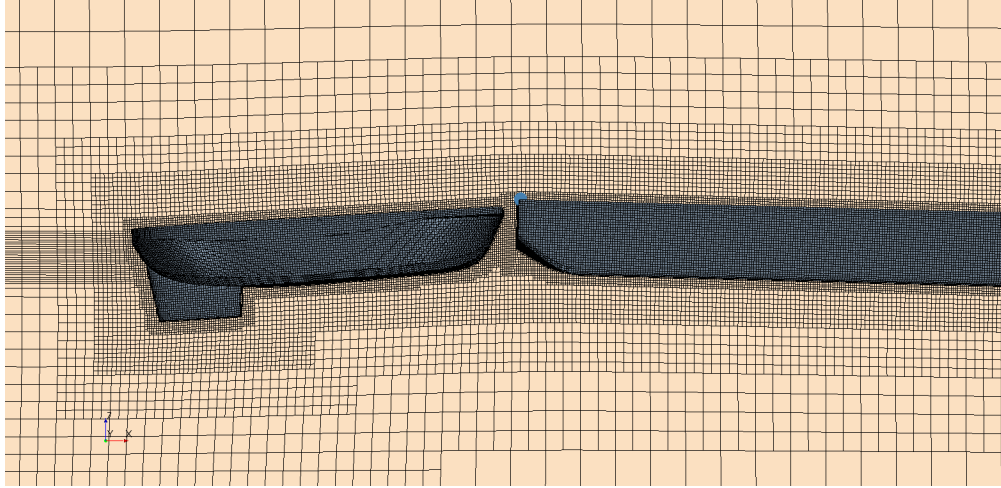


Figure 3.16: Side views of morphed mid grid at elapsed simulation time of 60 s.

The influence of discretization on the computation results are evaluated by examining the responses of heave and pitch motions as well as hinge force. Figure 3.17 shows the computed response amplitude operators of heave and pitch motions for the coarse, mid and fine grids. In order to consider wave damping, which especially affect short waves, the response amplitude operators are computed per amplitudes of actual computed waves, instead of target wave amplitudes. As obvious the results computed with different sized grids deviate relatively small from each other. It leads to the conclusion, that discretization does not significantly influence the response amplitude operators as long as the actual computed wave heights are applied. Studying the hinge force responses led to the same conclusion. Figure 3.18 shows representatively the time series of numerical waves and hinge vertical forces computed with examined grids for head waves of 6.0 s of period, ≈ 55 m of length and ≈ 0.5 m of amplitude at zero body forward speed.

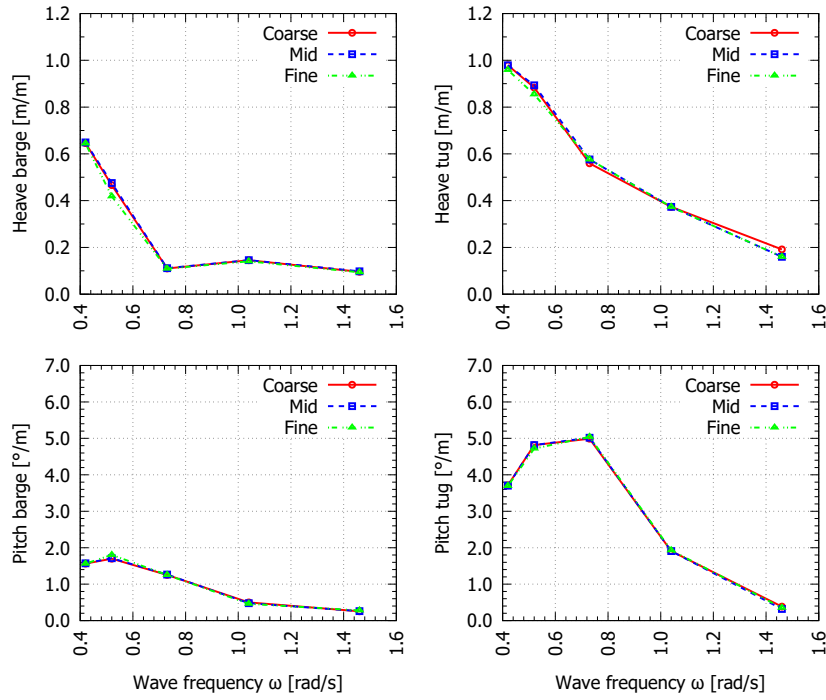


Figure 3.17: Heave (top) and pitch (bottom) motion responses per wave amplitude for head waves at zero body forward speed obtained from RANS simulations with fine, mid and coarse grids.

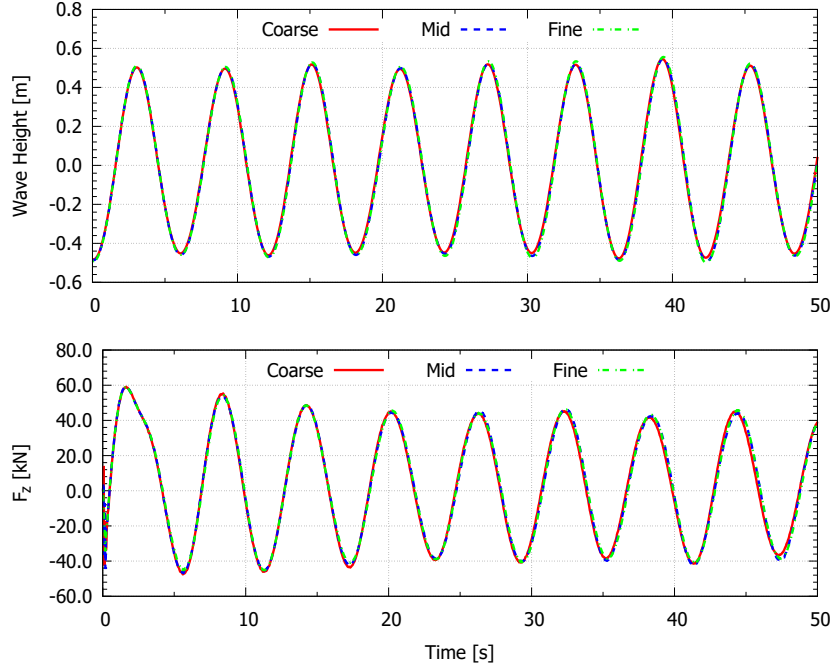


Figure 3.18: Wave height (top) and computed vertical, F_z , hinge forces (bottom) obtained from RANS simulations with coarse, mid and fine grids. The simulations are conducted for head waves of 6.0 s of period, ≈ 55 m of length and ≈ 0.5 m of amplitude at zero body forward speed.

3.3 Single Body Simulations

The same known workflow as in theory chapter is followed here, i.e. numerical simulations are done at first for single bodies, then for unarticulated multibodies and finally for the articulated ones. In this section, the bodies are simulated in their stand-alone conditions, in which they are exposed to encounter waves without any bodies in the vicinity. The computed motion responses are validated with model test measurements. The results from stand-alone simulations provide a robust foundation to construe the corresponding results from subsequent multibody analyses.

In the following, at first, the setup configurations for boundary element methods are stated. Afterwards, some descriptions on the configurations of RANS simulations are given. Finally, the results from different solvers are presented and compared with each other, and with the model test measurements. The model test setup configuration, the body geometries, and loading conditions applied in this section can be extracted from section 3.1.

All simulations in this work are done for full scale ships. Right-handed coordinate systems are applied. Body-fixed coordinate systems are defined at the centers of gravity of respective bodies. The free surface levels are positioned at $z=0$ m.

3.3.1 Computational Procedure and Simulation Setups of Boundary Element Method Solvers

3.3.1.1 Computational Procedure of Boundary Element Method Solvers

The seakeeping problem is addressed in frequency domain boundary element methods by solution of the periodical velocity potential, which is composed of six radiation and the diffraction potentials. Inserting the radiation and diffraction potentials into the respective boundary conditions, results in linear equation systems. After solving these systems, the resulting potentials can be used to calculate respective first order complex pressure amplitudes. Calculated pressure amplitudes yield the complex amplitudes of the radiation and diffraction forces and moments. The forces and moments are used to solve the linear equation of motion. Airy waves are employed for incident waves, whose potentials are used to determine diffraction potentials. Diffraction, radiation, and ship hydrostatic forces are all considered linearly.

In the time domain boundary element method used within this work, the diffraction forces are taken directly from prerequisite frequency domain simulation; hence,

treated linearly and the hydrostatic forces as well as Froude-Krylov forces are calculated for instantaneous wetted surface; hence, treated non-linearly. Again, Airy waves are applied for time domain simulations, whose kinematics are corrected by means of Wheeler stretching [139] to provide predictions of fluid velocity and acceleration at points above the mean water level. Convolution integral is approximated after approximation of memory function. Memory function is computed from frequency-dependent linear damping matrix (or frequency-dependent added mass matrix), which are in relation with radiation forces. In order to get the added mass in infinite frequency, an upper frequency limit, approximately 10 rad/s, is set in frequency domain calculations. By means of added mass at upper frequency limit and frequency-dependent added mass and damping matrices an asymptotic added mass matrix in infinity can be obtained [1]. After approximation of the convolution integral the motion equation is set up to compute the motion responses.

It is recapitulated that RSM, GLR, and AQW simulations are related to frequency domain boundary element methods. Additionally, AQW-simulations in time domain are performed, which require some input from respective AQW frequency domain simulations. RSM and GLR simulations are done in *GL Rankine* package, whereas AQW simulations in *Ansys Aqua* suite. In the following, the configuration applied in RSM, GLR, and AQW to accomplish simulations successfully are mentioned.

3.3.1.2 Simulation Setups of Rankine Source Method

In the Rankine source method used within this work (RSM), unstructured grids of triangular panels types are used to discretize the body surfaces up to steady flow waterline, whereas block structured grids consisting of rectangular panels are employed to discretize the free surface. The block structured grid consists of inner panels, on which the free surface boundary condition is imposed, as well as side and end panels. The side panels impose a free surface condition, which makes the waves propagate only away from the ship:

$$\phi_y^1 \pm \phi_z^1 = 0 \quad + \text{ for port and } - \text{ for starboard} \quad (3.2)$$

Additionally, waves are damped by multiplying the gravity acceleration with a complex factor of positive imaginary part. It is done by modifying the kinematic free surface condition to:

$$\nabla \phi^t (\vec{r} - \vec{d}) \vec{n} - ux = 0 \quad (3.3)$$

where $\vec{n}$ is the normal vector on the free surface; $\vec{d}$ is a vector in the direction of

longitudinal grid lines. In [21], it is demonstrated picturesquely how this condition results in wave damping. Free surface extremities and panel sizes are defined as a function of forward speed and ship's length. The free surface height of the grid generated for seakeeping is set to mean free surface position. New free surface grids are used for each wave frequency to optimize the calculation time because longer waves require larger and shorter waves finer grids. The applied Rankine source code is capable of considering the transom condition, which realizes the unwetted transom condition of a ship in (high) forward speed. The transom condition for periodical flow follows from the dynamic free surface condition with the additional statement that the free surface elevation is equal to the vertical component of the velocity shift in the inertial system of body-fixed points at the transom. Transom condition is applied for the barge but not for the tug. The tug's geometry and its main principals lead to a negligible amount of transom panels, which cause negligible contribution to seakeeping behavior of the tug. The sharp geometry edges undergo special treatment during mesh generation to bypass the problem of potential singularities at sharp edges. Body panels are dissingularized by positioning the adherent source points 20% of panel width outwards, whereas the sources of free surface panels are positioned at one panel width above the free surface panels. A horizontal bottom is defined at water depth of 17 m to be in line with model tests. At this bottom line, the bottom boundary condition, i.e. no flow through the bottom is fulfilled by using image sources. Basin side walls are remained unmodeled within simulations. Figure 3.19(a) and (c) show the free surface and body panels, and sources used for barge Rankine source simulations for wave length of ≈ 120 m and ship's forward speed of 6.0 knots. All the simulations are done for head waves, and with the bodies suppressed in roll (see table 3.2). It eliminates the necessity to take roll viscous damping into account. No corrections are done to consider the flow separation at the stern and rudder.

Linear seakeeping analyses require the solution of steady simulations as base solution. Due to the small forward speed, steady solutions are computed by means of double-body flows. The same unstructured triangular grids is used in the ship's hull as in the seakeeping computations. Figure 3.19(b) shows the body panels and sources used to compute the double-body flow around the tug.

3.3.1.3 Simulation Setups of Green Function Method

Zero-speed free surface Green functions provide accurate solutions for problems with the linearized free surface boundary condition at zero forward speed. Encounter frequency correction approach is applied for small forward speeds. Since Green function methods satisfy the linear free surface boundary condition analytically, no singularities are required on the free surface. Same triangular unstruc-

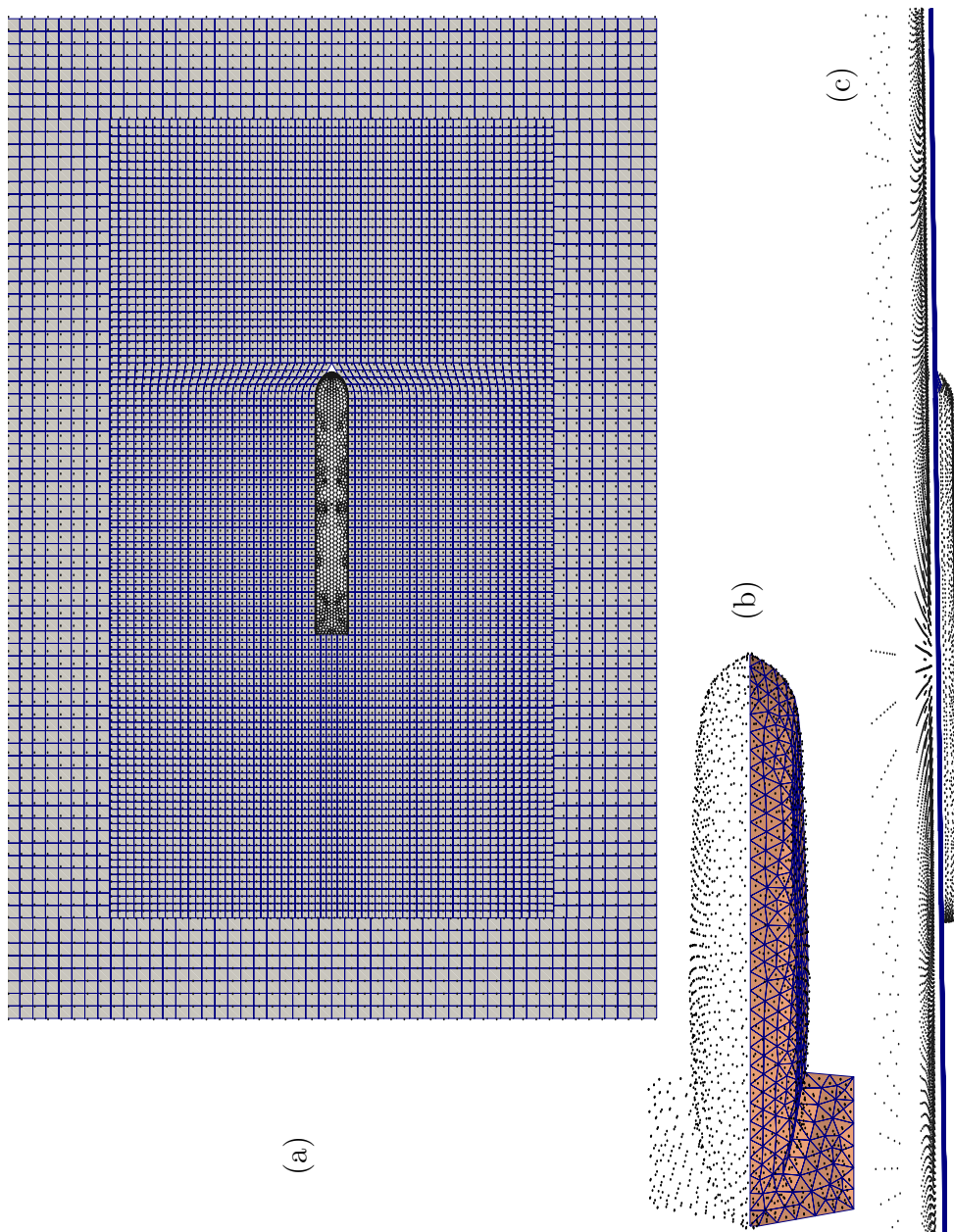


Figure 3.19: Free surface and body panel grids and sources shown from top (a) and side (c) used for stand-alone simulation of the barge for wave frequency of ≈ 0.6 rad/s at forward speed of 6.0 knots with Rankine source method. (b) shows the triangular tug panels and sources used for double-body as well as GLR simulations. Shallow water sources are not shown.

tured grids as in Rankine source method simulations are used for simulations of the first Green function method used in this work GLR. Irregular frequencies, which cause unphysical standing waves inside the body, are eliminated by generating a body internal boundary. Figure 3.19(b) shows the body and sources used for GLR simulations. In fact, the same body panel grids and sources are used both for doubly-body and GLR simulations.

In the second Green function method used within this work, named AQW herein, the same procedure as in GLR is used principally. Nevertheless, apart from the theoretical formulation distinctions mentioned in section 2.1.4, there exists some differences in configurations, too. The panels in AQW can be of either triangular or quadrilateral type with their associated source point located in panel collocation points. Not only the wetted body surfaces, but also the surfaces above the mean free surfaces are discretized. The panels above the mean water level are tagged non-diffracting to avert their consideration in the respective frequency domain simulations. These non-diffracting panels above mean water levels are then used in time domain simulation because the Froude-Krylov and hydrostatic forces are calculated nonlinearly by taking the instantaneous wetted surface into account. The question may arise why GLR and AQW are taken, although both are frequency domain zero-speed Green function boundary element methods with linearized equation of motion solvers: In section 3.5, it will be seen that GLR and time domain AQW are used for articulated multibody boundary element method simulations in frequency and time domain, respectively. Time domain AQW simulations require frequency dependent parameters from prerequisite frequency domain AQW computations. Considering incremental structure of validation chapter, from single body simulations, over hydrodynamically coupled multibody simulations, to articulated multibody simulations, frequency domain AQW provide additional comparative data, especially in qualifying and identifying the efficiency, accuracy and discrepancies of GLR.

3.3.2 Computational Procedure and Simulation Setups of Reynolds-Averaged Navier-Stokes Equation Solver

3.3.2.1 Computational Procedure of Reynolds-Averaged Navier-Stokes Equation Solver

Generally, the procedure shown in figure 3.20 is used to couple body motion solver with the fluid solver. Within each outer iteration, new velocity, pressure, volume fraction as well as turbulence parameter fields are computed via inner iterations. By means of updated flow forces, the new position of the floating body is computed

and the grids are updated accordingly. The procedure recurs until the convergence is achieved. Then, the time is advanced and the same procedure is repeated.

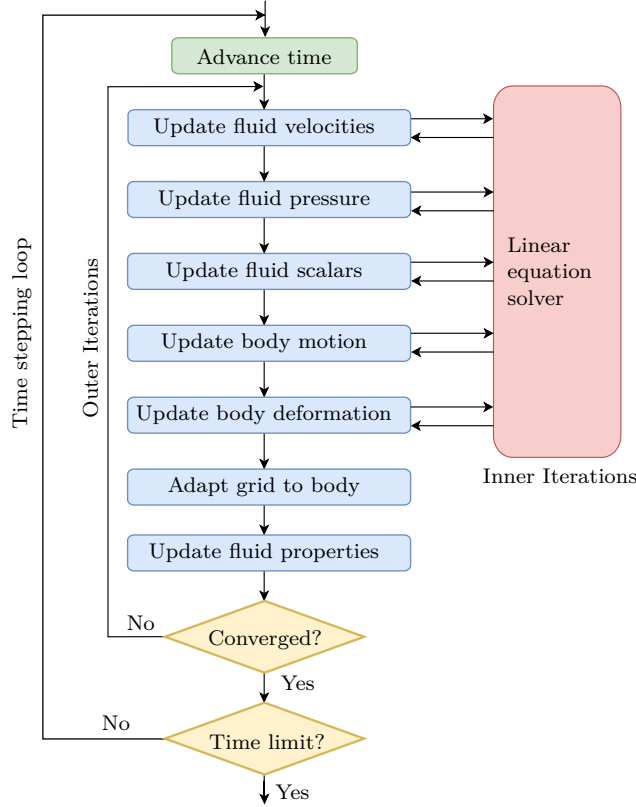


Figure 3.20: Schematic solution procedure of Reynolds-Averaged Navier-Stokes Equation solver [99].

3.3.2.2 Simulation Setups of Reynolds-Averaged Navier-Stokes Equation Solver

The RANS solver delivered in the software package *Starccm+* is used to perform simulations. Unstructured trimmed hexahedral control volumes are used for finite-volume RANS solver used to discretize the numerical domain. The conservation equations in integral form for mass and momentum are solved using a segregated iterative solution method based on SIMPLE algorithm to solve the fluid problem. All surface and volume integrals are approximated using midpoint rule. Interpolations are done by means of second order linear scheme. Gradient approximations are based on second order central differencing scheme. Second order implicit three time level marching scheme is used to approximate temporal derivatives. Free

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surface effects are modeled using the Volume-of-Fluid method approach to track the deformation of the initially flat free surface. This model increases the number of equations required to solve by one transport equation. The same second order implicit three time level marching scheme used in discretized momentum conservation equation is employed to approximate the time derivative of volume fraction transport equation. The spatial transport of volume fraction is treated by HRIC scheme [62, 63, 20]. SST- $k-\omega$ turbulence model with wall functions is used to describe the effects of turbulence on the flow. The turbulence model increases the whole number of equations required to solve the flow problem by two eddy-viscosity equations.

Local refinement in zones of interest are applied to obtain maximum accuracy at low cell counts. Hereto, the free surface and the body surrounding areas are locally refined. As the test case is conducted in shallow water (see section 3.1), the zone underneath the body surfaces is also refined to capture the shallow water effects.

The velocity components in the main flow direction within the boundary layer of non-slip body surfaces is much larger than the components in other directions. Furthermore, the pressure gradient across the flow is much smaller than that of the principle flow direction in this zone. Therefore, prism layers are used near non-slip body surfaces; otherwise, application of hexahedron cells may lead to long thin cells, that would create problems in the approximation of diffusive fluxes. The number of prism layer cells are reduced by associating the wall function into the turbulence model.

The simulations are all done for full scale models at various forward speeds (see table 3.2) in a time marching mode, starting from a flat water surface with 0.01s time steps. Both tug and barge are geometrically modeled without any appendages - in compliance with the model tests. Nonlinear ship motions are computed with an implicit six degrees of freedom dynamic solver, which is coupled with the flow solver. The models are free to heave and pitch, and are exposed to head waves. The simulations show that the tug undergoes relatively high motion responses, which have caused instability of mesh morphing technique. As a solution, the whole domain could be so arranged to follow the body motions. Since multibody hydrodynamics is the core topic of this work, this solution is not chosen because this solution is not necessarily appropriate for multibody simulations. As it will be seen in section 3.4, Chimera grid technique [14, 37] is the most flexible technique in RANS-simulation of unarticulated multibodies. Therefore, this technique is also applied for those stand-alone simulations, for which relatively high motion amplitudes are expected, otherwise mesh morphing are applied. Generally, more satisfactory results have been experienced with Chimera grid in the used RANS software package.

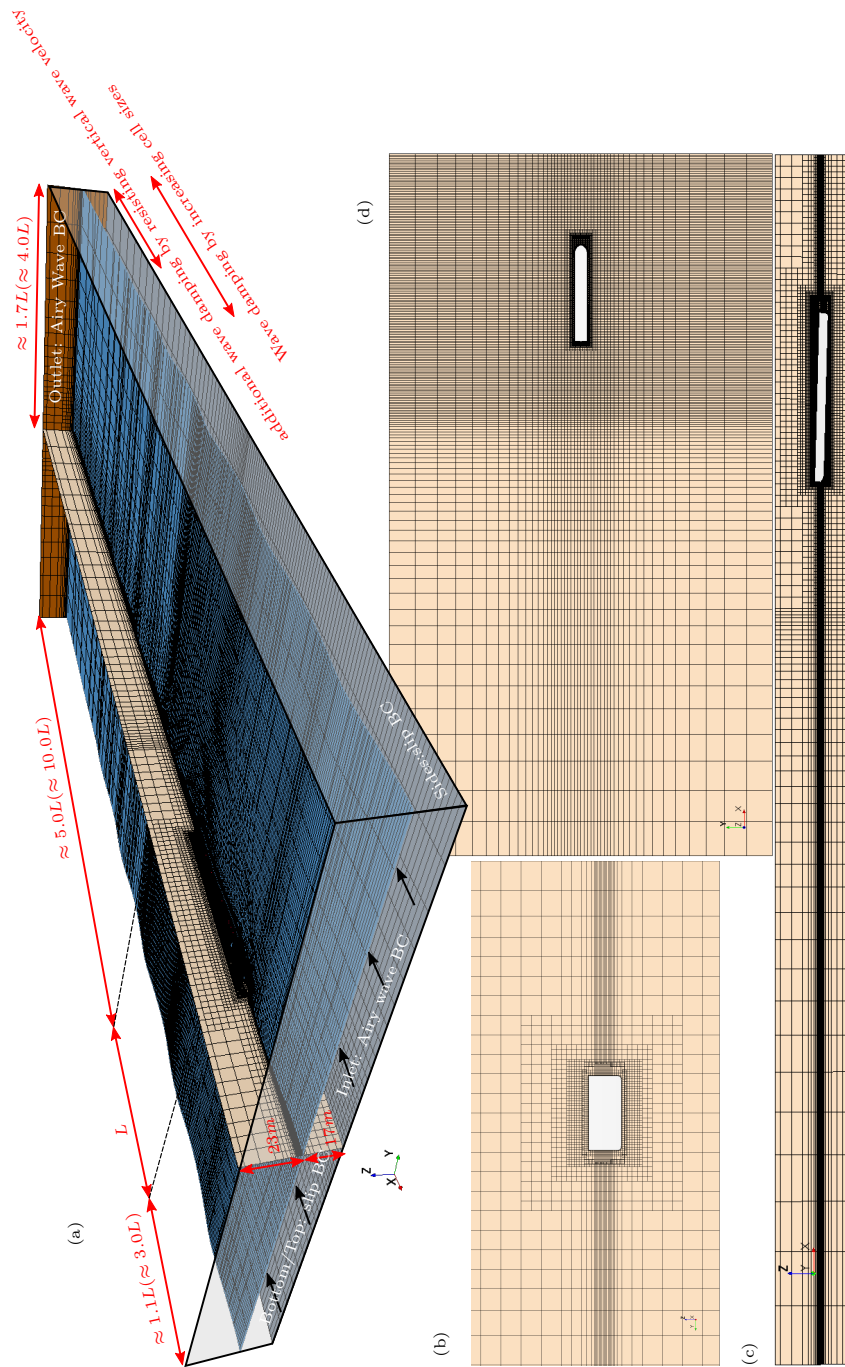


Figure 3.21: Computational domain for stand-alone barge simulations. The values within parentheses are related to tug simulations. (a) perspective view (b) front view (c) side view (d) top view.

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There is no predefined recipe to determine the extents of the computational domain. In [44], it is recommended to have $2L$ (L denotes ship's length) ahead, $3L$ abaft, $2.5L$ abeam and $2.5L$ atop of the ship. In [98], it is proposed to have $2 - 2.5\lambda$ (λ denotes wave length) ahead of the ship if potential waves are triggered into the domain. In [96], a compacter domain, namely $0.41L$ and $0.50L$, in front of, $1.35L$ and $1.00L$ behind the ship, $0.23L$ and $0.20L$ above design draft, and $1.0L$ from ship centerline to domain side are chosen for long waves of $0.6\frac{\lambda}{L}$ to $2.0\frac{\lambda}{L}$, and for short waves of $0.1810\frac{\lambda}{L}$ to $0.4782\frac{\lambda}{L}$, respectively, to compute the added resistance, and motions of benchmark KVLCC2 tanker with free heave, pitch, and fixed and free surge motions. In [96], satisfactory results are reported with such a relatively compact numerical domain. The computational domain used within this work, depending on λ/L , ship's speed and wave angle of attack, varies slightly from simulation to simulation, but in general the numerical domains used here for stand-alone bodies, in relation to the ship's length, follows principally figure 3.21. The total number of control volumes used for barge simulations is about 1.1 to 1.4×10^6 , whereas those of tug's is about 4.0 to 6.0×10^5 . These cell numbers are associated to the results gained from grid independence studies described in section 3.2. In all RANS simulations, right-handed coordinate systems are used, see figure 3.3, and body-fixed coordinate systems are defined at the centers of gravity of each respective body, see figures 3.1 and 3.2. The free surface at rest lies at $z=0$ m.

Several types of boundary conditions are required in RANS simulations. The applied boundary conditions are summarized in table 3.4. Principally, the symmetry boundary condition could be imposed so as to model only one-half of the ships and flow fields; nevertheless, this condition is not applied to be in compliance with the articulated multibody simulations. The no-slip condition is applied on body surfaces. The bottom, top and side walls are also of slip type. Wave boundary conditions are applied for inlet and outlet of the numerical domain. The generated waves are regular periodic sinusoidal linear Airy waves, see section 2.1.3. Mesh refinement is done in x and z directions to avoid strong numerical dissipation on wave propagation, and to capture wave lengths and heights efficiently. Waves are numerically damped behind the ships on the way to the outlet by stretching the control volumes in wave propagation direction. Additionally, the waves are damped mathematically as per [15] by augmenting a vertical resistance term to the wave vertical velocity. These damping measures allow definition of a relatively compact numerical domain, which offers stable simulations over a long simulation time without concerning reflecting waves.

Table 3.4: Boundary conditions in RANS single body simulations.

location	velocity inlet	pressure outlet	top, bottom, sides	body surface
type	wave	wave	slip	no-slip
u	equation (2.12)	0	0	0
v	0	0	$\nabla v = 0$	0
w	equation (2.13)	0	$\nabla w = 0$	0
p	0	equation (2.14)	$\nabla p = 0$	$\nabla p = 0$
k	0	0	$\nabla k = 0$	0
ω	0	0	$\nabla \omega = 0$	*
ζ	0	0	0	0

* constant turbulence intensity, turbulence viscosity ratio, and turbulence viscosity scale is used instead.

* u , v , w , p , k , ω and ζ notate longitudinal, transverse, and vertical wave velocities, wave pressure, turbulent kinetic energy, turbulent dissipation frequency, and wave elevation, respectively.

3.3.3 Results and Discussions

Figures 3.22 and 3.23 show barge and tug motion responses per wave amplitude. On these figures, GLR, RSM and RANS denote the results from GLR, RSM and RANS simulations, respectively. AQW freq and AQW time indicate the results from frequency and time domain AQW simulations, and measurements are denoted EXP.

The heave responses of the barge per wave amplitude for 0.0, 6.0 and 8.0 knots forward speeds, figure 3.22 top, are predicted well by all methods throughout the whole investigated wave frequency range. For example, all computations for head waves at wave frequency 0.47 rad/s and at zero speed body forward speed predict the measured heave response amplitude operators within 1% accuracy. Also, the computed pitch motion response amplitudes of the barge, figure 3.22 bottom, agree well with the measurements. GLR predicted responses tend to deviate slightly in forward speeds, which is ascribed to inaccuracies of its forward speed correction method, figure 3.22 bottom. At zero speed, AQW time results have underprediction tendency of barge heave motion amplitudes compared to AQW freq results. At 6.0 knots forward speed, all boundary element methods tend to slightly overpredict the measured pitch motion response amplitude at 0.52 rad/s wave frequency. This slight pitch response overprediction tendency mitigates at 8.0 knots forward speed.

Similarly, good agreements are achieved for tug's heave and pitch predictions with

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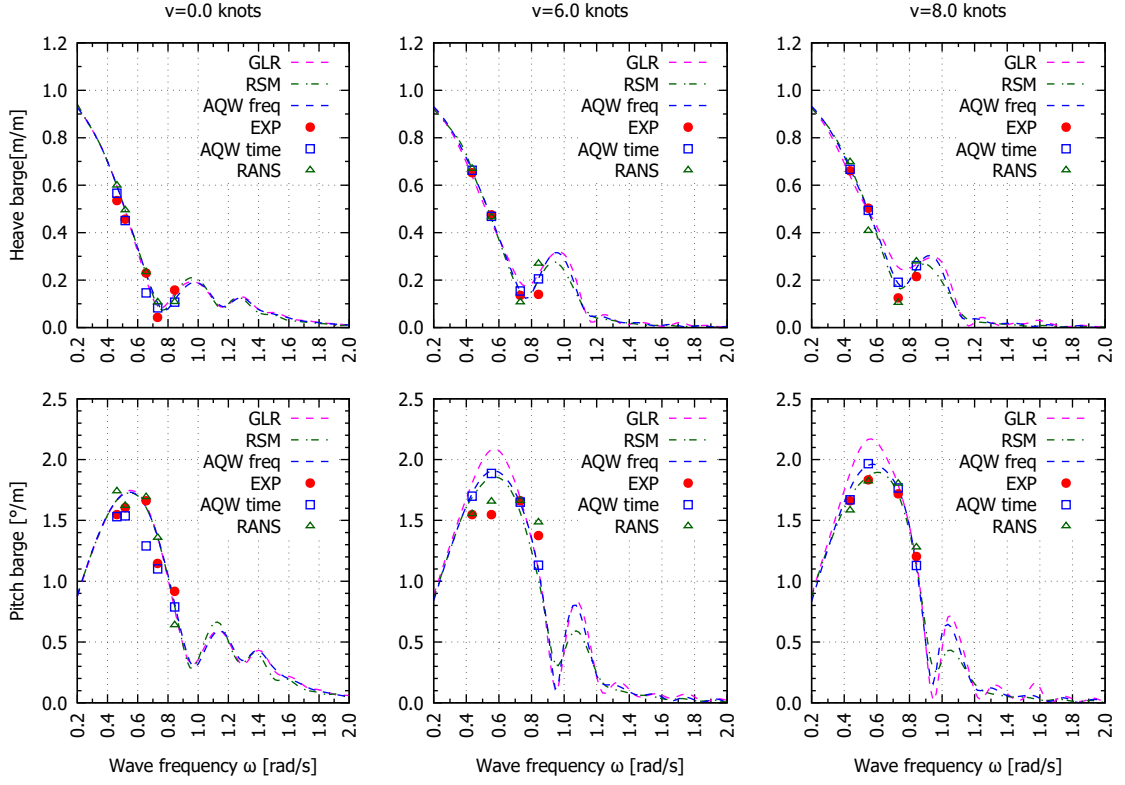


Figure 3.22: Heave and pitch motion responses per wave amplitude of stand-alone barge in head waves at $Fn = 0.0, 0.10$ and 0.14 body forward speeds.

measurements, figure 3.23. GLR heave predictions at 6.0 knots forward speed show an unreasonable, but a known peak. The peak is attributed to inaccuracy of GLR's forward speed correction method. Such a peak is damped significantly in AQW freq and AQW time simulations because their underlying forward speed correction method, published in [97] and [1], corrects additionally the frequency dependent parameters. RSM shows the smallest hump because it is capable of modeling forward speed without requiring such corrections. RANS heave and pitch motion response amplitudes at 1.46 rad/s were relatively overpredicted. The overprediction attenuated after halving the base grid size from 6.0 m to 3.0 m; thus, changing consequently all parametrically associated mesh refinement zones. This has led to a change of mesh cell numbers from 0.4×10^6 to 1.6×10^6 . Recall that the base grid size is just a governing parameter, to which most of the other mesh size parameters are referred. Figure 3.24 shows changes in motion responses of the tug after mesh refinement. Referring to figure 3.24, mesh refinement improved the results moderately but it seems improbable to meet the measurements of 1.04 rad/s wave frequency with further grid refinements. All heave and pitch responses shown

in figure 3.23 for 6.0 knots are related to the halved base grid size.

Figure 3.25 compares the numerically generated waves in RANS with the model test waves. The heave and pitch responses of the bodies, while encountering these waves at 6.0 and 8.0 knots are also presented. Figures 3.26 and 3.27 show wave patterns of the ships in two different waves.

Summary

Both barge heave and pitch responses were predicted well by all boundary element and RANS methods. The tug reacted very sensitive to hydrodynamic forces due to its geometrical characteristics and inertial properties. The tug pitched up to six times higher than the barge at non-zero forward speeds. Altogether, satisfactory predictions were achieved for tug's heave and pitch motion response amplitudes by all methods.

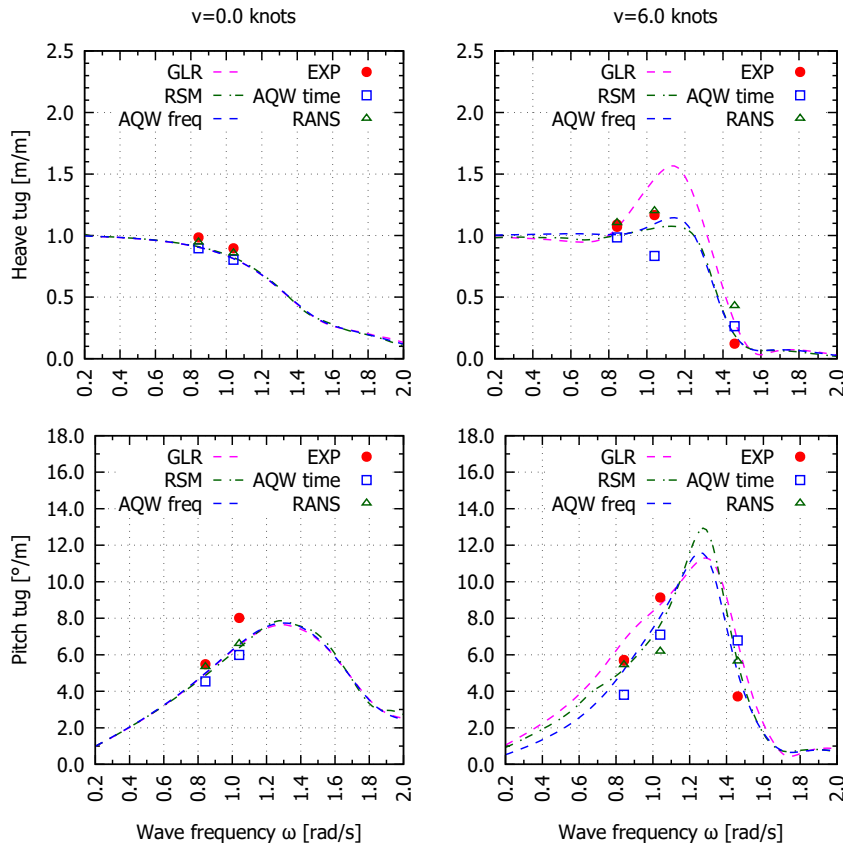


Figure 3.23: Heave and pitch motions per wave amplitude of stand-alone **tug** in head waves at $Fn = 0.0, 0.20$ and 0.26 body forward speeds.

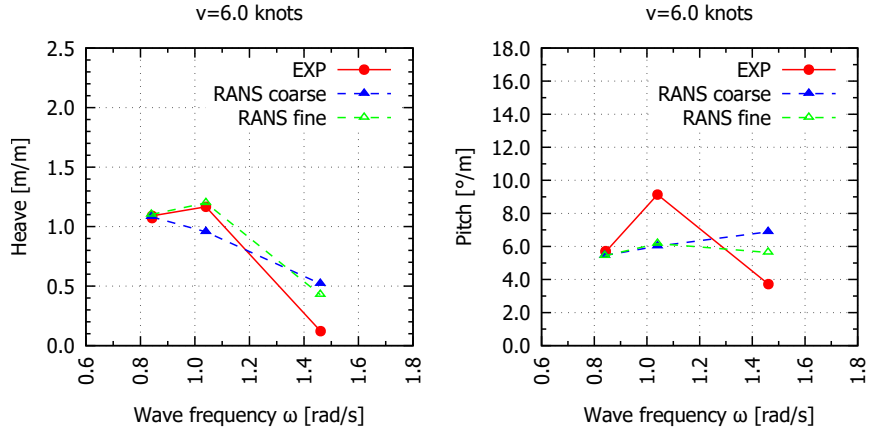


Figure 3.24: Changes in tug's heave and pitch responses per wave amplitude for RANS simulations after halving the governing base grid size from 6.0 m (denoted RANS coarse) to 3.0 m (denoted RANS fine). Model test measurements are denoted EXP. Simulations are done for head waves at $Fn = 0.20$ body forward speed.

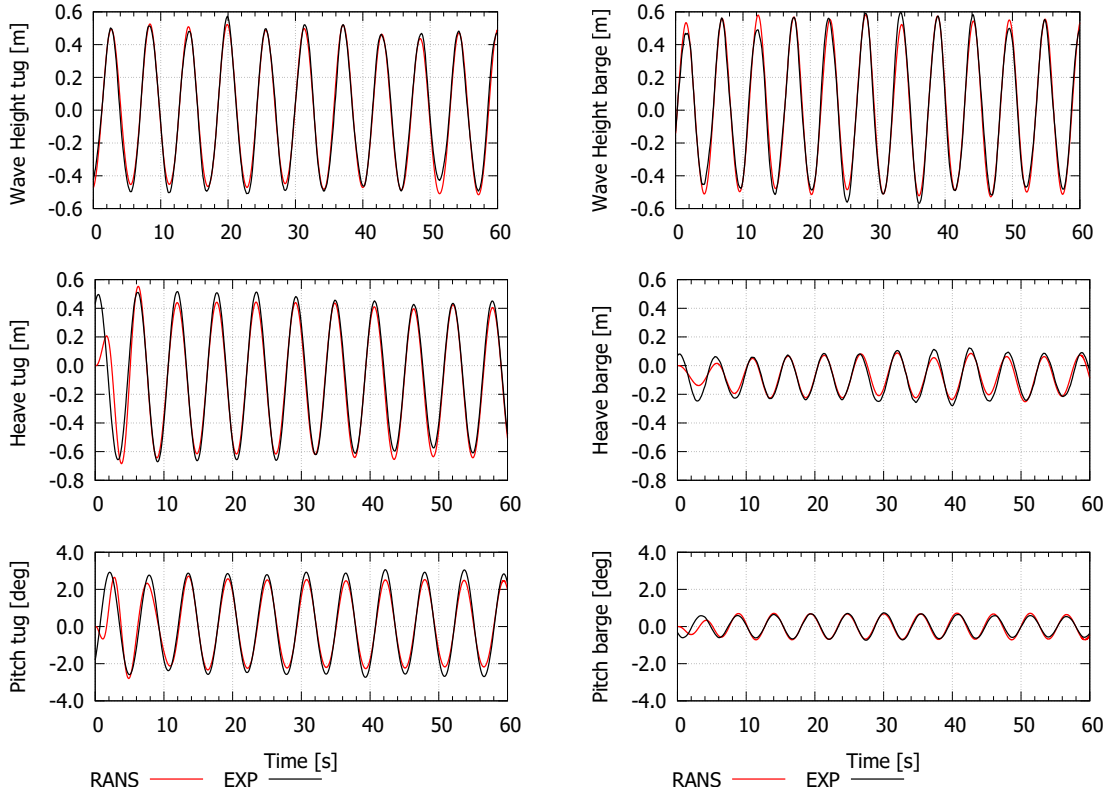


Figure 3.25: Comparison of time series of wave height as well as heave and pitch motions related to measurements and RANS simulation results for the tug (left) and the barge (right) in head waves of ≈ 77 m of length, 7.5 s of period and 0.49 m of amplitude. Tug and barge advance with 6.0 ($Fn = 0.20$) and 8.0 ($Fn = 0.14$) knots forward speeds, respectively. Computed RANS results and measurements are denoted RANS and EXP, respectively.

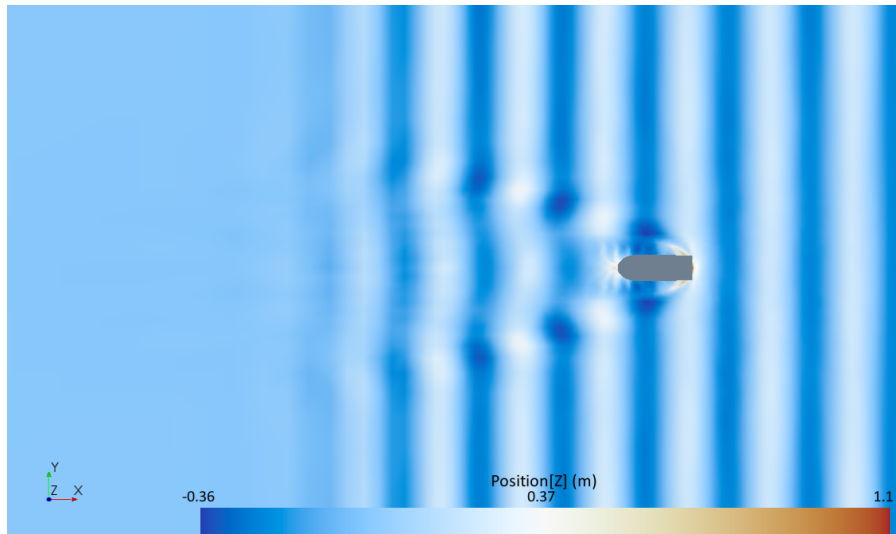


Figure 3.26: Tug's wave pattern with forward speed of 6.0 knots ($Fn = 0.20$) in head waves of ≈ 29 m, period of 4.3 s and amplitude of 0.22 m. The legend $position[Z](m)$ stands for wave elevation.

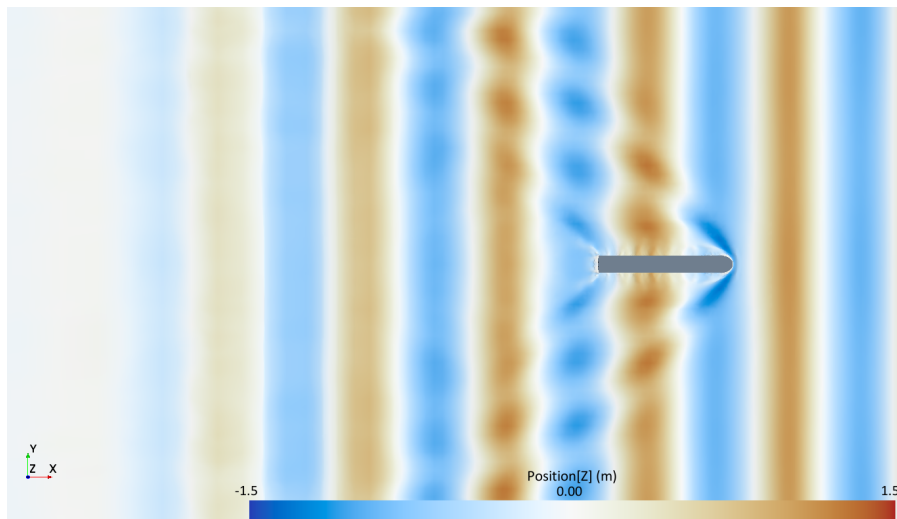


Figure 3.27: Barge's wave pattern with forward speed of 6.0 knots ($Fn = 0.10$) in head waves of ≈ 90 m, period of 8.6 s and amplitude of 0.98 m. The legend $position[Z](m)$ stands for wave elevation.

3.4 Hydrodynamically Coupled Multibody Simulations

Following the simulation of single bodies in section 3.3, hydrodynamic coupling of multibodies will be discussed in this section. *Hydrodynamic coupling* refers to multibody hydrodynamics in absence of any mechanical interconnections. Both tug and barge used in stand-alone simulations are taken for multibody simulations. Similar to the stand-alone simulations in section 3.3, both zero-speed Green function methods (AQW and GLR), Rankine source methods (RSM), and RANS will be used to assess the motion response of the bodies. GLR and RSM simulations are performed in frequency domain, whereas RANS simulations are done in time domain. AQW simulations are done both in frequency and time domains. Theoretical background of multibody hydrodynamics, and the challenges in its realization with different numerical methods have already been discussed in section 2.2.2. In the following, a short description on the configurations applied in the employed boundary element and RANS methods is given. Afterwards, the motion responses obtained from different simulation methods are presented and compared with the model test measurements.

3.4.1 Simulation Setups

The examined tug and barge (figure 3.1) are positioned in a convoywise fore and aft arrangement in compliance with the respective model test runs (figures 3.5 and 3.8-right). Data related to the mass bodies are set in accordance with table 3.1. A series of simulations following the unarticulated multibody test runs, as stated in table 3.2, are performed both in frequency domain and time domain with the boundary element and RANS solvers described in section 3.3.

For the multibody boundary element simulations, the same body boundary grids used in single body simulations are taken. The tug and barge surfaces are discretized by about 1000 panels and 3700 panels, respectively, in GLR simulations. Totally 3800 diffracting and non-diffracting panels are used for both vessels in AQW simulations (see section 3.2). The time domain boundary element simulations follow the Cummins' approach, and are backboneed by the frequency domain results from AQW simulations. Convolution integrals are evaluated directly. Time step of 0.01s was chosen for these simulations.

The small distance between both vessels made it difficult to discretize the free surface in the applied frequency domain Rankine source method RSM. Free surface discretization for computation of steady velocity potential, ϕ^0 , is circumvented by

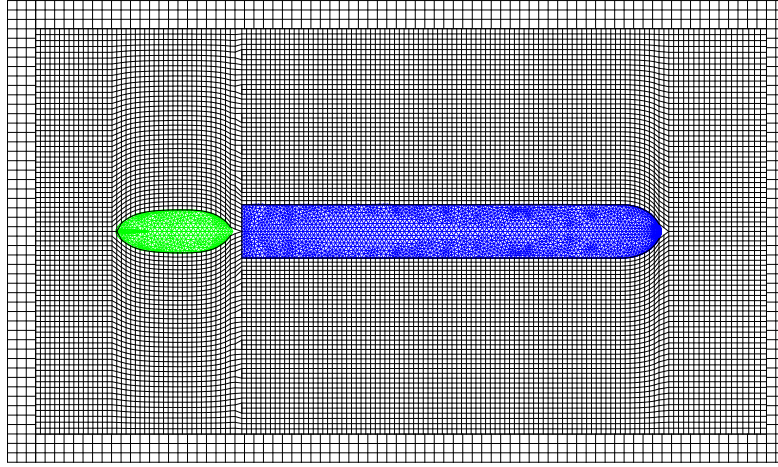


Figure 3.28: Body and free surface grids used in the multibody Rankine source method simulations.

application of double-body. However, no bypasses to avert the free surface grid for computation of unsteady velocity potential, ϕ^1 , is available for RSM simulations. Therefore, a free surface grid, that works well with the small gap between both vessels has manually been generated, and input in the Rankine source sea-keeping program. This free surface is used for all analyzed encounter frequencies. The heave and pitch motions per wave amplitude of the tug and the barge from RSM simulations are tagged RSM, and shown in figure 3.29. Comparative results and measurements from GLR simulation, and model test are also presented for reference. Comparing to GLR simulations, RSM has predicted the motion responses better, especially at forward speeds. However, RSM results are acceptable only for the wave frequencies greater than 0.7 rad/s ($\omega > 0.7$ rad/s). For $\omega < 0.7$ rad/s, the predicted motion responses show strong oscillations (see gray zones in figure 3.29). The oscillations intensify at higher speeds and are quantitatively much higher for the tug than the barge. The Rankine source method employed within this work treats total velocity potential specially, in such a manner, that the steady velocity potential is not developed about the inertial reference frame but about the body-fixed coordinate systems. This development allows to allude m -terms, [82, 76, 127], on the body boundary condition to predict hydrodynamic forces more accurately. Since in frequency domain, the excitation and responses are harmonic, ϕ^0 is also harmonic in body-fixed reference frame. The influence of harmonic motion on ϕ^0 is realized via interconnecting coupling terms of steady and unsteady potentials [110, 102]. Therefore, a small ship motion may

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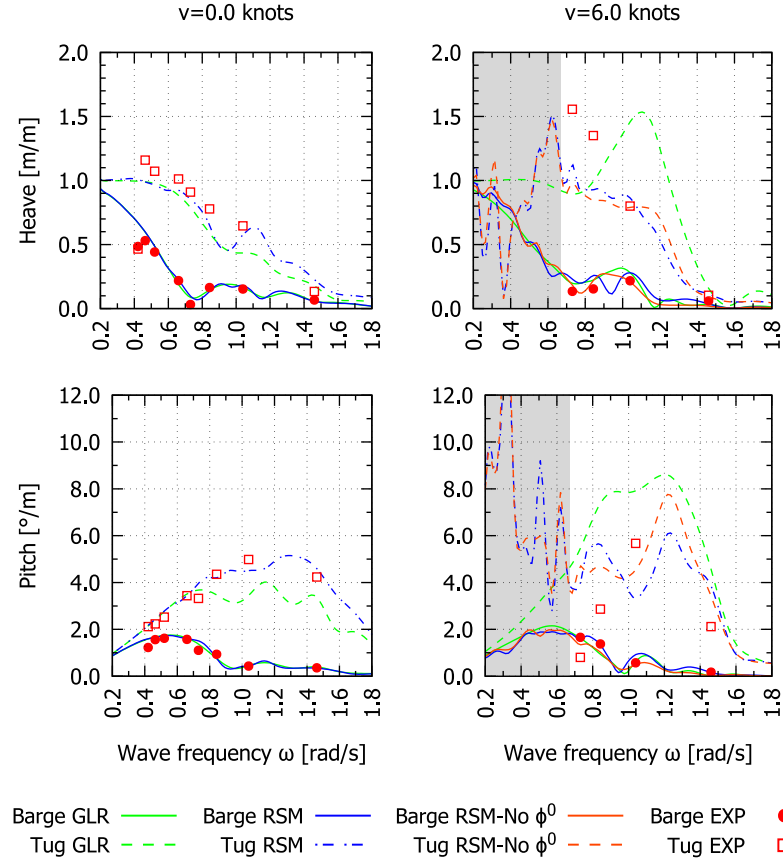


Figure 3.29: Heave and pitch motions per wave amplitude of hydrodynamically coupled tug and barge in head waves with RSM under consideration and elimination of stationary velocity potential field (RSM and RSM-No ϕ^0 , respectively). GLR results and measurements, EXP, are plotted for comparison. The oscillation zone is highlighted in gray.

cause extreme oscillation of ϕ^0 in far field. Depending on the distance between bodies of a multibody system, such ϕ^0 -development may be burdensome, because ϕ^0 -field of a body may excite significant forces on the other ships. In [127], this unphysical phenomena has been recognized in simulation of an overtaking maneuver in the roll response amplitude operators. Consequently, it is recommended in [127] to ignore the ϕ^0 -field in solving roll radiation problem. Intuitively, the unphysical oscillations of heave and pitch motion were ascribed initially to the ϕ^0 -field. The intenser oscillation of the tug than the barge, and the similarity of the symptom to the case in [127], in which the smaller vessel was more affected by ϕ^0 -field of the bigger vessel, fortified this rationalization. Subsequently, the proposed ad hoc solution of ignoring the steady velocity potential field has also been employed during solution of pitch and heave radiation problems. These resultant heave and pitch

3.4 Hydrodynamically Coupled Multibody Simulations

response amplitude operators are tagged RSM-No ϕ^0 in figure 3.29. Neglecting the unsteady velocity potential field shows some effects at the forward speed, but the improvements are negligible. It makes it evident that another cause, most probably the free surface grid, is the main source of oscillation. In long waves, greater free surface panel grids in size are numerically more damping. Therefore, application of greater-sized free surface grids may help to reduce or eliminate the oscillations. Nevertheless, further studies are required to clarify this matter. On the contrary, zero-speed Green function methods, GLR and AQW, have provided relatively good results with less effort (cf. figure 3.29 with figure 3.31 and figure 3.32); henceforth, RSM is excluded in the rest of this work, and all boundary element simulations are done with GLR and AQW hereinafter.

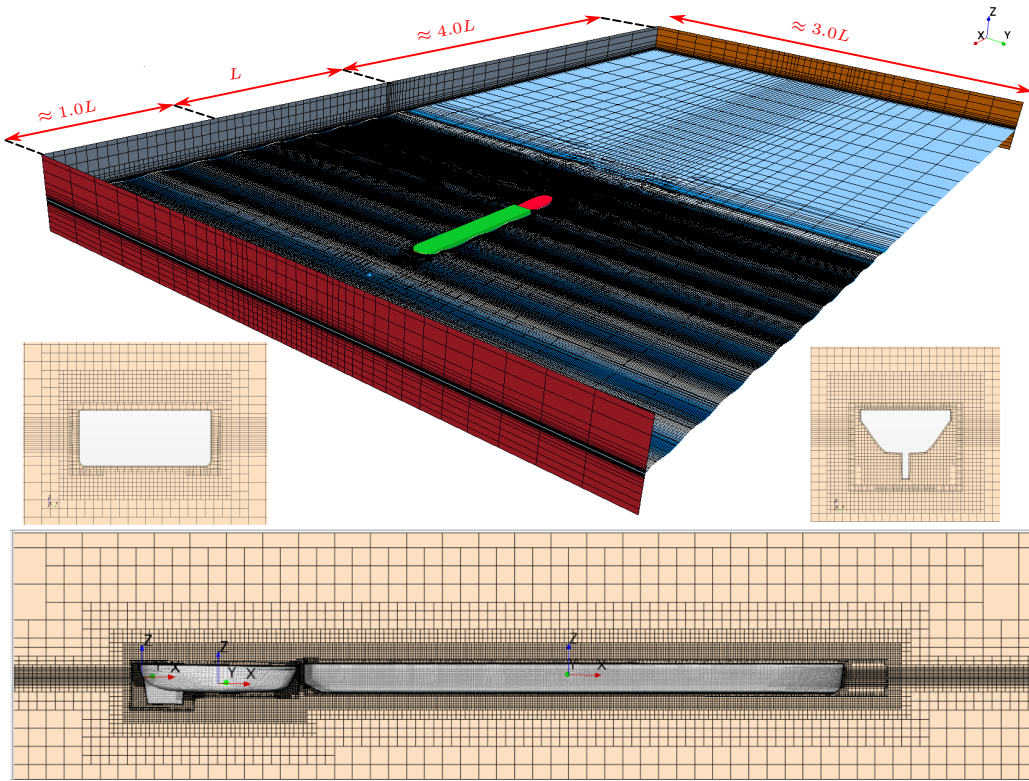


Figure 3.30: Computational domain used in multibody RANS simulations. Perspective view (top), tug's cross section (middle right), barge's cross section (middle left), and side view (bottom).

RANS simulations are setup mostly in the same manner as stand-alone simulations with full scale geometries. The same computational domain size is used for RANS in all simulations. The inlet is positioned about one convoy length ahead of the

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barge's forward perpendicular; the outlet boundary, four convoy length behind the tug's aft perpendicular; the side boundaries, one and a half convoy length to port and starboard of the convoy centerline. Symmetry condition has not been employed. The values imposed on the boundary conditions are specified in table 3.4. Unstructured grids consisting hexahedron control volumes are used throughout the whole computational domain. Finer grids are defined locally in the vicinity of the free surface and around the hulls to resolve the wave patterns and the incident wave fields. The numerical grid is generated in accordance with the results gained from grid independence studies, see section 3.2. The grid is retained all over the whole multibody simulations with only one exception, that the wave refinement zone is refined as a function of wave height and length. I tried to resolve the wave amplitude and the wave length with about 8 to 10 and 25 to 35 cells, respectively. Depending on the wave parameters the total amount of grid control volumes is between 1.4×10^6 and 1.8×10^6 . It should be noted that the simulation setups are relied on the setups of single body simulations, and will act as basis for articulated multibody simulations. Therefore, the refinement zones are so arranged to resolve the body hydrodynamic responses efficiently, and less attempt is done to capture the bodies' Kelvin wake optimally. Nevertheless, agreeable wave patterns are registered. Figures 3.33 to 3.35 show the wave patterns of the the convoy at 0.0, 6.0 and 8.0 knots in the waves of 77.0 m of length and 7.5 s of period (wave frequency of 1.04 rad/s). The incident and hull-induced waves start to damp at about one convoy length behind the tug's aft perpendicular by means of extruded control volumes. Additionally, wave amplitudes are mathematically damped by resisting vertical wave velocity starting from two convoy length ahead of the domain outlet. The origin of the inertial reference frame is located at the aft perpendicular of the tug at mean water level (see figures 3.2 and 3.30). The origin of the body-fixed reference frames are positioned at their respective centers of gravity. Since the bodies can pitch out-of-phase relative to each other, Chimera grids are used for these simulations. However, refinement of the background and dedicated body grids in the small gap between both vessels has been the most challenging part in the mesh generation phase because the intersection handling of all three grids, i.e. two body and one background grids at the gap area has proven to be the chief source in simulation break-downs.

3.4.2 Results and Discussions

In figures 3.33 to 3.35, the times series of numerically generated waves for RANS simulations, and the related heave and pitch motion responses of bodies in head waves of 7.5 s of period, ≈ 77.0 m of length and 0.93 m of amplitude are shown for three different forward speeds. Respective model measurements are also drawn for

comparison. The amplitudes of numerically generated waves as well as the computed heave and pitch motion responses tend to underestimate the corresponding measurements. The tug is somewhat sheltered behind the barge in head waves; thus, comparing to stand-alone tug's motion responses (see figure 3.25), relatively smaller heave and pitch responses are observed.

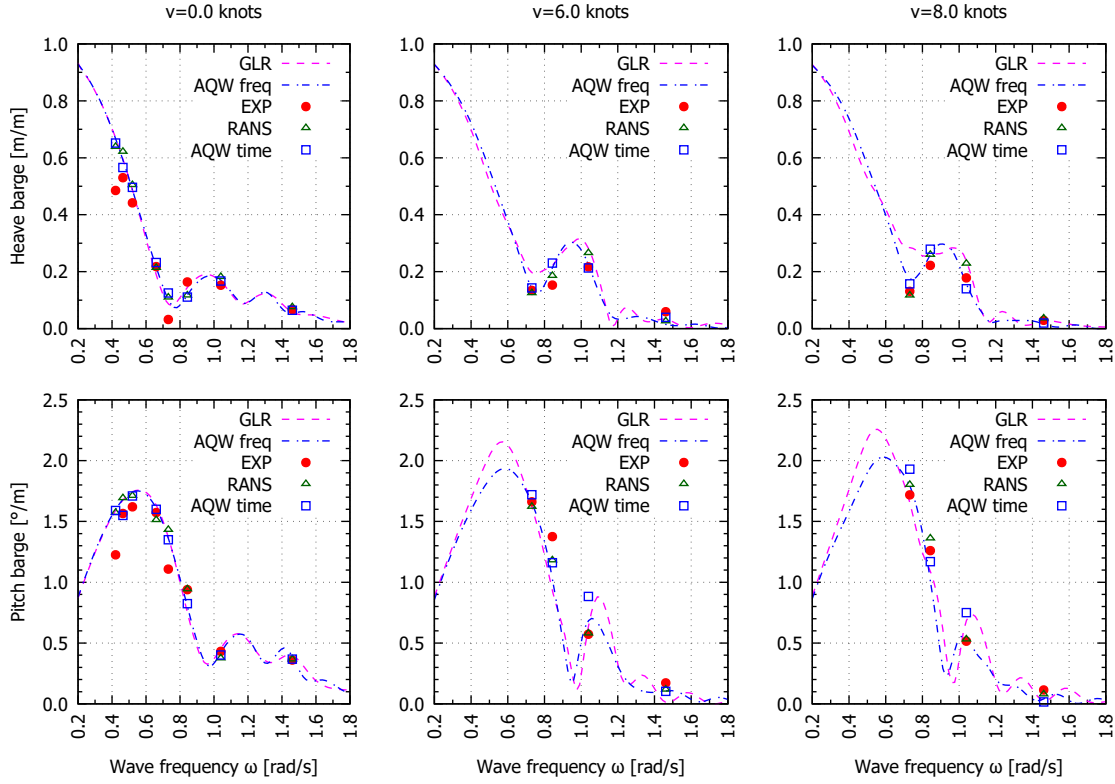


Figure 3.31: Barge's heave and pitch motion responses per wave amplitude for hydrodynamically coupled condition for head waves at $Fn = 0.0, 0.10$ and 0.14 body forward speeds.

Figures 3.31 and 3.32 show responses per wave amplitude of computed and measured pitch and heave motions. On these figures, GLR, RSM and RANS denote the results from GLR, RSM and RANS simulations, respectively. AQW freq and AQW time indicate the results from frequency and time domain AQW simulations, and measurements are denoted EXP. With reference to figure 3.31, barge motions predicted by all numerical methods agree well with each other and with model test results at all forward speeds over the entire frequency range. For example, at wave frequency 0.47 rad/s , computed heave motion response amplitudes of AQW time, AQW freq, GLR and RANS deviate respectively $\approx 2\%$, 4% , 4% , 8% from corresponding measurement.

3 Validation

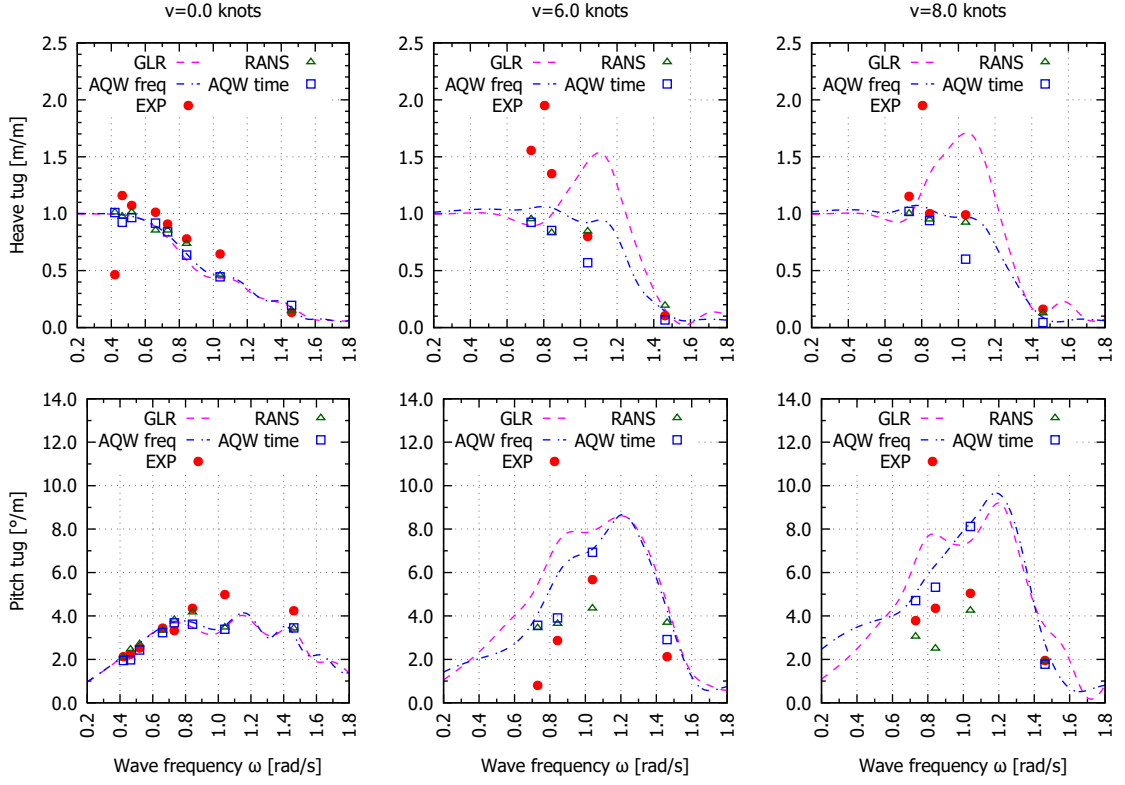


Figure 3.32: Tug’s heave and pitch motion responses per wave amplitude for hydrodynamically coupled condition for head waves at $Fn = 0.0, 0.20$ and 0.26 body forward speeds.

Opposite to the corroborating motion predictions of the barge, tug motions show some differences in some cases (see figure 3.32). Observing tug’s heave measurements for zero speed for $\omega < 0.66$ rad/s reveals that the heave response amplitude operators of measurements are all above 1.0 m/m. Correcting all these measured heave responses for zero forward speed with average offset value bring the measurements nearly at simulation result levels. Therefore, the slight difference between measurements and the tug’s predicted heave responses for zero forward speed might originate from inaccuracies in measurements.

Referring to figure 3.32, at non-zero forward speeds, GLR has overpredicted the tug’s heave motion at the wave frequencies around 1.0 rad/s, which is ascribed to inaccuracy of its underlying wave encounter frequency approach in correcting the responses for moderate and high forward speeds. This overprediction does not occur in AQW freq results due to the application of a more accurate forward speed correction method. RANS, AQW time and AQW freq have predicted the tug’s

heave responses at non-zero forward speeds fairly well. For example, at the wave frequency of 0.84 rad/s, the computed heave response amplitudes of the tug at 8.0 knots forward speed, figure 3.32 top, deviate about 1% from the corresponding measurement. The overpredicted measurements of tug's heave responses at 6.0 knots forward speed at wave frequencies 0.73 and 0.84 rad/s are ascribed to defect sensors. One effect of hydrodynamic coupling becomes evident by investigating the hump of tug's heave responses for forward speeds in figure 3.32. Such a hump, that appeared in stand-alone tug heave responses (cf. figure 3.23) does not occur in hydrodynamically coupled responses. Instead, much smaller humps but shifted towards lower frequencies are emerged at both 6.0 and 8.0 knots speeds.

The tug pitch motions at non-zero forward speeds computed by boundary element methods clearly overestimate the measurements, see figure 3.32 bottom. On the contrary to boundary element responses, the RANS predictions agree better with the measurements but underestimate them. For example, at the wave frequency 0.73 rad/s at 8.0 knots forward speed, 12% and 9% over and underprediction are computed by AQW time and RANS, respectively. Again, tug pitch responses at 6.0 knots forward speed at wave frequencies 0.73 and 0.84 rad/s are attributed to defect test sensors.

Summary

Both heave and pitch responses of the barge in hydrodynamically coupled condition were predicted very well. Good estimations of motion responses were also registered for the tug at zero forward speed. The computed tug's heave responses for non-zero forward speeds were also generally in a good agreement with the respective measurements. On the other hand, the tug's pitch responses for forward speeds by boundary element methods were overpredicted. It might be attributed to shortcomings of zero-speed Green function methods in efficient capturing of interaction scenarios. Tug's pitch responses for forward speeds by RANS, in comparison with those from Green function methods were more satisfactory, but underestimated the measurements. As a whole, heave and pitch motion responses of bodies per wave amplitude were predicted well by numerical methods. Not only the inaccuracies in measurements, but also shortcomings in numerical methods contributed to the differences between measurements and simulation results. During model tests as well as after assessing the simulations, it became evident that the tug is very susceptible to external forces.

3 Validation

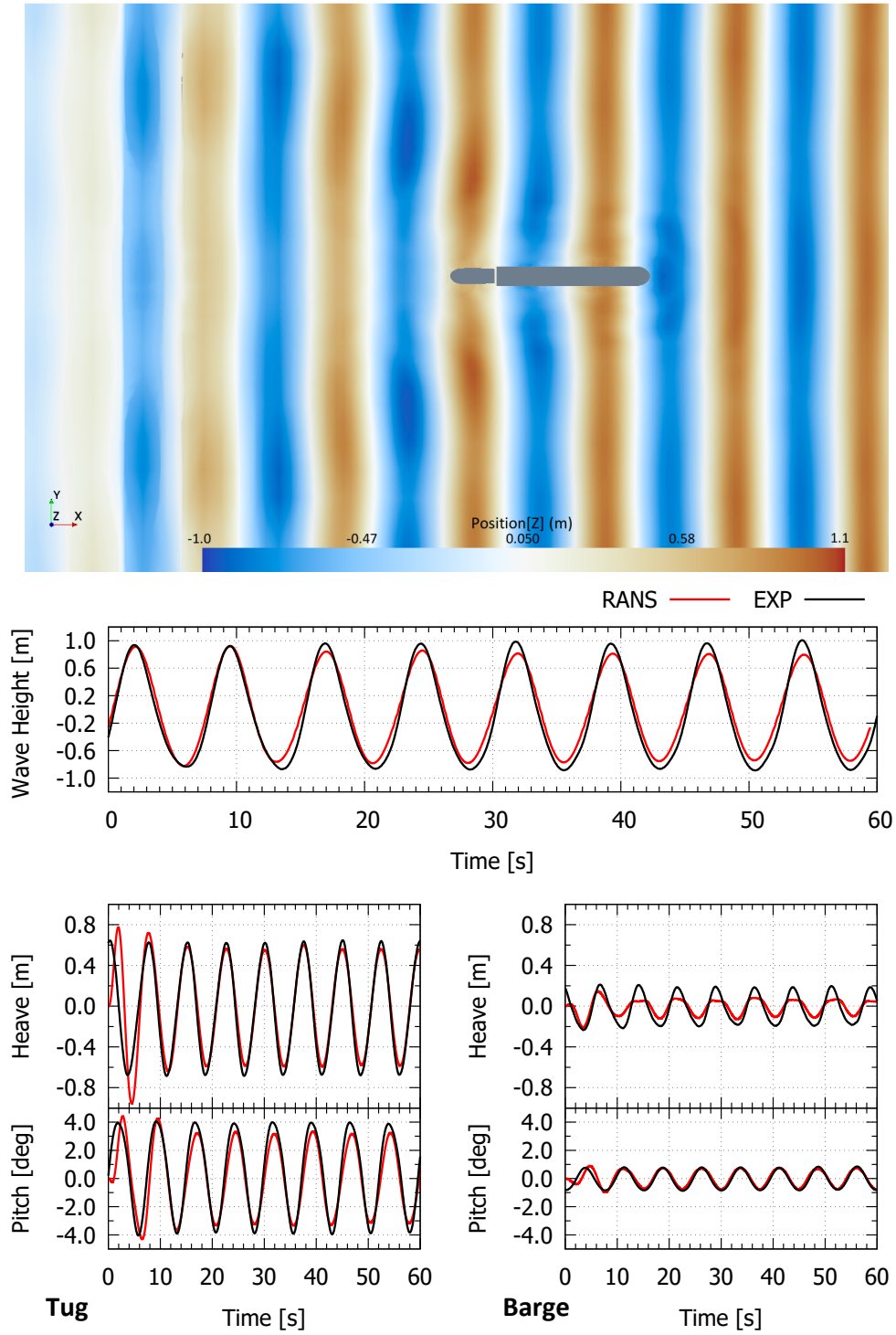


Figure 3.33: Wave patterns of convoy bodies for zero forward speed in head waves of ≈ 77.0 m of length, 7.5 s of period and 0.93 m of amplitude. The legend $Position[Z](m)$ stands for wave elevation. Wave height as well as bodies' heave and pitch time series are also drawn.

3.4 Hydrodynamically Coupled Multibody Simulations

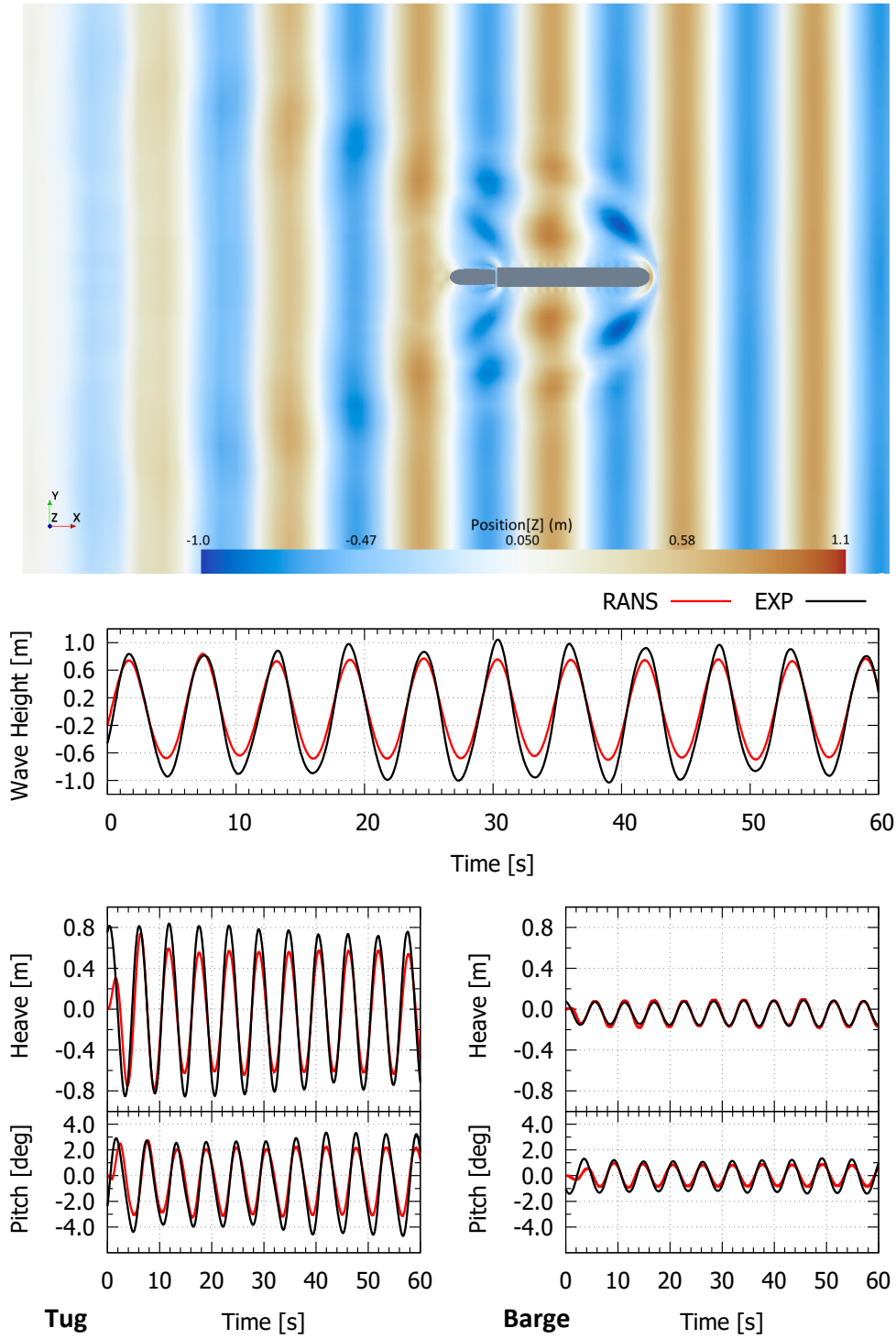


Figure 3.34: Wave patterns of convoy bodies for 6.0 knots forward speed in head waves of ≈ 77.0 m of length, 7.5 s of period and 0.74 m of amplitude. The legend *Position[Z](m)* stands for wave elevation. Wave height as well as bodies' heave and pitch time series are also drawn.

3 Validation

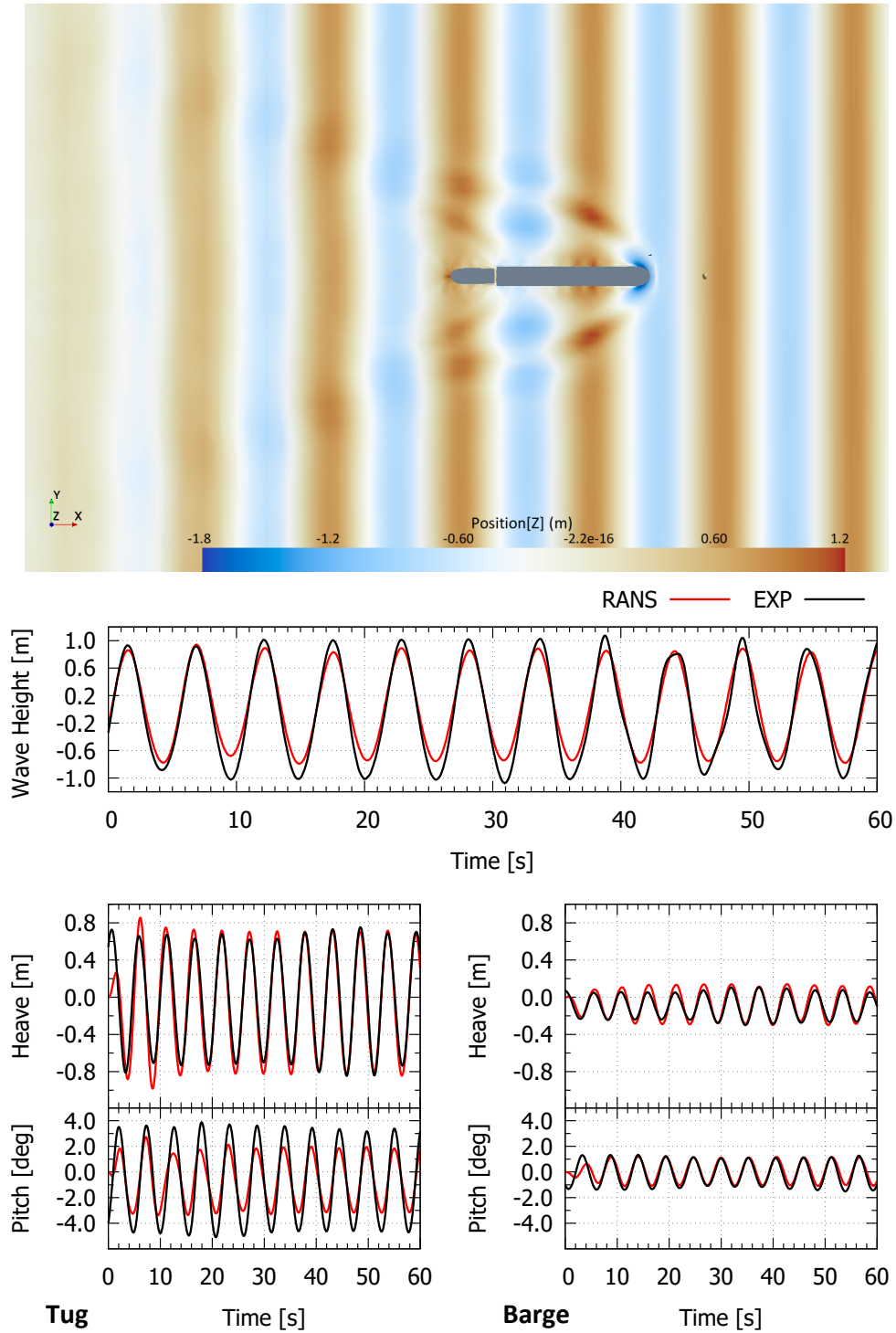


Figure 3.35: Wave patterns of convoy bodies for 8.0 knots forward speed for in head waves of ≈ 77.0 m of length, 7.5 s of period and 0.85 m of amplitude. The legend *Position[Z](m)* stands for wave elevation. Wave height as well as bodies' heave and pitch time series are also drawn.

3.5 Articulated Multibody Simulations

In this section, the simulations on the articulated convoy bodies are presented. The emphasis is put on application of different numerical approaches, not only from the aspect of articulation treatment, but also from the aspect of influence of different hydrodynamic solvers on articulation forces. Especially, for the latter, the analyses of hydrodynamics of bodies in their stand-alone and hydrodynamically coupled conditions presented in the preceding sections provide a solid backbone to get a better insight in the effects of different numerical hydrodynamic approaches on the responses of articulated ships.

As known from section 3.4, several different seakeeping solvers are applied: two linear seakeeping codes based on zero-speed Green function boundary element fluid solvers in frequency domain (AQW, GLR), a seakeeping code in time domain as per Cummins' approach with a linearized equation of motion, and a RANS solver, which is coupled with a nonlinear equation of motion solver. In this section, GLR, time domain AQW and RANS are used, whereas frequency domain AQW simulations are merely used to submit hydrodynamic coefficients to time domain AQW simulations.

Two different articulation types are studied in this section, i.e. hinge and rigid mechanical couplings. The hinge joint is the joint used for validation purposes to prove the suitability of contact elements in representation of articulations. However, the measurement data related to rigid joints are available, which are used in this section to support the validation process.

The hinge and rigid articulations are incorporated in the boundary element method solvers, whereas only hinge joints are modeled in RANS simulations. Articulations are integrated in these solvers by means of kinematic constraints. Additionally, hinge joints are represented with contact elements in the time domain boundary element method solver.

Both tug and barge known from simulations in the preceding sections are taken for the forthcoming simulations. The simulations were performed with the bodies being connected to each other at one articulation point located at their centerlines, or with two articulation points at port and starboard sides. The approaches are verified with code-to-code comparisons, and model test measurements. The connections between two bodies are defined as constraints and integrated as additional terms in the equation of motion. The articulation (reaction) forces are assessed by computing sectional loads at the articulation section, as well as by direct evaluation facilitated via constraint formulation. Additionally, contact elements were used to simulate hinges. In this case, the forces on each contact element contribute to the total forces on hinges.

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The measurements used for validation purposes are related to the model tests described in section 3.1. The bodies are positioned fore and aft of each other in a pushing convoy arrangement, and interconnected with each other either hingedly (figure 3.6) or rigidly (figure 3.7). Two model test runs were conducted with the hinged arrangement, one in different wave attack angles at zero forward speed, and one in head waves at various forward speeds (see table 3.3).

In the following, at first the configurations set in different methods are presented, and subsequently the results are shown and discussed.

3.5.1 Simulation Setup of Frequency Domain Boundary Element Method Solver

The seakeeping code specified as GLR within the course of this work is the code used to simulate the hingedly and rigidly interconnected vessels in frequency domain. The seakeeping solver procedure is already described in the preceding sections. The hinge and rigid connections between both vessels are realized by augmenting the motion equation with kinematic constraint equations and application of Lagrange multipliers. Equations (2.86) and (2.87) are used to simulate hinge and rigid joints in GLR. Two different approaches are chosen to validate correct definition of kinematic constraints, i.e. validation of reaction forces and motion responses.

For validation of the reaction forces, the constraint formulation elaborated in section 2.3.1.2 are validated by referring to D'Alembert-Lagrange principle, that the reaction forces are in the same direction as the direction of the constrained degrees of freedom. As a consequence, the total articulation reaction forces calculated from equation (2.70) can theoretically be equal to the forces provided by a sectional cut at the articulation section. The ability of GLR in computing the sectional loads was used for validation purposes. Reference is given to [129] for theoretical formulation of sectional loads in this package.

Sectional loads are applied to validate the correct formulations and implementation of the defined constraints, and to prove if the constraint reaction forces are computed correctly. Additionally, three pairs of distinct evaluation points for both bodies at articulation positions (port and starboard), as well as at the common centerline of both vessels are defined. The motion of these distinct points in each pair must be maintained equal if the constraints are defined correctly. Figure 3.36 shows a comparison of longitudinal and vertical articulation forces, when corresponding constraints are imposed on one or on two connection positions to articulate both bodies hingedly. The forces are determined by computing the loads at articulation section, as well as via Lagrange multipliers (see equation (2.70)).

3.5 Articulated Multibody Simulations

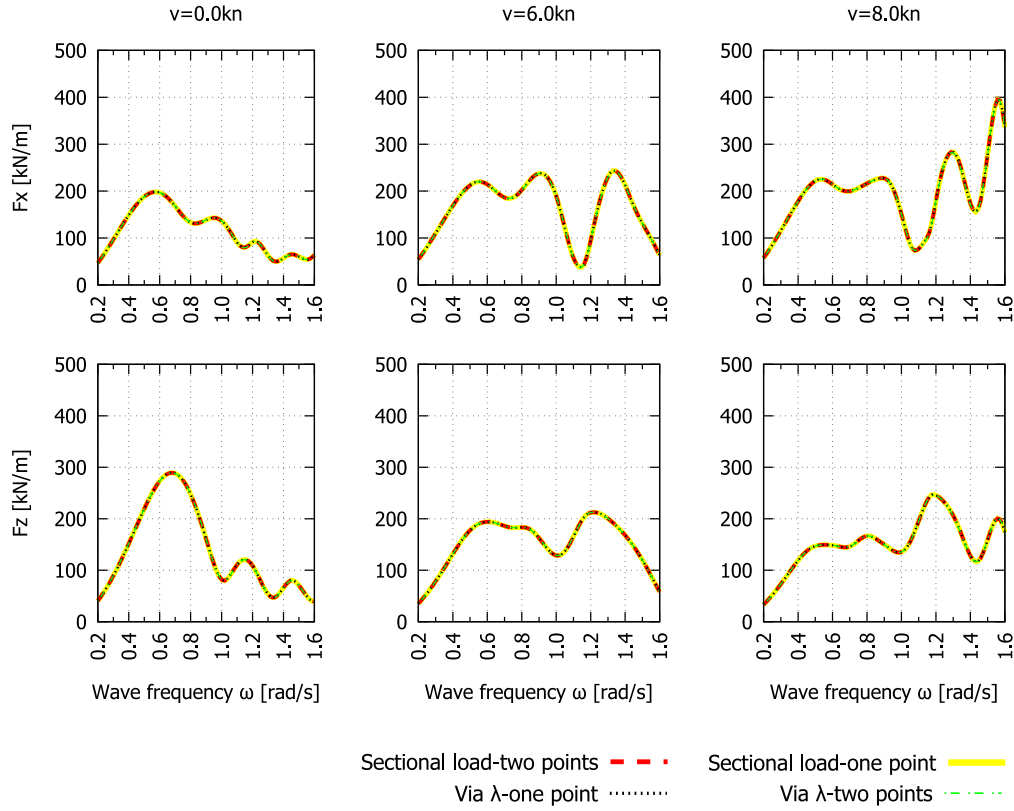


Figure 3.36: GLR computed longitudinal (top) and vertical (bottom) joint forces using sectional loads at articulation section and Lagrange multipliers while the bodies are hinged with each other by means of imposing constraints at two or one articulation point(s). Bodies encounter head waves with 0.0, 6.0 and 8.0 knots forward speeds.

These results justify the equivalence of these approaches. Certainly, the motions of respective bodies computed from these different approaches overlap each other, in analogy to the shown forces.

The comparison of absolute motions at the pairs of evaluation points with each other, as well as the sectional loads with constraint reaction forces have confirmed the successful definition and implementation of the constraints in both hinged and rigid articulation conditions. It can further be justified that the number of articulation in a simulation may be reduced as long as the imposed constraints in the reduced arrangement lead to the same coupling situation as the imposed constraints in the original arrangement. In other words, hinged and rigid articulation of both bodies with the help of two points with imposed $g_1(\vec{r})$ to $g_6(\vec{r})$ constraints (equations (2.86) and (2.87)) can be represented by constraining both bodies at only one point, but with different constraint conditions. For instance,

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the same hinged articulation achieved by $g_1(\vec{r})$ to $g_5(\vec{r})$ (equation (2.86)) with reference to figure 2.3 can be achieved if with reference to figure 2.2, $g_1(\vec{r})$, $g_3(\vec{r})$, $g_5(\vec{r})$ constraints are applied on articulation points at the centerline, and in addition, roll and yaw rotation constraints, $g_7(\vec{\alpha}) = \phi_1 - \phi_2 = 0$, and $g_8(\vec{\alpha}) = \psi_1 - \psi_2 = 0$, are imposed at respective body-fixed reference frame origins. Likewise, augmenting these constraints with $g_6(\vec{\alpha})$ (equation (2.87)) make both vessels articulate rigidly.

In summary, in the frequency domain boundary element method hinge and rigid articulations via kinematic constraints as per equations (2.86) and (2.87) are defined (see figures 2.2 and 2.3). For this purpose, the ability of *GL Rankine* to impose holonomic constraints at position level on a specific point is used.

3.5.2 Simulation Setup of Time Domain Boundary Element Method Solver

The same seakeeping time domain boundary element method AQW known from preceding sections is used here. The articulations are realized in AQW by means of constraint articulations and contact elements. Articulation of the vessels via constraint formulations is done for both hinge and rigid connections, whereas contact element are employed merely for the hinge joint. The kinematic constraint approach follows equation (2.88) (see figure 2.4), and the contact element employment corresponds to the configuration in figure 2.5.

3.5.3 Simulation Setup of Reynolds-Averaged Navier-Stokes Equation Solver

In the RANS simulations, only hinge joints are incorporated. Comparing to hydrodynamically coupled multibody simulations, except the algorithm to compute the equation of motion, no other algorithms are essentially changed in articulated multibody simulations (cf. figure 3.20). The numerical instabilities caused by numerical discretization of the constraint equation are stabilized by fictive damping, α , and spring constants, β , (see equation (2.73)). Generally, there is no rules to determine fictive damping and spring constants. Since great α -values result in the loss of energy, small α -values together with great β -values are used within this work. Respective α and β -values of 5.0 and 20.0 have shown to be appropriate for numerically stable simulation of hinges in RANS simulations.

The same computation domain used in section 3.4 is employed for head waves (see figure 3.30). In one model test run series, the bodies were exposed to fol-

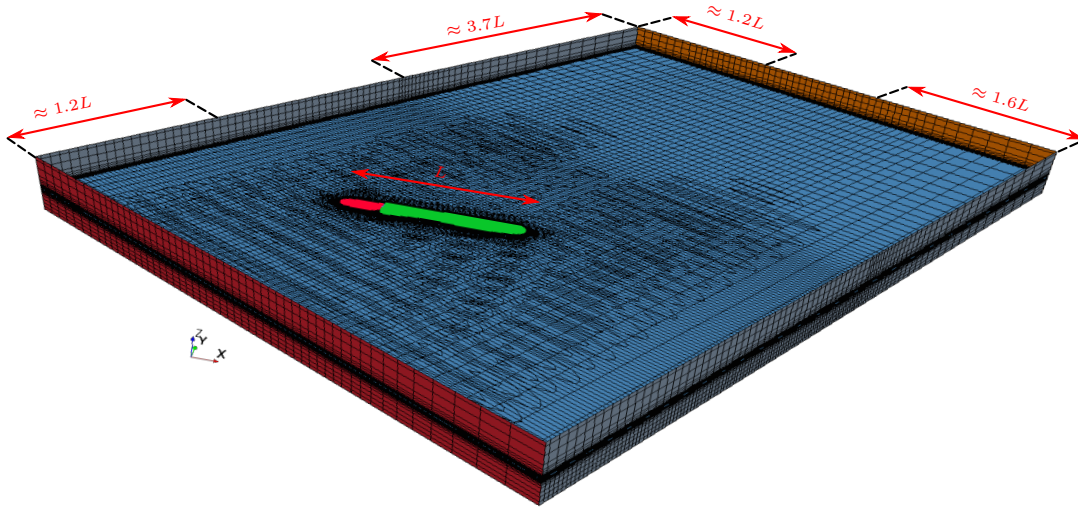


Figure 3.37: Computational domain used in stern-quartering waves ($\mu = 60^\circ$).

lowing, head, bow-quartering, and stern-quartering waves (see table 3.3). In case of following waves, minor modifications are done on the head wave computation domain and the bodies are rotated 180° . On the hand, fundamentally different computational domains are considered for stern- and bow-quartering wave simulations. Figure 3.37 shows the computational domain used for stern-quartering wave simulations.

Since both vessels are articulated at the barge transom, no high relative pitch motions at the articulation section are expected. Consequently, the mesh morphing method can be used efficiently to propagate the body motions throughout the domain. Therefore, the critical small gap between the vessels, which is a concern in the Chimera-method, will not be an issue anymore. Figure 3.39 shows a case, in which both bodies are pitching out-of-phase. This figure shows how well the consequent body motions are propagated via cell morphing.

3.5.4 Results and Discussions

Figures 3.41 to 3.43 illustrate the development of RANS simulations at elapsed time of 60 s for hingedly articulated convoy bodies at 0.0, 6.0, 8.0 knots forward speeds in head waves of different wave amplitudes, and of 77.0 m of length and of 7.5 s of period. This wave condition has intentionally been chosen because associated model test data for both hydrodynamically coupled and articulated

3 Validation

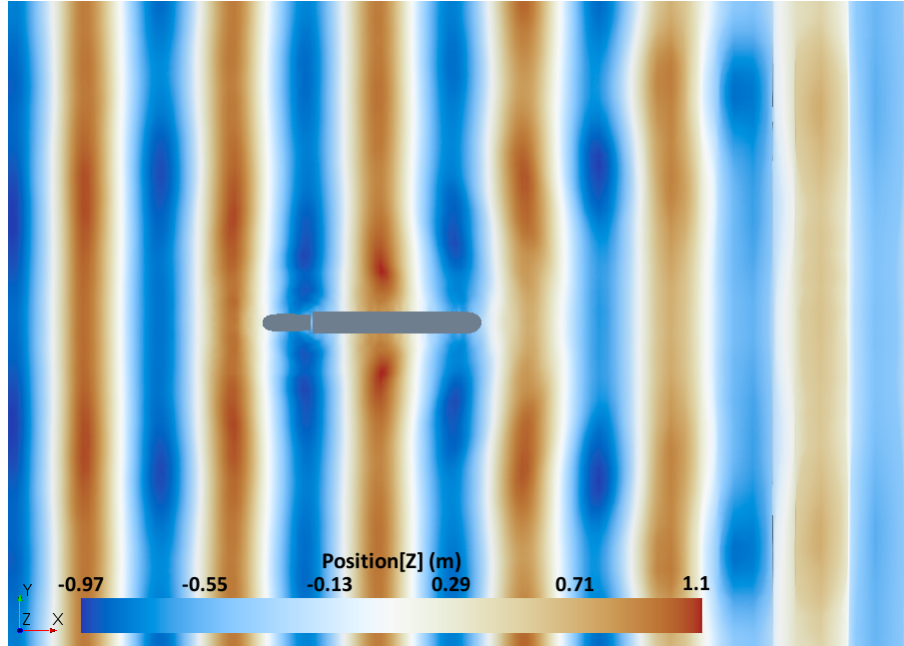


Figure 3.38: Convoy wave pattern for zero forward speed in following waves of ≈ 77 m of length, 7.5 s of period and 0.94 m of amplitude. The legend *position[Z](m)* stands for wave elevation.

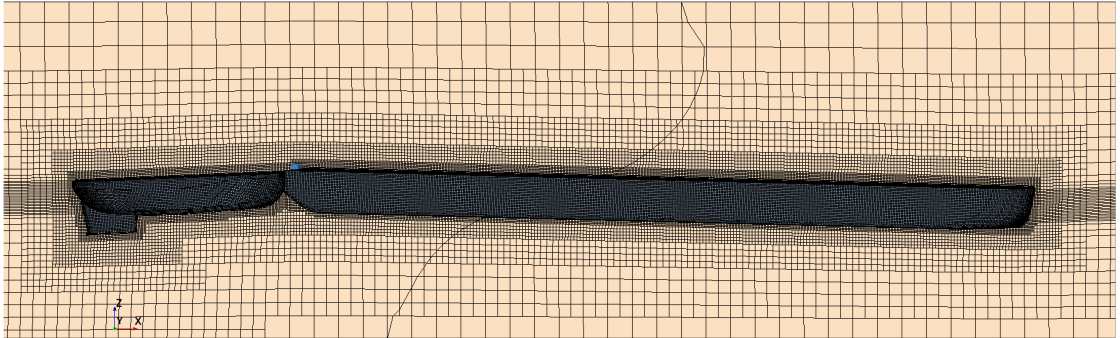


Figure 3.39: A longitudinal section of morphed mesh at $t = 60.0$ s. The hinge position is depicted at the barge transom.

cases are available (cf. figures 3.33 to 3.35 and 3.41 to 3.43).

Observing zero speed cases (figures 3.33 and 3.41) makes it evident that the simulation waves have tracked the model test waves very well. Heave and pitch motions of the tug are estimated well in both cases. Similarly, the pitch motions of the barge are predicted satisfactorily. Heave motions of the barge are slightly under and overpredicted in unarticulated and articulated cases, respectively. However, considering the small heave motion amplitudes, predicted simulation results are in good agreement with the measurements. Simulation and measured hinge forces are also in good agreement with each other. Similar favorable behavior is maintained for 6.0 and 8.0 knots forward speeds, and numerical and model tank waves correspond well with each other (figures 3.42 and 3.43). Tug's heave responses for 6.0 knots forward speed showed differences with the measurements. Therefore, new simulations were set up with a finer mesh with the refinement factor of $\sqrt{2.25}$ for all simulations related to 6.0 knots forward speed. The responses shown in figures 3.42 and 3.44 to 3.46 for 6.0 knots are related to these refined simulations. Indeed, no significant improvements of the responses could be registered.

As a whole, the time series of body motions, and hinge forces show agreeable results underlying RANS success in simulation of articulated multibody systems.

Before proceeding to investigate the response amplitude operators of body motions and forces, an outline of the order of the related figures is provided. As mentioned, hinge and rigid joints are studied in this section. Result of the hinge joint simulations are used for validation purpose of contact element approach, and the results from rigid joint simulations have a supporting function for the validation process. The results related to hinge joint simulations precede the result of rigid joint simulations. Two different model test run series have been conducted for hinge arrangement, one with 0.0, 6.0 and 8.0 knots forward speeds in regular head waves, and one series with zero speed in following, stern-quartering, bow-quartering and head waves. Simulation results and measurements related to the run with various speeds are presented first. Motion responses are always presented before articulation force results. Frequency domain GLR simulations apply the kinematic constraints as per figure 2.3 to articulate both bodies. Likewise, time domain AQW simulations utilize kinematic constraints with nonlinear contributions from Froude-Krylov and restoring forces. Time domain AQW solver is also taken to incorporate contact elements as per figure 2.5 to model hinge joints. RANS simulations use also kinematic constraints, and not only involve nonlinear contributions from the fluid solver but also solve the nonlinear equation of motion. In the following figures, EXP, GLR, RANSE denote measurements, computed results from GLR and RANS, respectively. The predicted responses from AQW while modeling the hinge joints with kinematic constraints and contact elements

are denotes kin. const. AQW and cont. ele. AQW, respectively.

Figures 3.44 and 3.45 show respectively the heave and pitch response amplitude operators (RAO) of the bodies in head waves at various speeds. The barge heave and pitch responses show, like the unarticulated counterparts, outstandingly good agreement of all simulations with measurements. In comparison to figure 3.31, the hinge joint has not affected the heave and pitch RAOs of the barge notably, except that the barge pitch responses are comparatively mitigated in articulated cases. Contrarily, motions of the tug are considerably affected by the link to the much larger barge. With reference to figures 3.32 and 3.45, the pitch behavior of the tug is so drastically affected by the articulation that almost no similarities between tug's pitch RAOs of unarticulated and articulated cases can be found. Tug's pitch RAOs resemble those of the barge's after articulation. It is still possible to recognize similarities in tug's heave RAOs for unarticulated and articulated cases. However, the local curve troughs in heave RAOs of non-zero forward speeds in unarticulated cases (figure 3.32) have become strikingly pronounced after articulation (figure 3.45). Additionally, articulation has attenuated the intensity of GLR's incorrect predictions of tug's heave responses at forward speeds.

With reference to figure 3.46, predictions by GLR and kin. const. AQW show small deviations from each other but tend to overpredict the measured longitudinal hinge forces, in particular at zero forward speed. The results from cont. ele. AQW and RANS simulations agree better with measurements comparing to GLR and kin. const. AQW results. For example, at the wave frequency 0.52 rad/s at zero forward speed, the amplitudes of longitudinal hinge force responses computed by GLR and kin. const. AQW deviate 8.7% and 8.3% from measurements, respectively; whereas, the corresponding response amplitudes computed by cont. ele. AQW and RANS deviate 0.9% and 1.7% from measurements, respectively. GLR frequency domain based results and the result time series from kin. const. AQW are harmonic. Therefore, the results can easily be read or transposed. The result time series from RANS and cont. ele. AQW, as well as model test measurements are all harmonic-alike, have noises, local maxima and minima, and may have their means not at zero. Fourier transformation technique is used to convert the time signals to their representation in frequency domain. Especially, the force response time series gained from contact element simulations are very unrefined. Only after evaluating the moving average or filtering the time series harmonic responses may be obtained. It is interesting that, for longitudinal force responses, these simulation results gained via Fourier transformation (cont. ele. AQW and RANS) are closer to Fourier transformed measurements. Experimentally, some cont. ele. AQW simulations were resumed up to 5000 s simulation time. It did not definitely made the results approach respective GLR or kin. const. AQW results. As a whole, it can be concluded, that kinematic constraint formulation associated with the frequency

and time domain BEM solvers (GLR and kin. const. AQW, respectively) tend to overpredict the longitudinal hinge forces.

At this stage, the results of simulations and measurements related to run series with hinge joints, and with various wave attack angles at zero forward speed will be investigated. Figures 3.47 and 3.48 present barge heave and pitch motion response amplitude operators. Like the barge heave and pitch results in the previous test run figure (figure 3.44), the measurements and simulation results show small deviations from each other. The tug's heave and pitch computed responses also show good agreements (figures 3.49 and 3.50). Reviewing simulation results of stand-alone, hydrodynamically coupled, and articulated tug, it is believed that the offset is the result of some mal-calibration of model test measuring devices; hence, the offsets are ignored. Figure 3.51 shows longitudinal hinge force responses per wave amplitude. Similar results to those from previous test run (figure 3.46) are obtained, i.e. for head waves, frequency domain and time domain simulations with kinematically constrained hinge provide relatively overpredicted results throughout the most range of studied wave frequencies. The forces gained for constraint hinge from RANS simulations, and for the contact element hinge from time domain boundary element method are closer to measurements. Such computation and measurement relationship is also seen for other wave attack angles. Figure 3.52 shows the hinge transverse forces per wave amplitude. All numerical methods have underpredicted measured transverse forces, while predicting the same overall trend. It is unclear why the measured lateral forces do not converge to zero at decreasing wave frequency; the offset of the measured lateral forces may be caused by the measurement setup.

Figure 3.53 presents the hinge vertical forces per wave amplitude. The results provide qualitatively similar relationship with those from previous tests figure 3.46, i.e. vertical forces of the hinges modeled by kinematic constraints in RANS or boundary element methods show small deviation from each other and are slightly overpredicted. The hinge represented by contact elements show better agreement for vertical forces with measurements. Such relationships are seen for all studied wave attack angles. Again, differences between simulation results and measurements are experienced in these test runs, which are ascribed to shortcomings or inaccuracy of numerical simulations, as well as inaccuracies in model test measurements.

The effect of nonlinear Froude-Krylov forces on the predicted hinge forces are examined by doubling the wave amplitude in AQW. Figure 3.40 shows the vertical hinge force per wave amplitude in head waves from time domain simulations in comparison with measurements, and linear results from GLR frequency domain computations. The hinge is modeled by means of kinematic constraints and contact

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elements in AQW, denoted **kin. const.** and **cont. ele.**, respectively in figure 3.40. The hinge in GLR simulations is modeled by means of kinematic constraints. Results from simulations with amplified wave amplitudes are denoted **Amplified amp.** in figure 3.40. The comparison shows that nonlinear forces have small effect on the hinge vertical forces. Generally, nonlinear Froude-Krylov forces slightly reduce the response amplitudes. This effect is also reported in [55], in which the sectional loads of stand-alone barge is investigated.

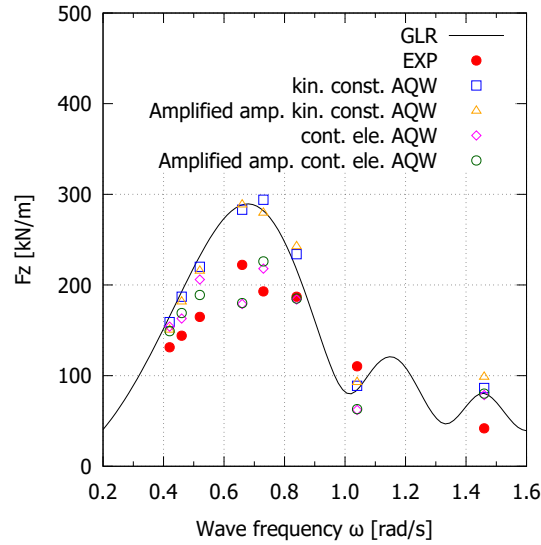


Figure 3.40: Effect of nonlinear Froude-Krylov forces on hinge vertical forces per wave amplitude in head waves at zero forward speed. **kin. const.** and **cont. ele.** denote kinematic constraint and contact element hinges, respectively, and **Amplified amp.** indicates simulations with amplified wave amplitudes.

After studying the hinge joint cases, the cases related to the test run with the rigid joint are studied. The cases of this test run are of auxiliary rank in this work because neither contact elements are involved nor RANS simulations are performed for rigid joints. Nevertheless, boundary element simulations with kinematic constraints as per equations (2.86) and (2.87) are setup in frequency domain, and according to equation (2.88) in time domain. These simulations give more insight on the quality of kinematic constraint BEM results and model test measurements. Figures 3.54 and 3.55 shows the corresponding motion and force responses per wave amplitude. In fact, a rigid connection can, if the circumstances of the case permit, be modeled by incorporating both connected bodies as one single body. Then, the result forces and moments at the connection can be calculated by computing the loads at the articulation section. D'Alembert-Lagrange principle justifies this procedure and its evaluation was concisely demonstrated in figure 3.36. The mo-

tion responses of the bodies agree well with measured responses. The measured longitudinal and vertical force responses are slightly inconsistent. For instance, at $\omega=0.84$ rad/s, two test runs are done but two divergent measurements are delivered. Investigating the other measurements underline the inconsistency of some measurements. It emphasizes that not only numerical simulation of articulated multibodies but also their tank towing modeling are challenging. Generally, like simulations with hinge joints, boundary element methods tends to overpredict joint forces. Overprediction of used boundary element methods for rigid articulations is also confirmed in [55], in which loads at different sections along the stand-alone barge are calculated by different linear, weakly nonlinear, and nonlinear sectional load computation methods. In the study reported in [55], nonlinear force responses are the most agreeable in comparison with measurements. This is also in compliance with nonlinear effects shown in figure 3.40.

For both rigid and hinge couplings, the longitudinal and transverse relative motions between bodies are constrained, therefore, the curves of longitudinal wave-induced forces look similar (figures 3.46, 3.51, 3.53 and 3.55). On the other hand, the curves of vertical forces for hinge and rigid joints look significantly different. The hinge is free to rotate around the transverse axis, which relieves the hinge from taking the load caused by the vertical bending moment. Therefore, the vertical coupling forces on the rigid connection are up to four times larger than the hinge vertical forces.

3.5.5 Concluding Remarks

Hinge and rigid joints were simulated in this section to articulate the previously introduced tug and barge in a fore and aft convoy-like arrangement. The rigid joint was of secondary rank, for which kinematic constraints were defined in boundary element methods of frequency and time domain (in GLR and AQW). Similar to rigid joints, GLR and AQW were taken for application of kinematically constrained hinges. Contact element were also used to represent hinge connection in AQW. RANS simulations applied constraints to articulate both bodies hingedly. The results affirmed the challenging nature of modeling and simulating articulated multibodies because some inconsistencies in some simulation results and measurements are evident. As a whole, satisfactory results were achieved for motion responses for different wave attack angles at zero forward speed. No simulation solvers proved any superiority to the others. Generally, the resultant longitudinal hinge forces were consenting. The kinematically constrained hinges in association with AQW and GLR submitted slightly overpredicted longitudinal joint forces, whereas constraint related hinge pertained to RANS solver, as well as the hinge represented

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by contact elements in conjunction with AQW showed closer longitudinal force responses to measurements. Similarly, good agreement between simulation results and measurements for vertical hinge forces at zero forward speed was achieved. Prediction from the methods based on kinematic constraints (GLR, AQW, RANS) showed small deviations from each other, but tended to slightly overpredict the vertical hinge forces. On the contrary, simulations based on contact elements predicted the results more favorably. As a whole, computed force and motion responses are deemed satisfactory at zero speed forward and for various wave attack angles.

In case of forward speed, the predicted barge motion responses showed good agreement with the measurements. Similarly, the tug's motion and hinge force responses at forward speeds agreed fairly well with each other, and with measurements. However, it is recommended for future studies to couple preferably two bodies of the same principals. In such as way, articulation interaction of the bodies are bilateral and not like here so one-sided, that one body is violently reactive to the actions of the other body.

The rigid connections were simulated with kinematic constraints in conjunction with GLR and AQW in frequency and time domains. The computed motion responses of rigidly articulated bodies corresponded generally well with the measurements. With the help of supporting data from [55], it was concluded that boundary element methods generally tended to overpredict the measured force responses.

Intentionally, I keep distance from stating any statistics on computation time and resource usage. Software data structure, CPU overhead, multi-threading, usage of different memory parts, optimum consideration of hardware structure, time required for pre- and post-processing, and reusability of a configuration are among some parameters why mere consideration of simulation time or required resources cannot be representative on efficiency of a method. Altogether, frequency dependent GLR was very handy in both pre- and post-processing phases, and computations were finished within minutes. AQW with kinematic constraints could be placed in the second rank from computation time point of view, followed by AQW simulations with contact elements. Requirement of AQW simulations with contact elements to have relatively small time steps (here $\Delta t=0.001$ s for AQW contact element simulations vs. $\Delta t=0.01$ s for AQW kinematic constraint simulations) caused these simulations to be computationally more expensive. Expectedly, RANS simulations were the most time consuming method used here.

3.5 Articulated Multibody Simulations

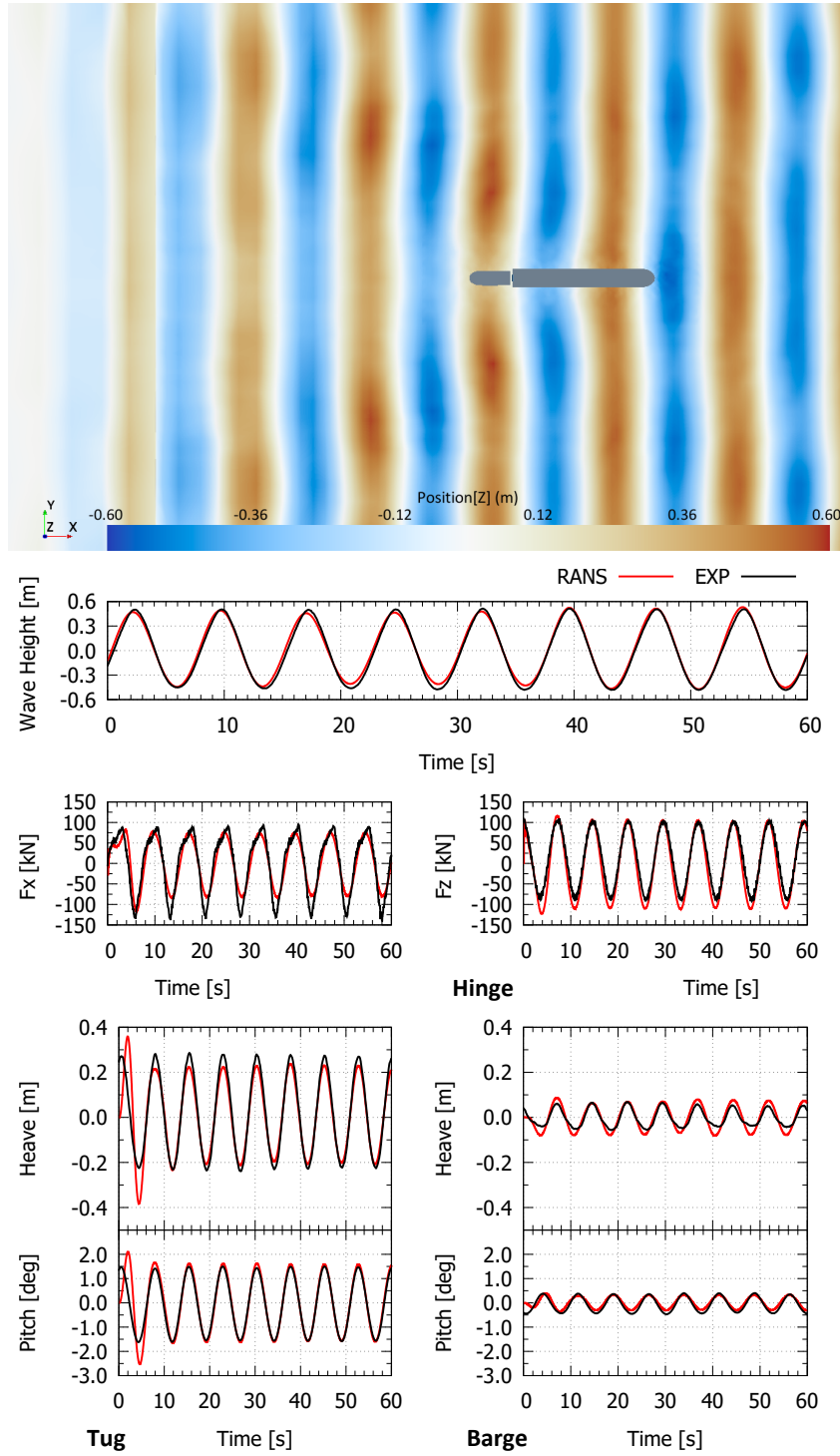


Figure 3.41: Wave patterns of convoy bodies for zero forward speed in head waves of ≈ 77.0 m of length, 7.5 s of period and 0.47 m of amplitude with hinge joint. The legend $Position[Z](m)$ stands for wave elevation. Time series of wave height, bodies' motion responses and induced forces at the joint are also drawn.

3 Validation

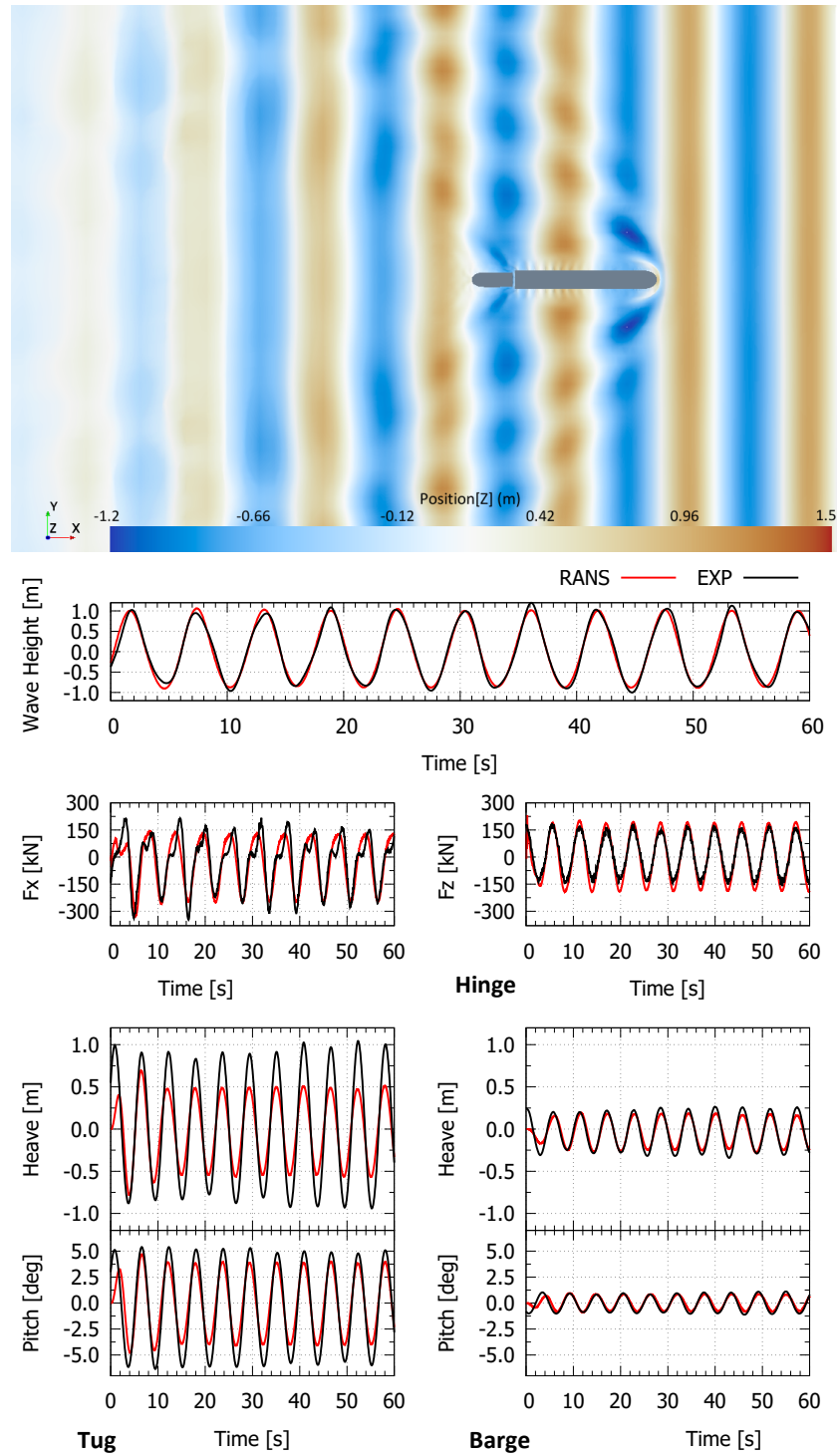


Figure 3.42: Wave patterns of convoy bodies for 6.0 knots forward speed in head waves of ≈ 77.0 m of length, 7.5 s of period and 0.99 m of amplitude with hinge joint. The legend $Position[Z](m)$ stands for wave elevation. Time series of wave height, bodies' motion responses and induced forces at the joint are also drawn.

3.5 Articulated Multibody Simulations

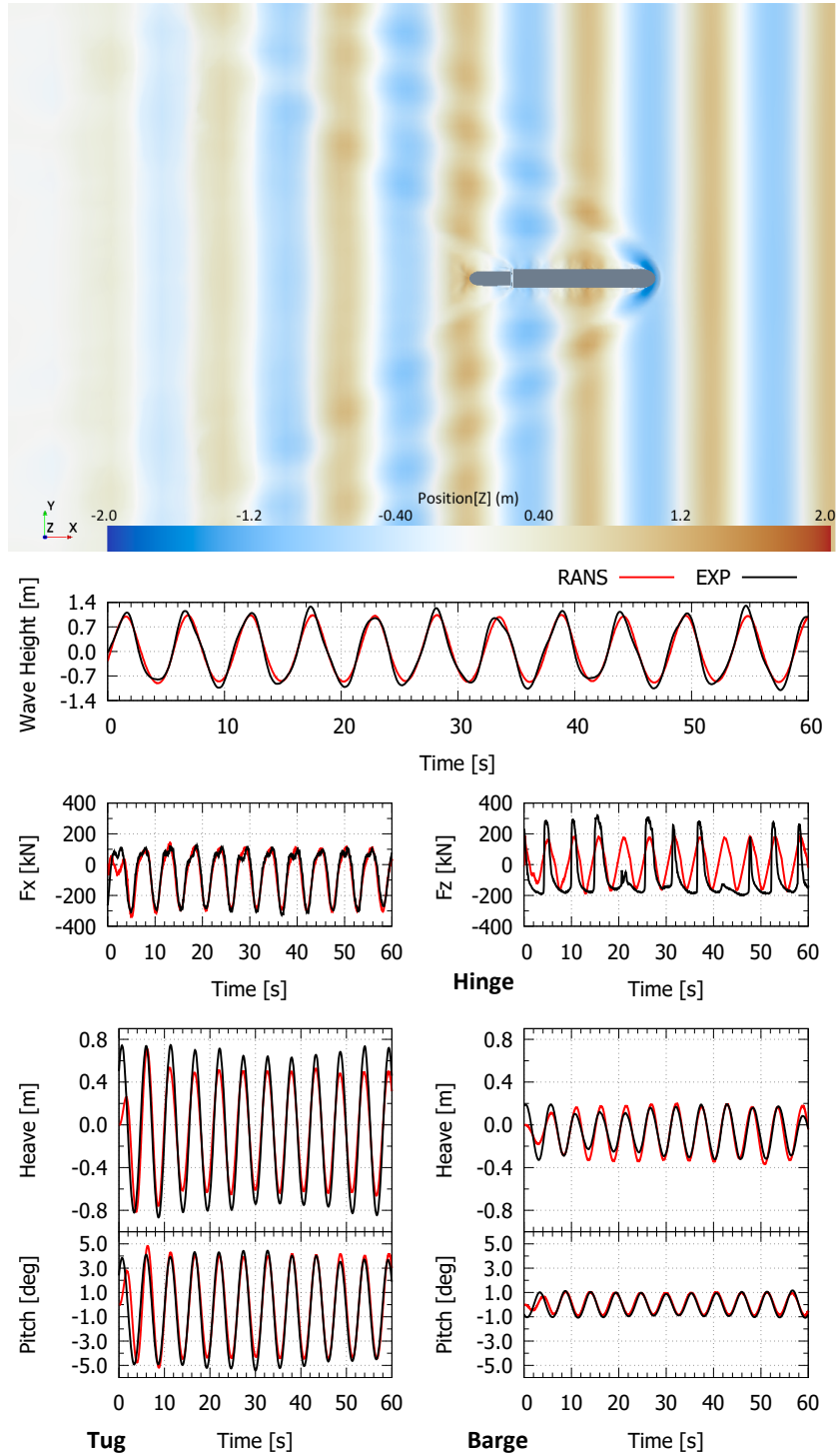


Figure 3.43: Wave patterns of convoy bodies for 8.0 knots forward speed in head waves of ≈ 77.0 m of length, 7.5 s of period and 0.98 m of amplitude with hinge joint. The legend $Position[Z](m)$ stands for wave elevation. Time series of wave height, bodies' motion responses and induced forces at the joint are also drawn.

3 Validation

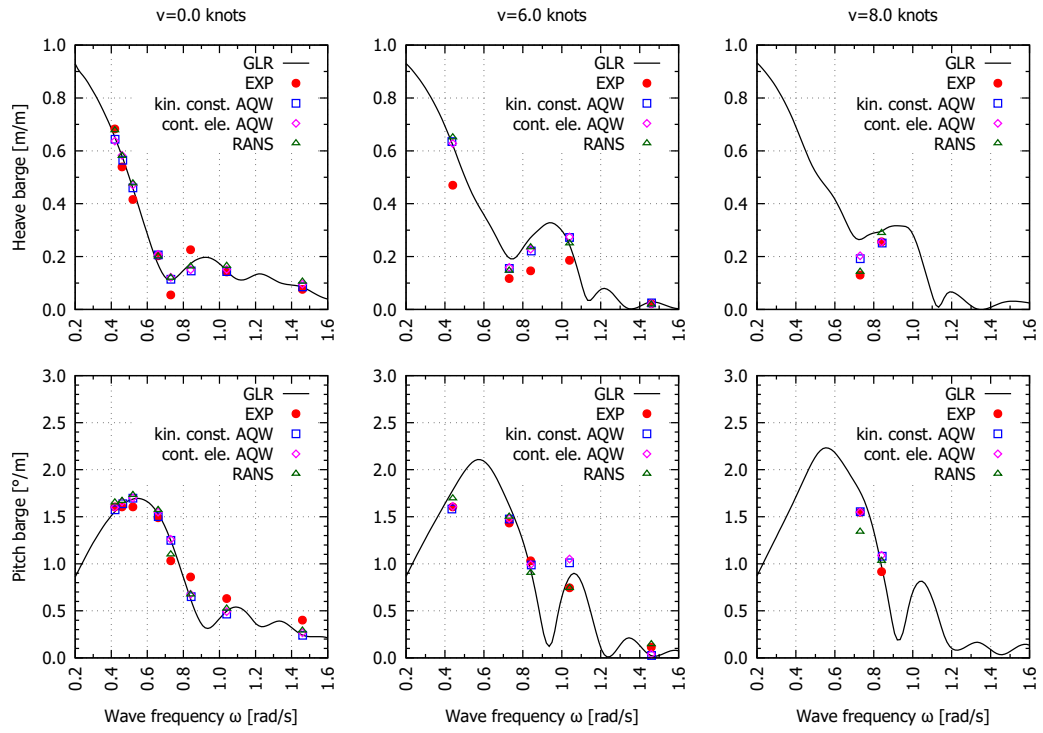


Figure 3.44: Motion responses per wave amplitude of hingedly articulated **barge** in head waves at $Fn = 0.0, 0.10$ and 0.14 forward speeds.

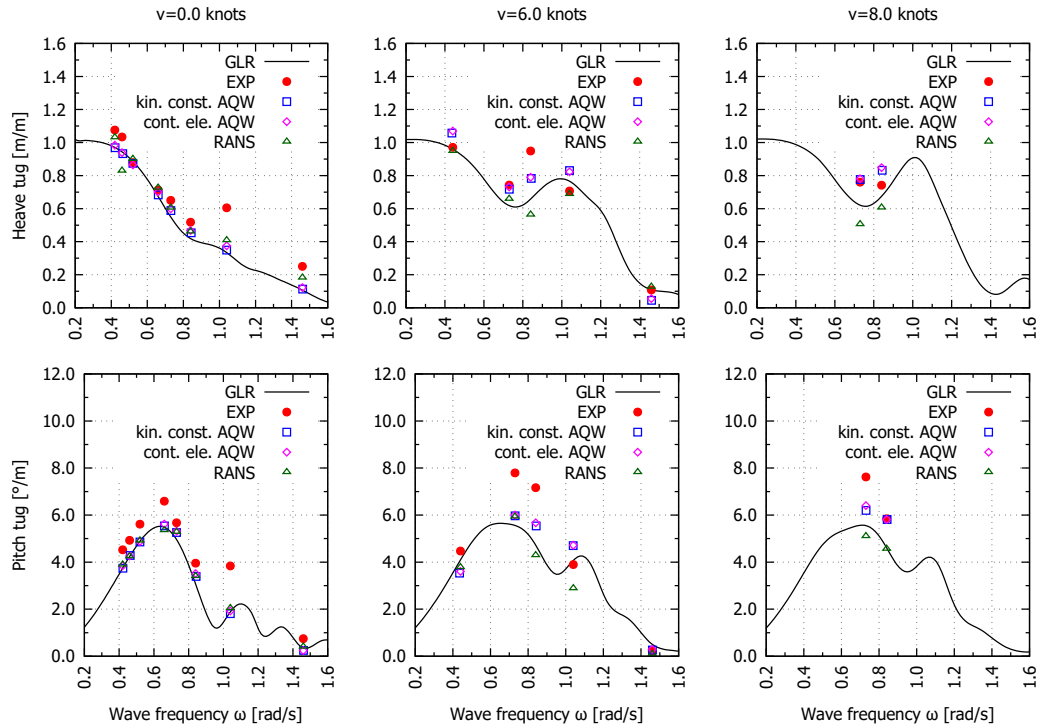


Figure 3.45: Motion responses per wave amplitude of hingedly articulated **tug** in head waves at $Fn = 0.0, 0.20$ and 0.26 forward speeds.

3.5 Articulated Multibody Simulations

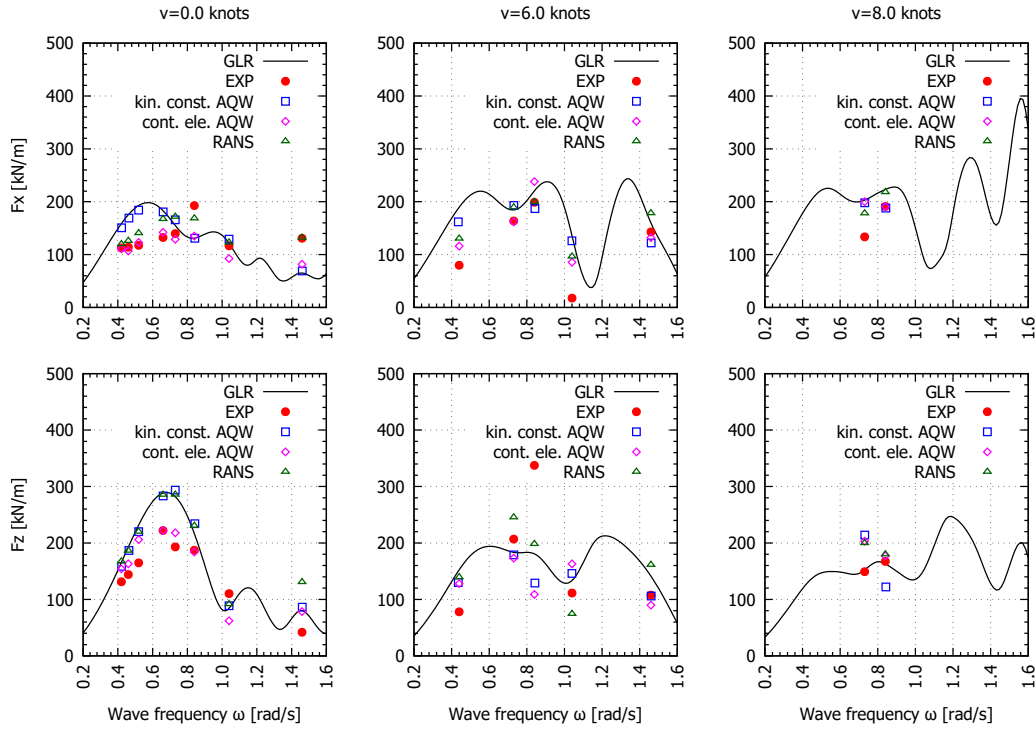


Figure 3.46: Hinge force responses per wave amplitude in head waves at various forward speeds.

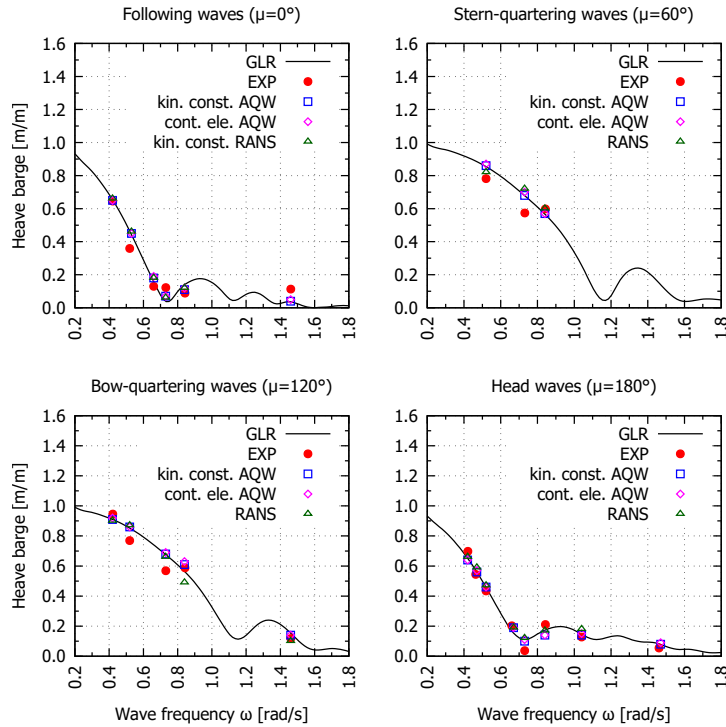


Figure 3.47: Heave responses per wave amplitude of hingedly articulated barge at zero forward speed in waves from various directions.

3 Validation

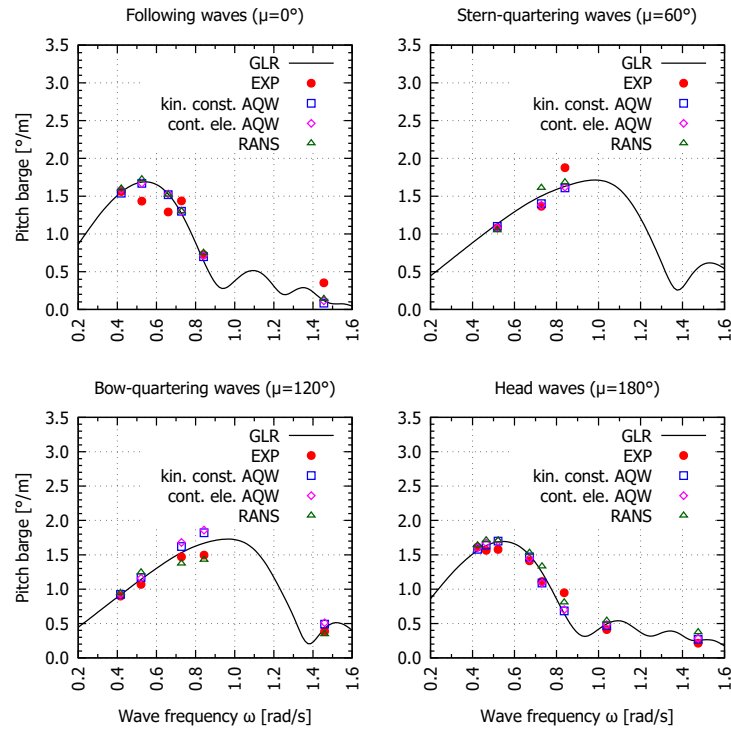


Figure 3.48: Pitch responses per wave amplitude of hingedly articulated **barge** at zero forward speed in waves from various directions.

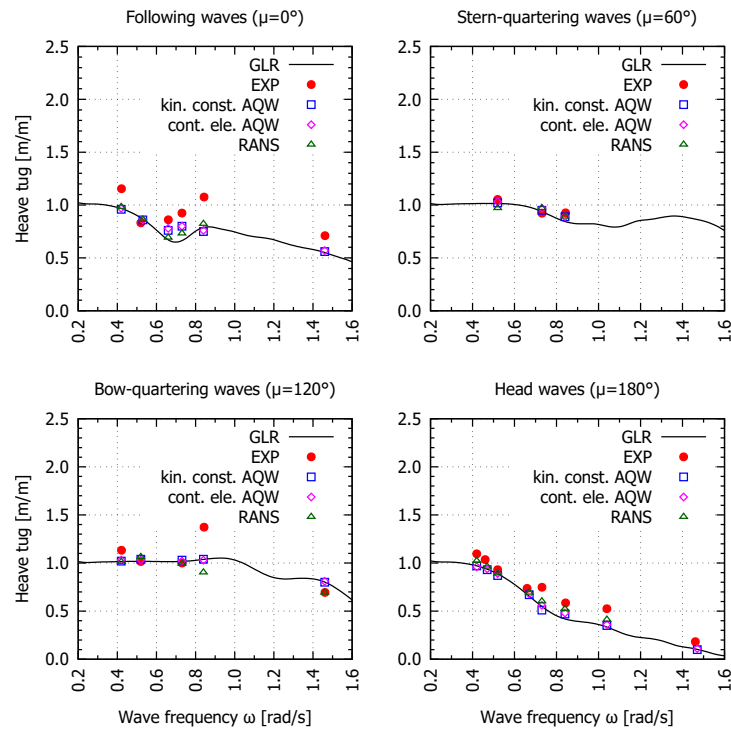


Figure 3.49: Heave responses per wave amplitude of hingedly articulated **tug** at zero forward speed in waves from various directions.

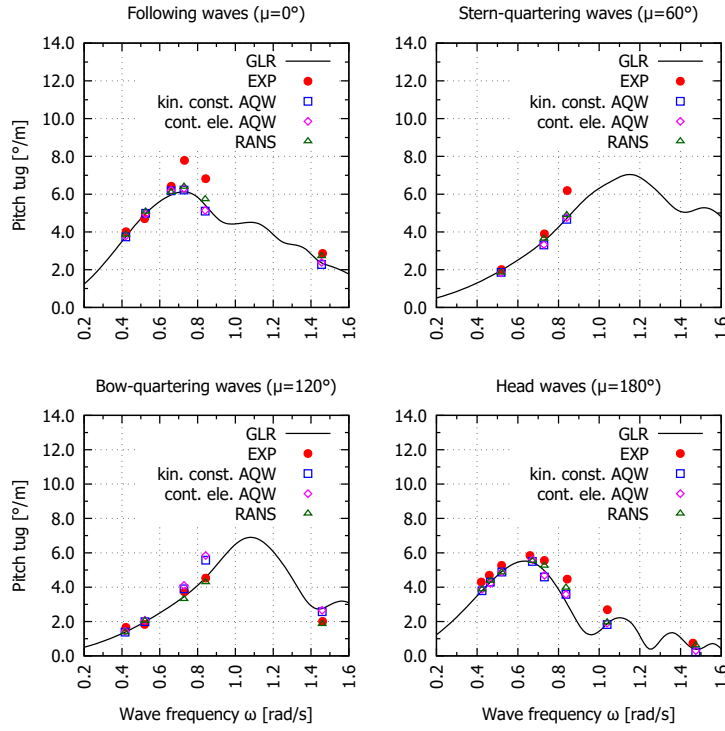


Figure 3.50: Pitch responses per wave amplitude of hingedly articulated **tug** at zero forward speed in waves from various directions.

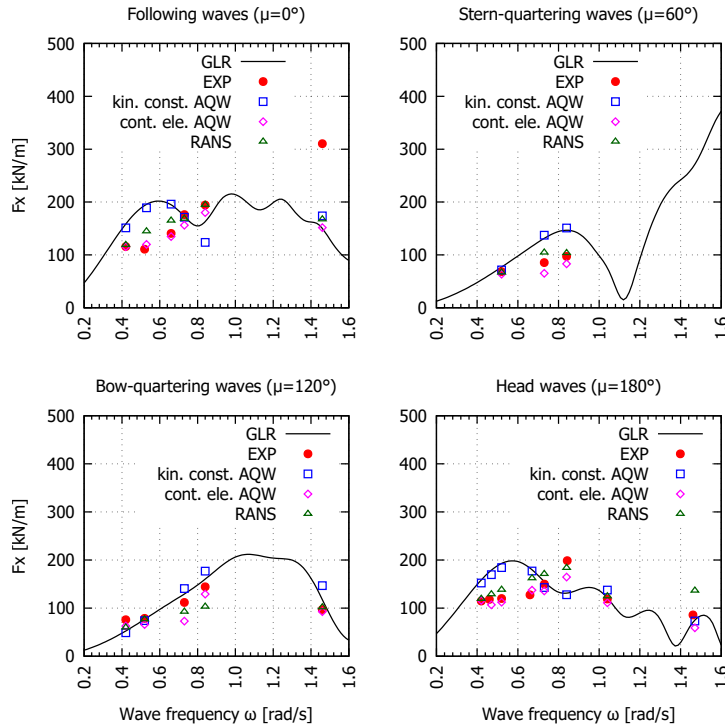


Figure 3.51: Hinge longitudinal force responses per wave amplitude at zero forward speed in waves from various directions.

3 Validation

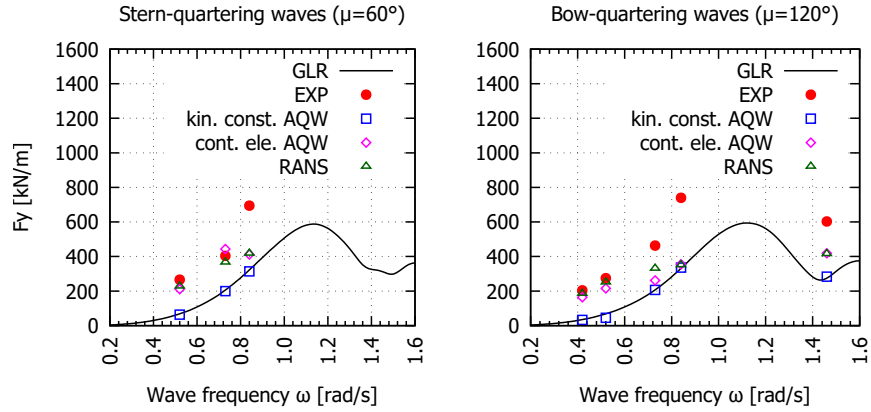


Figure 3.52: Hinge transverse force responses per wave amplitude at zero forward speed in waves from various directions.

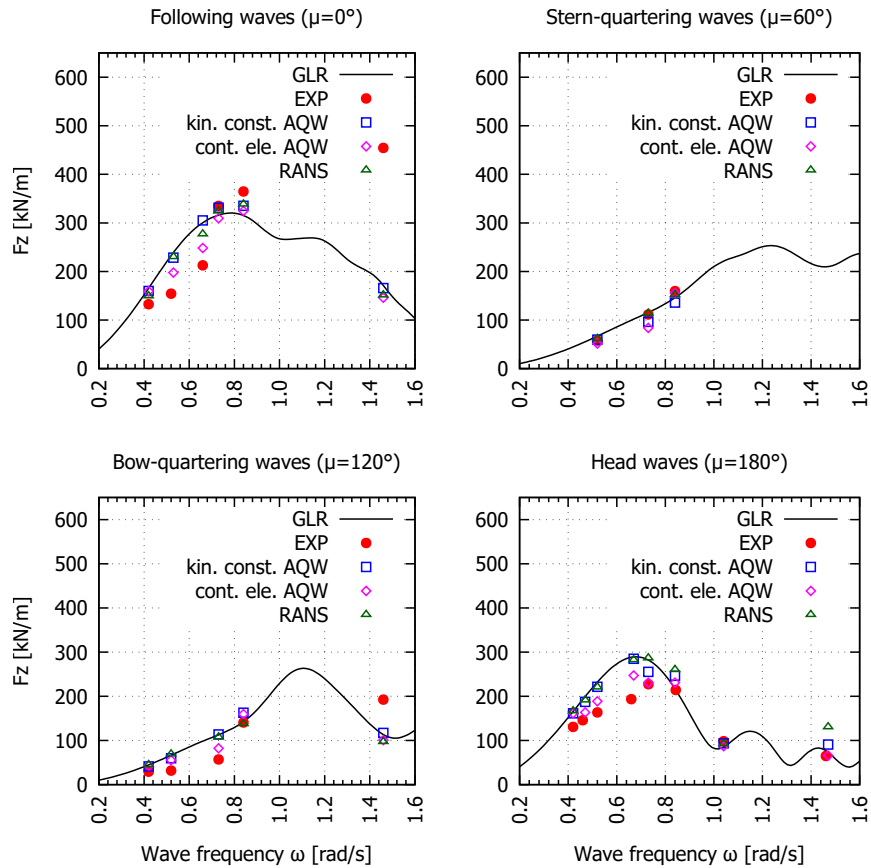


Figure 3.53: Hinge vertical force responses per wave amplitude at zero forward speed in waves from various directions.

3.5 Articulated Multibody Simulations

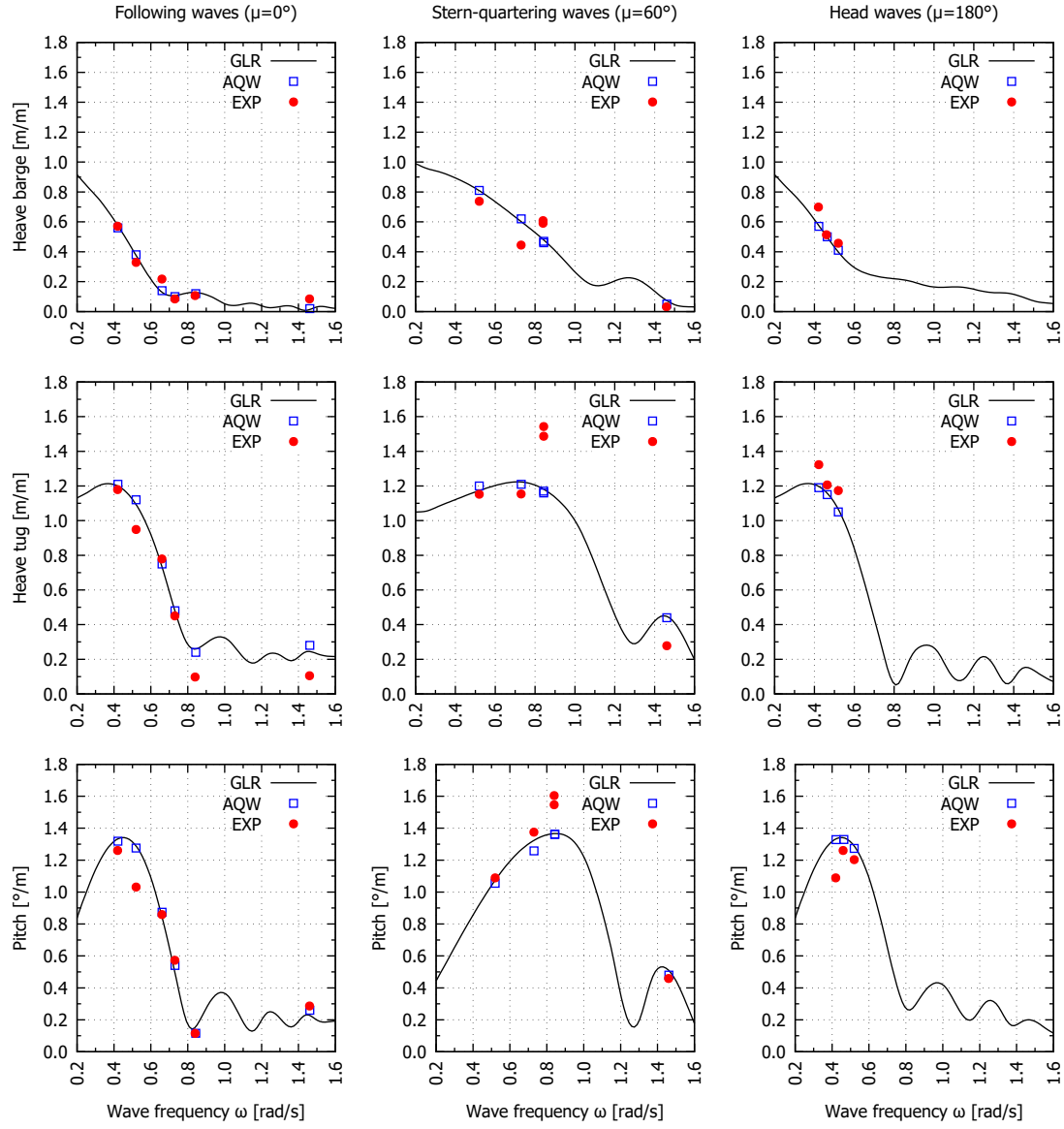


Figure 3.54: Motion responses per wave amplitude of rigidly articulated tug at zero forward speed in waves from various directions.

3 Validation

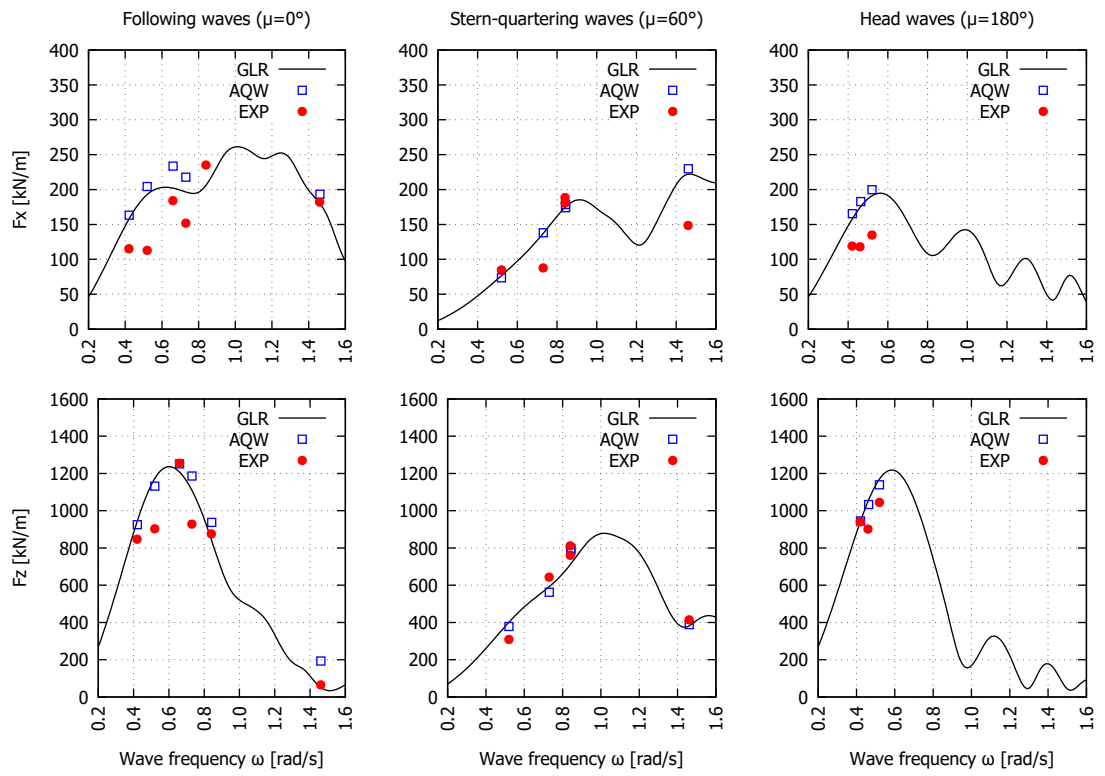


Figure 3.55: Rigid joint force responses per wave amplitude at zero forward speed in waves from various directions.

4 Method Application

In the preceding chapter, it was implied that contact elements are eligible to represent articulations. In this chapter, a practical case study is presented, in which induced joint forces and motion behavior of articulated bodies are studied. The bodies are articulated by a dedicatedly designed coupling mechanism. The coupling mechanism consists of prismatic joints to allow relative vertical motions of the bodies. The mechanism possesses a machinery to enforce pretension at the coupling points to reduce relative motions. The contact element model is associated with the zero-speed Green function boundary element method **AQW**. The simulations are done in regular waves at zero forward speed. The simulation results are compared with respective model test measurements. The measurements are related to the model tests, which are run some time after the first campaign model tests (cf. section 3.1). Therefore, these tests are called second campaign model tests in this work. In the following, at first, the coupling mechanism, to which the model tests are referred, is described. Then, the coupling replica applied in the model tests is presented, as well as the model test conditions are given. Subsequently, the configurations of numerical simulations are shown and discussed.

4.1 Coupling Mechanism and Test Cases

After the first model test campaign, in which tests with single, multibodies, and articulated multibodies were conducted, another model test campaign was performed (see section 1.1). The second campaign of model tests followed the intention of modeling the convoy mentioned in section 1.1 as realistically as possible in almost every aspect, such as coupling mechanism, environmental condition, and convoy bodies. In the following, the coupling mechanism conceived for the convoy is described. Then, the model test conditions are presented.

4.1.1 Coupling Mechanism

The pushing convoy consists of a pushing tug, and a pushed barge. These two bodies are coupled with each other by means of a mechanical coupling mechanism (figure 4.1). The coupling comprises a driving part on the tug and a receptacle system on the barge. The receptacle part consists of two runner chambers carved onto port and starboard sides of barge transom to accommodate the ram sledgehammers. The rams are driven by hydraulic winches on the tug. Each ram is made up of a jacket with a supportive steel slab (painted in black in figure 4.1-right) and a T-shaped sledgehammer settled inside the jacket. At the connection time, the two rams are nudged out of the tug and lodged into the runners. The rams are finally pretensioned to sandwich the runner after walls between the sledgehammers and the jacket support slabs; thus, keeping both vessels attached to each other during the voyage. Sledgehammers are rotated by 90° prior to insertion to allow the T-shaped heads to pass through the runner openings. To ensure the free rotation of the hammers inside the runners, enough lateral space must be provided. The relation between the required spaces on each side of the hammer is shown in figure 4.2. The required free side space depends merely on the hammer geometry; hence, the greater the hammer height, the larger is the lateral freedom between the bodies to move. Additionally, the hammers are equipped with a mechanism at both ends that works like a ball-bearing permitting both vessels to pitch a small amount relative to each other. The to-some-degree rotatable hammer parts are shown in light green in figure 4.1-right. Moreover, the vessels are free to heave along the runners.

The machinery on the tug is capable of exercising pretension on the rams. The pretension exerts a constant tension force along ram axis to attenuate the intensity of relative body motions. Furthermore, the hydraulic pumps are equipped with a safety system that slacks the rams immediately if a jerk of greater than a specific magnitude is exerted in the opposite direction of the pretension. The safety system prevents the machinery from damage.

4.1.2 Model Tests - The Second Campaign

The second campaign of the model test was executed by *Development Centre for Ship Technology and Transport Systems* (DST). A coupling model to replicate the complex real coupling mechanism was conceived. Due to the open-topness of the barge, the test priority was set on fulfillment of requirement of International Maritime Organization (IMO) for sea-going open-top container vessels. Therefore, self-driven convoy exposed to JONSWAP-Spectrumed seas was prioritized. Two

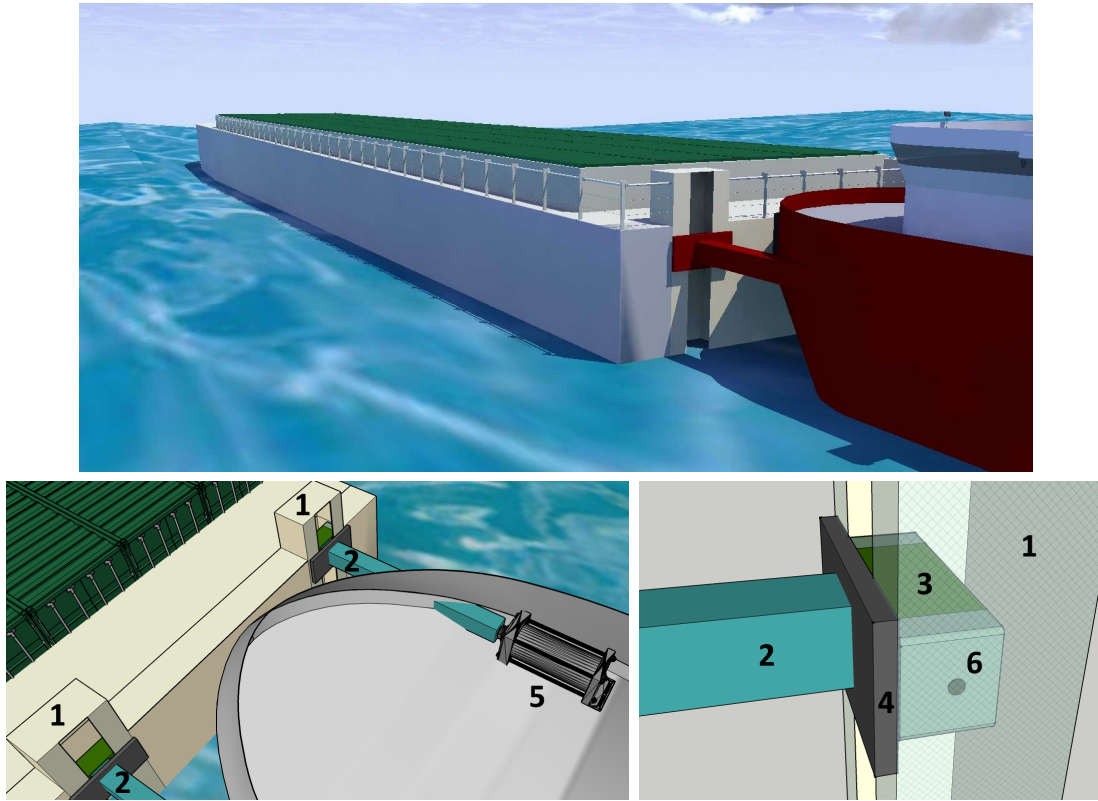


Figure 4.1: Convoy overview (top and left) and the zoom-out of the connection ram after insertion (right). The runner walls in the right figure are translucent to demonstrate the arrangement. (1) runner chamber (2) ram jacket (3) ram sledgehammer (4) ram supportive slab (5) hydraulic winch (6) rotatable hammer part.

other additional test series were conducted firstly with regular waves to provide validation data for computed coupling forces, and secondly with a so-called design wave to provide required data to determine the structural design loads on the coupled open-top barge. In the following, the coupling mechanism conceived and applied in the model tests to replicate the real one depicted in section 4.1.1 is presented. Subsequently, description of relevant test runs referenced within this work are given.

4.1.2.1 Coupling Mechanism Replica

The original design of the coupling mechanism allows both vessels to be fully connected in surge. Relative Sway and roll motions can be considered as suppressed as long as the friction between contact parts is not excessive. Otherwise, sway

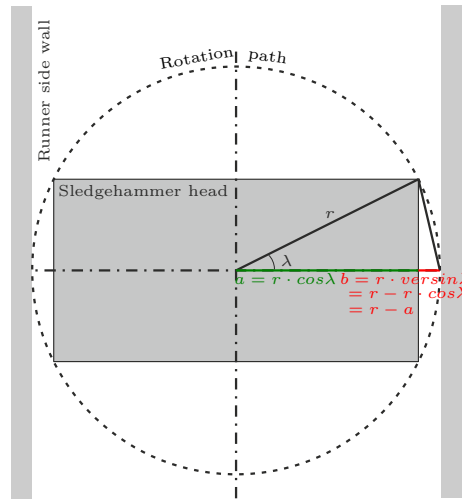


Figure 4.2: Relationship between free spaces required on each side of a ram sledgehammer to ensure its free rotation as seen from the ram rotation axis.

and roll are free to some extent. Relative heave and pitch motions are allowed - their movement is influenced by the pretension in the system. In other words, relative heave and pitch can be considered fully or partly suppressed, depending on the intensity of pretension and hydrodynamic responses of the bodies under environmental conditions. The availability of safety system, which frees the rams in case of excessive loads is fully disregarded both in model tests and simulations for brevity.

The modeled coupling consists of several components, with each component fulfilling one specific purpose (figure 4.3). The relative heave motion is realized by a prismatic joint, but restricted in its extremities by intervening mechanical stops. The freedom of rotation of both vessels at the sledgehammer about their transverse axis is ensured by a hinge at the point where the tug is connected to the prismatic connector. Arbitrary pretension can be modeled by governing the friction along the prismatic joint axis via screws on the prismatic joint (figure 4.5). The greater the pretension, the tighter are the prismatic joint surfaces fastened; thus, increasing the friction between these surfaces. The contact surfaces of the prismatic joint where the friction is applied are of steel-rubber type to increase its effects and to decrease the noise produced by the system. In all test runs pretensions, i.e. friction forces along the free prismatic joint axis, are applied. Relative surge and sway motions in combination with the application of friction force can be considered suppressed. Relative roll motion characteristics of such a configuration is influenced by the system's geometric configuration and by the friction force.

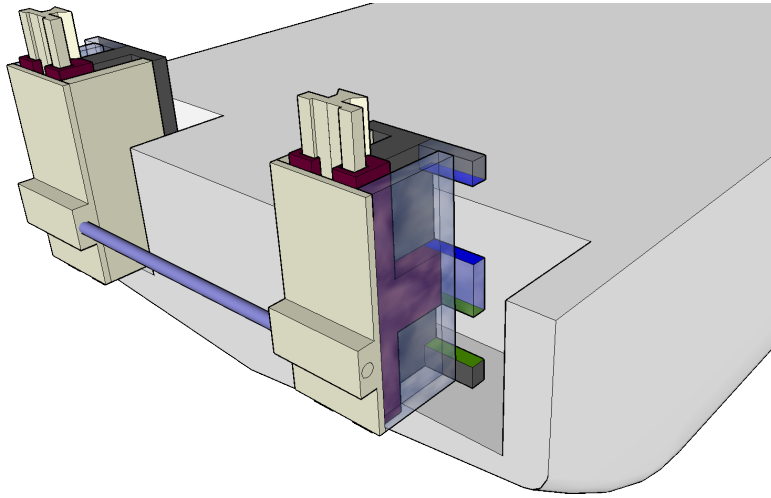


Figure 4.3: Coupling mechanism employed in model tests.

4.1.2.2 Convoy Bodies and Environmental Conditions

The test was conducted in the shallow water tank of 200 m of length, 10 m of width, and 1.2 m of depth for the vessel models with the scale factor of 20. Wave maker at one end of the tank generated the waves, and iron grid wave damping zone at the other end of the tank absorbed them. The measurements from the first model test campaign (see also sections 3.3 to 3.5) proved the sensitivity of the tug to external forces. Although only the design of a seagoing barge was the objective of the funded project, the effect of tug in the multibody system could be neglected. Regarding that the tug used in the first model test campaign was really not a dedicated tug design for the designed barge, no realistic multibody analyses, as well as force assessment on the coupling might be achievable. Consequently, a dedicated tug was designed preliminarily by DST for the second model test campaign to allow better prediction of forces on the relevant coupling mechanism. Figure 4.4 shows the body and sheer plans, as well as the perspective view of the new tug design. Table 4.1 presents the tug's main particulars, and test loading condition. The barge design employed in the first test campaign was maintained, with the exception of its height increment. Right-handed coordinate systems were used and body-fixed coordinate systems were defined at the centers of gravity of each respective body, see figure 4.4. The free surface at rest lies at $z = 0$.

Three test run series were conducted. Since the barge could be classified as an open-top container vessel, model test instructions prescribed by IMO were followed to ensure the vessel intactability. Results of open-top tests were demanded by the IMO representative administration, participated in the sponsored research project.

4 Method Application

Table 4.1: Main particulars of the tug. Units are in [m] if not otherwise stated.

	L	B	T	∇ [m ³]	LCG	VCG	GM	r_{xx}	r_{yy}	r_{zz}
Full scale	27.8	8.8	3.0	402	14.78	-1.07	2.67	3.60	7.50	7.50
Model scale	1.39	0.44	0.15	0.05	0.74	-0.05	0.13	0.18	0.38	0.38

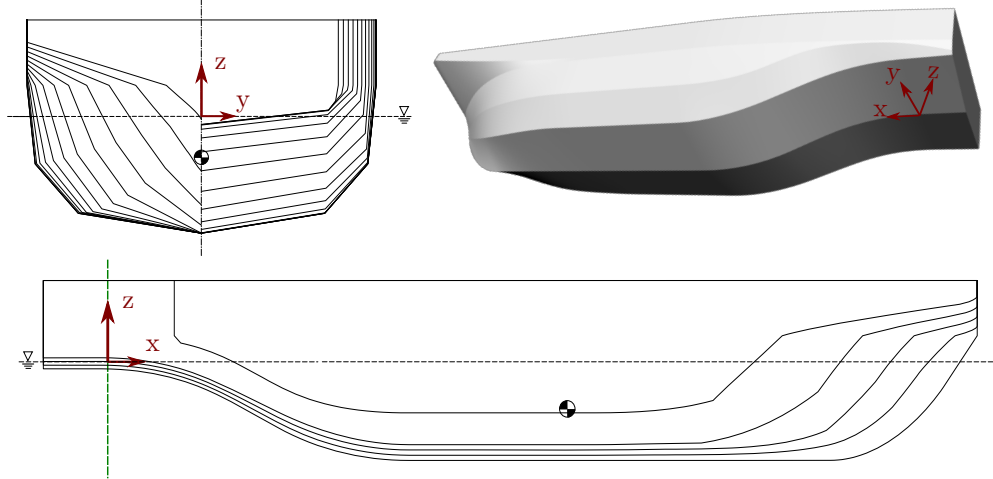


Figure 4.4: Perspective view (top right), body plan (top left), and sheer plan (bottom) of the tug employed in model tests.

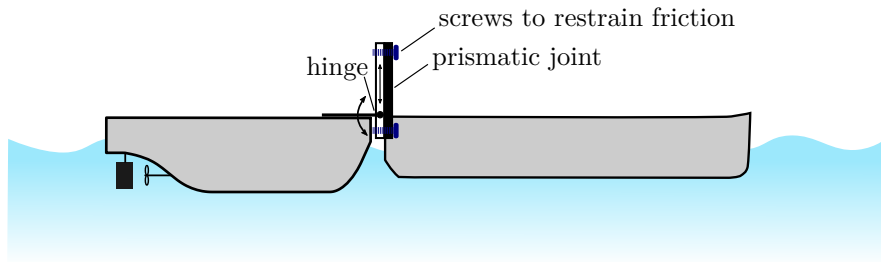


Figure 4.5: Arrangement of model test setups.

Therefore, the open-top run gained the primary status; thus, the convoy was made self-driven. Figure 4.5 illustrates the convoy test arrangement schematically.

In addition to open-top container tests, dedicated test runs were performed with head design waves to determine construction design load of the barge. A design wave corresponds to the Most Likely Response Wave (MLRW) a structure encounters during its designed life time [80].

The third run series were done in regular waves with the purpose of providing validation data for numerical simulations. The tests were conducted in head, beam

4.1 Coupling Mechanism and Test Cases

and following waves at convoy zero forward speed. The bodies were positioned with moorings. A hawser was fastened tightly across the tank, to which the barge's forward end in head waves, and tug's aft end in following waves were connected by means of spring links to the taut hawser. The other end of convoy body was fastened to an elastic hawser, which was also drawn across the tank (figure 4.6). In beam waves, two spring links connected the barge to taut hawser. In this run series screws to restrain friction were so adjusted to model 0.01 kN model scale (80 kN full scale) pretension (table 4.2).

Open-top simulation data as well as the data from design wave test runs are not addressed in this work.

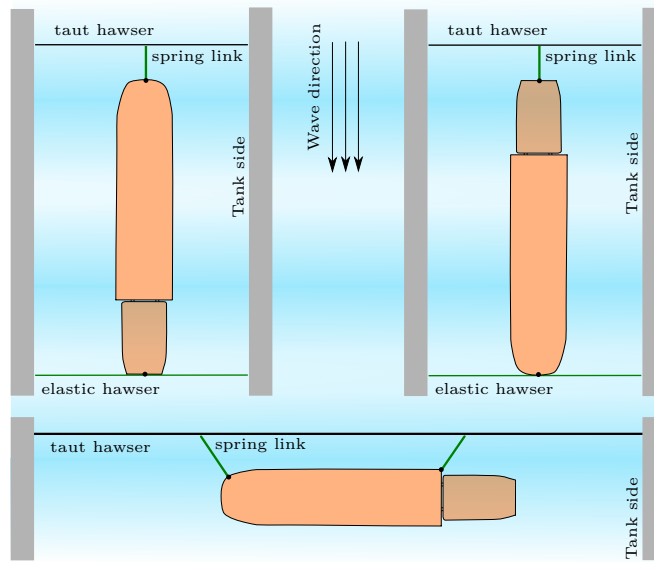


Figure 4.6: Test arrangements for runs in regular waves.

Table 4.2: Overview of model test configurations for runs in regular waves.

wave direction [deg]	pretension (full/model scale) [kN]	speed [knots]
0 (following wave)	80 / 0.01	0.0
90 (beam wave)	80 / 0.01	0.0
180 (head wave)	80 / 0.01	0.0

4.2 Numerical Simulations

Following preceding section regarding description of real coupling mechanism and its model test replica, as well as different model test arrangements, associated numerical simulations are presented here. Firstly, numerical model used to simulate the coupling is described followed by the presentation of simulation setups and the results. Emphasis of numerical simulations is set on the eligibility of contact element models in presentation of prismatic joints, and ability of contact elements to model the complex coupling used in the model tests. All simulations have been done in *Ansys Aqua* suite, in which boundary element methods are employed to calculate the hydrodynamic forces. The motion equations with integrated contact element formulation are solved in time domain (see equation (2.94)).

4.2.1 Numerical Coupling Models

Since the model test measurements are taken for validation purposes, the coupling model of the test cases rather than the real coupling is referred to. The numerical coupling model employed should be able, with reference to figure 4.3, to simulate

1. the prismatic joint of the model allowing relative translation along the prismatic axis, and
2. the hinge joint allowing relative pitch motion of both bodies, and
3. arbitrary pretension capabilities.

The model illustrated in figure 2.6 is extended to fulfill the first and second items. With reference to figure 4.7, four contact elements, $A_{1..4}^1$, build a runner chamber, and two contact elements one at the top and one at the bottom, $A_{5,6}^1$, close the chamber at its higher and lower extremities. All contact elements are part of the barge and follow its motion. A virtual point, P_1^2 , is enclosed by four contact elements, $A_{1..4}^1$, which permits P_1^2 to slide only in upward/downward directions. This combination is equivalent to the translation freedom of a prismatic joint. Figure 4.3 shows that free translation along the prismatic joint in the model test coupling is not restricted by the end of the rail, but by the plate riveted at the back of the rail (shown in black in figure 4.3). Obviously, the upward translation is restricted if the lower blue surface in figure 4.3 touches the upper one. Similarly, the downward translation is restricted if the upper green surface reaches the lower one. The position of the red virtual point, P_1^2 , in figure 4.7 is nearly midway between the lower blue and upper green surfaces in figure 4.3. Therefore, P_1^2 has to trace a greater distance to reach A_6^1 and A_5^1 than its model test counterpart point - and this situation causes wrong results. A virtual point at the lower blue

surface, and a point at the upper green surface complete the simulation model. The blue point P_2^2 is theoretically related to the blue contact element A_6^1 , the green point P_3^2 to the green contact element A_5^1 , and the red point P_1^2 to four side elements $A_{1..4}^1$. With this configuration, the lateral motion restriction of the model test prismatic joint is modeled by tug's red point P_1^2 in conjunction with the barge's red contact elements $A_{1..4}^1$, the upper end stop of the model test prismatic joint by tug's blue point P_2^2 and barge's blue contact element A_6^1 , and the lower end stop by the green point P_3^2 and element A_5^1 . The prime sign of the notations in figure 4.7 denotes port side, and the upper indices show the surface or point relation to the tug (body 2) or the barge (body 1).

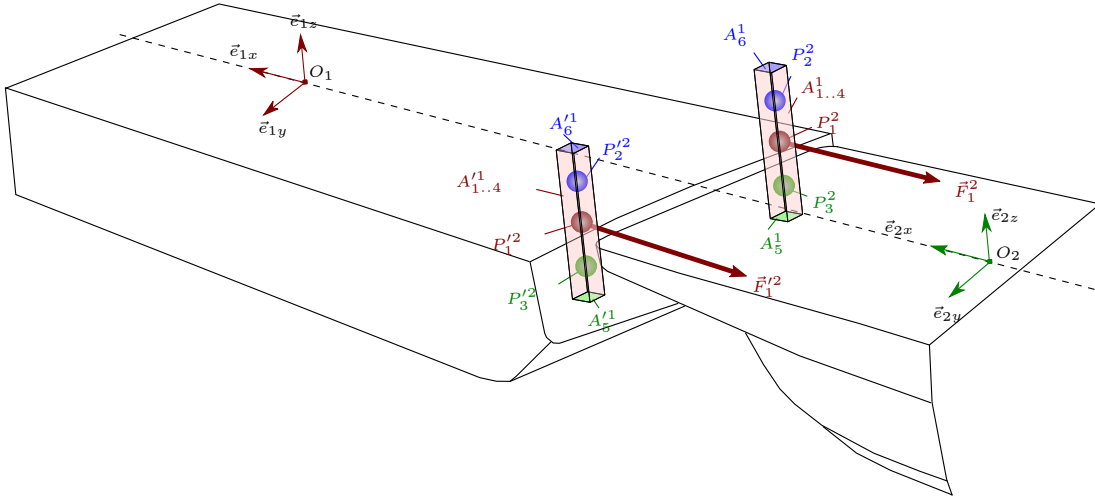


Figure 4.7: Coupling model used in numerical simulations. Forces $\vec{F}_1^2$ and $\vec{F}_1'^2$ represent the directionally constant force model to simulate arbitrary pretension forces.

Two different models idealized the pretension, namely directionally constant force model and friction model.

Directionally constant force model represents the real coupling pretension system, in which the pretension is exercised by a hydraulic winch via the ram at the sledgehammer head (figure 4.1). Similarly, pretension is exercised by a directionally constant force model at the red point. The direction of the force is always constant; hence, the pretension force is sufficiently approximated as long as relative pitch motions are small. The pretension force component in the z-direction increases with increasing pitch motion, whereas the directionally constant force maintains its initial z-component. This discrepancy yields inaccurate simulations, in which high pitch motions are expected. Therefore, the directionally constant force model is employed only in beam waves. This

model provides a comparison model to the friction model explained below.

Friction model represents the pretension employed in the model test coupling arrangement, in which the pretension is realized as friction force along the prismatic joint axis. Friction force is direction-dependent; hence, the motion at a specific time needs to be compared to the motion at its previous time step to determine the friction force direction. Hereto, one pair of control point is defined in this model for each coupling; thus, two pairs for starboard and port couplings have to be considered. Both points of each pair are initially coincident and are located at the red point position (figure 4.7). One point of each pair follows the barge's and one the tug's motions. At each time step the new positions of control points of a pair are evaluated, and compared to their respective positions in the previous time step, as well as with the position of the other point of the pair at the time being. Since the points are defined in each body's reference frame, they are transformed to initial reference frame for comparison. The comparison of a point position at two successive times is used to determine the direction of friction force, which is to be applied. The comparison of positions of both points of a pair with each other for the time being is used to determine which point(s) take(s) the friction force(s). The determination is based on the following criteria:

- if both points of the pair have moved upwards or downwards, the friction force is assigned to the body point with the greater distance Δs in the time span Δt . It indicates that in case of same directional movement of bodies at these specific points, the body with the higher point velocity takes the friction.
- if the points have not moved in the same direction, friction force is assigned to both bodies.

The friction forces at points are realized as moments at the center of gravity of their respective bodies. Only the vertical component of the friction force is taken and the other components are neglected for brevity. A motion tolerance of 1×10^{-6} has been applied to avoid exertion of the force for any minute changes of point positions. This tolerance has been determined after monitoring the motion development. Although this tolerance is small, it has enhanced the simulation stability, especially at the beginning of the simulation, when the bodies start moving away from their initial positions gradually. The tolerance avert insertion of large forces into the system by any negligibly minute movement of the studied points. Friction model is implemented in the Application Programming Interface, API, of *Ansys Aqwa*.

4.2.2 Simulation Configurations, Results and Discussions

It was endeavored to associate the numerical coupling model with contact elements described in section 4.2.1 with RANS solver *Starccm+*. Instead of virtual points above the water level, coupling geometries with the rams and sledgehammers were integrated. Zero gap interfaces were used between body meshes to allow close approaches and contact of bodies with each other. Contact elements were used at runner walls to exert forces at and during contact times and duration. The stability of simulation demanded extremely small cell size at the coupling zones. Relatively small time step was also required. The experiments showed that the implemented Chimera method was not able to come along with the complexity of geometrically defined coupling, and all attempts failed. Indeed, this complex model is the proof of field method immaturity in some fields. No need to mention, that this model is challenging even in association with boundary element solvers, in which the unwetted surfaces may not be required, and the tug coupling connections are defined as virtual points. Ley et al. [55] tried to bypass the restrictions of RANS model by taking the bodies apart to reduce the interaction of bodies' dedicated meshes. The authors realized the connection of bodies during simulations as well as the pretension by means of two moorings connected at coupling positions. They used a contact element to prevent the tug being dragged by the pretension force inside the barge. This combination allowed free movement of bodies in vertical direction, as well as relative pitches. The authors reported agreeable results but did not study how much the increase of distance between two bodies could influence the final results. Additionally, mooring lines influence the responses in low frequencies. This effect was not addressed, too. Nevertheless, the model presented in [55] is still a rare example of application of RANS solvers in combination with the application of contact elements and mooring lines in offshore field. Scenarios with combined application of contact elements and moorings are common in offshore field but are still handled in the industry by means of boundary element solvers. The model in [55] simulates the real coupling niftily, but the application of the contact element as a fender does not comply with the objective of this work, i.e. application of contact elements as joints.

Consequently, all the simulation done in this section were performed using boundary element methods. The Green function hydrodynamic solver for zero-speed with forward speed correction provided by *Ansys Aqwa* is used. Time domain motion equation is used to solve the body motions. The contact element forces used in the system augments the excitation forces on the right hand side of the motion equation (equation (2.94)). Frequency dependent added mass and damping terms are calculated in advance. Convolution integral is approximated numerically. The diffraction forces are transposed directly from corresponding frequency domain

4 Method Application

analyses. Nonlinear restoring and wave excitation forces are computed nonlinearly; hence, the body geometries above mean water level are panelized, too. On the contrary, the coupling model is not defined geometrically, because the coupling model described in section 4.2.1 circumvent this demand. The virtual points are assessed in each time step by the intervening API-code.

The simulation conditions are configured as per table 4.2. Each contact element has its own damping, C_d , friction, μ , and stiffness, C_k , coefficients (see section 2.3.2). In case of application of friction model for realizing pretension, another factor is required, which governs the ratio of pretension force to the associated friction force. Determining these parameters requires understanding the relationship, and effect of each parameter on the others. Opposite to contact element hinge joints, for which merely stiffness coefficients of elements were decisive, almost all element coefficients are influential here. Therefore, a systematic study has been conducted, which finally resulted in dozens of simulations. Such a great number of simulations is still discouraging from practical point of view.

The pretension force is already known, 80 kN per coupling in full scale. The parameter used to calculate the friction force via pretension, *friction model coefficient*, can be, but is not necessarily consolidable with each contact element friction coefficient, μ . The friction model coefficient acts indirectly on the coupling forces because it is related to body motions, and ultimately it affects the coupling forces in the next time step. On the hand, element friction coefficients, μ , are related directly to coupling forces, and are effective at time t . Nevertheless, the friction model coefficients are maintained, and element friction coefficients, μ , are varied. Friction coefficients, μ , of top and bottom elements, A_6^1 , $A_6'^1$, A_5^1 and $A_5'^1$ in figure 4.7, are zero because no friction forces along longitudinal axis are expected. All friction coefficients of $A_{1..4}^1$ and $A_{1..4}'^1$ are equal in each simulation. Experiments show that the element damping coefficient contributes negligibly because numerical instabilities related to sudden motion changes are controlled by the force trigger tolerance defined in the API-code. Therefore, the element damping coefficients are zero in all simulations. Consequently, the friction coefficients remain as the principal variables required to be determined. These variables are related to the rubber-steel contact surfaces of the model test prismatic joints, their tightness and wetness.

In the following, simulation results are presented and discussed.

4.2.2.1 Convoy in Beam Waves

Instead of attaching the barge with two spring links to a taut hawser, its sway and yaw motions are suppressed. The tug motions and the other barge motions

are free. Although this arrangement eliminates the effect of mooring lines, it reduces the simulation complexity. After studying the tug and barge roll motions per wave amplitude (figure 4.8-right), it becomes evident that both bodies roll almost synchronously, and that there is less relative translational motion along the prismatic joints. It originates from the fact that the vertical motions of the bodies, in absence of pitch motion consist of heave and roll vertical components. The relative vertical motion is restrained by the friction along the prismatic joint axis. This has caused the measured roll responses of both bodies to be nearly equal for all tested wave frequencies in regular beam waves. Consequently, it is concluded that the conditions can be simulated ideally with hinges, such that no relative vertical motion at the hinged position is allowed.

Furthermore, considering that the bodies' pitch responses are negligibly small in beam waves, and that the the bodies' vertical motion responses caused by heave and roll motions are nearly synchronous at the connection level, the application of the directionally constant force model of figure 4.7 is justified.

In summary, three models are implemented in beam waves, two models with the contact elements, one of which has the pretension represented by directionally constant forces model, and one by API-intervening friction model. The third model is a simplified model with a kinematic constraint hinge joint. All these models are compared with experiment measurements.

Figure 4.8 and figure 4.9 show the computed and measured roll and force responses. The results from simulations with kinematic constraint hinge, contact element simulations with directionally constant force and friction models, and measurements are denoted kin. const. hinge, cont. ele.-tension, cont. ele.-friction and EXP, respectively. Considering that all simulation settings are maintained for each directionally constant force and friction models, except the pretension models, the equivalence of the motion and force responses imply that both pretension models are equivalent. Thus, it provides confidence on the accuracy of the API friction model to represent pretension. Both models have predicted the force and motion responses relatively well in long waves, but the predictions deviate with decreasing wave lengths.

One of the most important dynamic phenomenon recognized as dangerous to seagoing ships is the resonance gain of rolling. When the encounter wave frequency is close to the natural frequency of ship roll motion, the roll motion gains resonance and the ship undergoes synchronous rolling. When synchronous rolling occurs, the ship will heel over and may be rolled further over by the action of the wave [50]. Consequently, there is a serious danger that the vessel will heel beyond a point or angle of heel, from which it cannot return to an upright condition. The ship ends up having negative stability, and will capsize. Natural roll frequency, ω_ϕ , can be

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calculated by means of gravity acceleration, g , metacentric height GM , and wet radius of gyration, r_{xx} , so that $\omega_\phi = \sqrt{gGM}/r_{xx}$ [104]. The effective metacentric height may strongly depend on the motion amplitudes in waves. Additionally, wet radius of gyration is frequency dependent. Depending on the wave frequency and roll response amplitude, the barge's natural roll frequency is ≈ 1.3 rad/s, whereas the tug's natural wave frequency is between 1.75 rad/s to 1.92 rad/s; hence, the natural roll frequencies of bodies are out of the studied wave frequency range shown in figure 4.8.

Similarly, the hinge model has also underpredicted the roll responses slightly. The hinge model acts as a reference model and has a less decisive role here. Nevertheless, the motions compared reasonably for the wave frequencies below 0.84 rad/s. The model predicted extremely high roll motions for 1.47 rad/s wave frequency. Studying the almost linear behavior of roll responses between 0.84 and 1.47 rad/s reveals that the convoy is prevented from rolling more than about $6.5^\circ/\text{m}$ because of attached spring links. The simulations lack such roll limits; hence, disagreements originate from different model test and simulation configurations.

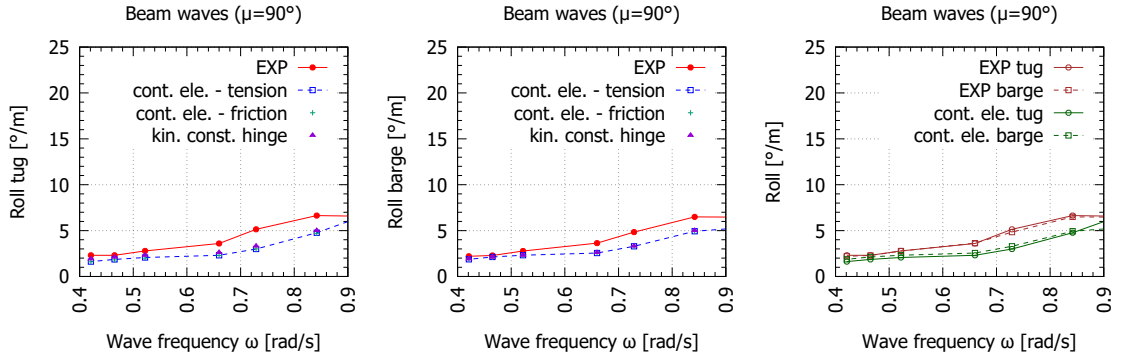


Figure 4.8: Roll motion responses per wave amplitude of convoy bodies at zero body forward speed in beam waves. Directionally constant force model is notated tension and the friction model friction.

The transverse component of total force response, F_y , is generally overestimated by the contact element models for all examined wave frequencies, except for 0.84 and 1.47 rad/s frequencies. On the other hand, all vertical force components, F_z , are slightly underestimated. This causes the calculated total forces consisting of both transverse and vertical components to agree favorably to measurements. The two bumps at 0.47 and 0.74 rad/s for transverse and vertical coupling forces are captured by simulations. Taking the relatively smooth track of roll responses into account, it is believed that these two bumps are related to inaccuracies in measurements, or in assessment of measurements. Nevertheless, further measurements for

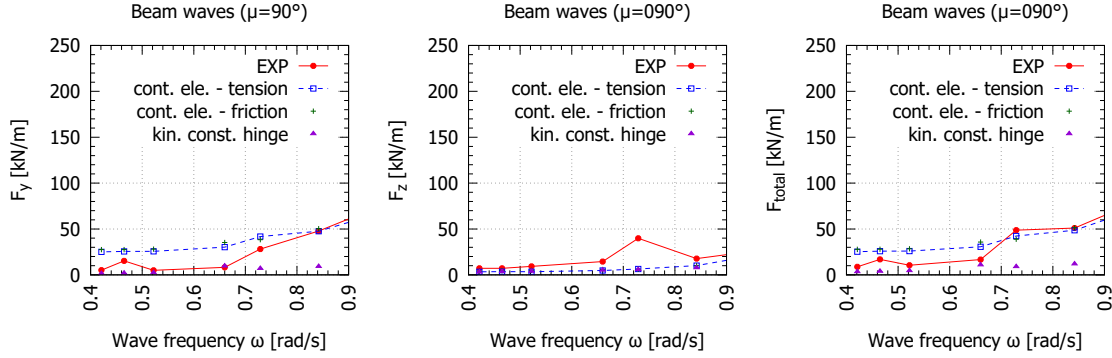


Figure 4.9: Coupling forces per wave amplitude in beam waves at zero body forward speed. Directionally constant force model is notated **tension** and the friction model **friction**.

these two wave frequencies, and preferably also for wave frequencies in their vicinities, are required before giving a final statement related to the bumps. Relatively extreme forces are found for 1.47 rad/s. Although the predicted vertical component at 1.47 rad/s still agree, the transverse component underpredicts the measurement unacceptably. Considering that the spring links prevent the vessels from rolling beyond a threshold, the links forces are transmitted to the system, and this is reflected accordingly in the resultant coupling forces. Therefore, the discrepancies of measurements and simulations originate from different configurations. These differences are believed to have fewer effects on the results at $\omega \leq 0.84$ rad/s because of reduced effects of spring links on the responses. Subsequently, it can be concluded that the responses for 1.47 rad/s cannot be valid neither for simulations nor measurements, and are subject to further analyses and tests.

Figure 4.10 compares time series of coupling forces for $\omega = 0.66$ rad/s (wave period of 9.5 s, wave length of ≈ 140 m and amplitude of 1.22 m) for different coupling models. The time series from simulations with kinematic constraint hinge, contact element simulations with directionally constant force and friction models, and measurements are denoted **hinge**, **tension**, **friction** and **EXP**, respectively. These time series show good capturing of force development over time with the contact element models. Although the hinge model has simplified the simulation configuration to a great extent, and has delivered relatively good results after Fourier transformation, its time series show inaccuracies. Therefore, this model is inappropriate to idealize the complex model test coupling arrangement. Both contact element models have yielded satisfactory results even though results from both models are relatively close to each other. Attention is drawn to measured time series of vertical forces and the existence of local extrema, which impede the analyses

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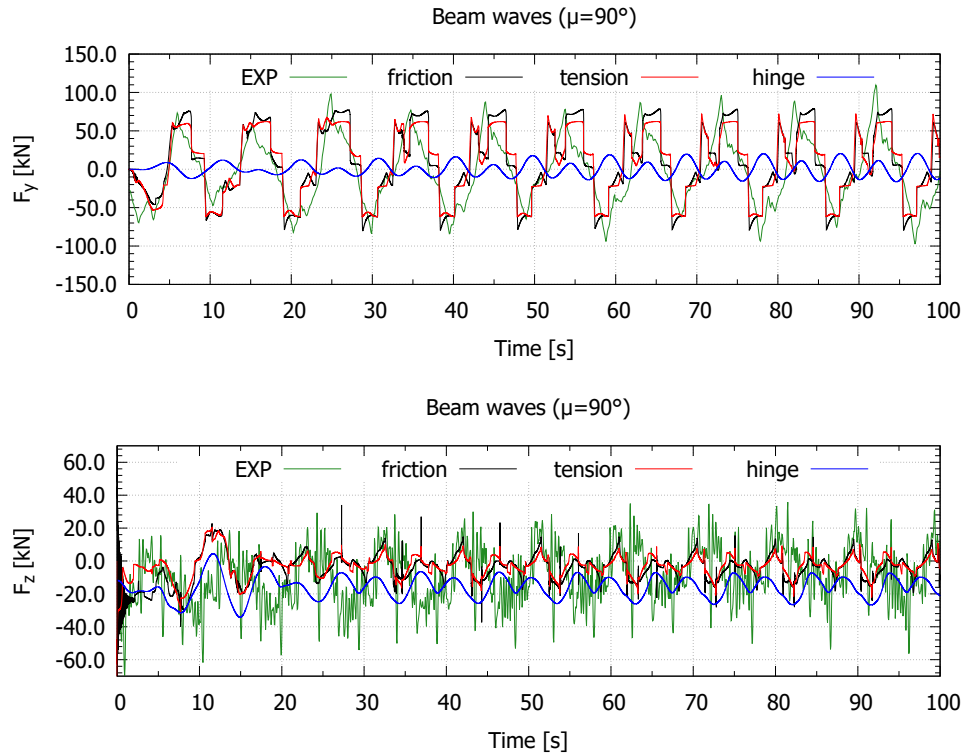


Figure 4.10: Time series of total connection force in transverse (top) and vertical (bottom) directions in beam waves of 9.5 s of period, ≈ 140 m of length and amplitude of 1.22 m at zero body forward speed.

of time series, and are a source of departure of measurements from simulations.

Figure 4.11 shows time series of relative motion between the two bodies related to both connection points of a pair for both starboard and port pairs defined for each coupling. The green line discloses the superiority of tug's point to its barge counterpart. At $t=0$ s, i.e. at initial geometrical positions, both points of a pair coincide. This is evident on both graphs of figure 4.11. The negligibly small distances between points show almost no movement of both bodies along prismatic joint axis. The blue deeper peaks imply the lagging tug movement, such that the barge moves first, engages the tug, and drags it along with itself.

4.2.2.2 Convoy in Head and Following Waves

Simulations in following waves have numerically been the most difficult from stability point of view. The reasons have again been the tug's sensibility to external

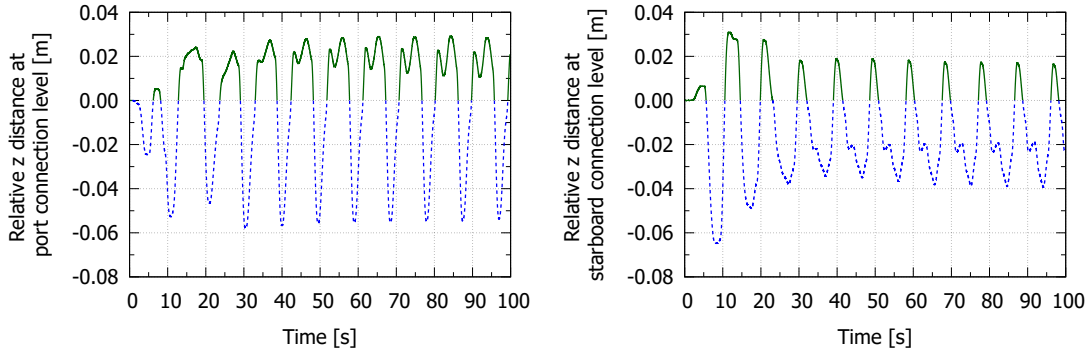


Figure 4.11: Time series of relative vertical distance between bodies at port (left) and starboard (right) connections. Positive relative distances indicate that the tug moves vertically above the barge with respect to the initial connection level. Tests are conducted in beam waves of 9.5 s of period, ≈ 140 m of length and amplitude of 1.22 m at zero body forward speed.

forces, and finding appropriate contact element configurations to keep both vessels stably attached. Particularly, in following waves, the wetness of prismatic joints has played a major role. The wetness is difficult to capture experimentally and not readily available per se, and only general statements are given whether the coupling points are submerged. It is also doubtful whether stringent arguments about splashed water on the coupled points can always be found. The wetness of prismatic joints is highlighted here because simulations could not be conducted with the same friction model and contact element coefficients. The only reasons would be either different wetness grades or pretensions. It is presumed that tightness of prismatic joints (pretension) was double checked before each test run; hence, the wetness grades were presumably different.

Figures 4.12 and 4.13 compare the computed pitch motions and force responses per wave amplitude in aft waves from the contact element model, denoted **cont. ele.** with the the corresponding measurements, denoted **EXP.** Availability of heave responses would have been helpful in construing the results, but heave measurements are not recorded. Similar to beam wave simulations, no moorings are involved in simulations and suppression of degrees of freedom is preferred. The tug's surge and yaw modes are suppressed and all other barge and tug degrees of freedom are free.

As relatively high pitch motions are predicted, pretension is realized by the friction model. For the directionally constant force model, the vertical component of pretension must be increased with increasing pitch motion. This cannot be realized by a force, which is constant in magnitude and direction (here in longitudinal

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direction) because, instead of increasing vertical component of the force, (here) a constantly zero vertical vector is returned. Additionally, depending on the force magnitude, the constant force in the longitudinal direction causes large moments about the center of gravity that makes the tug turn over. The directionally constant force model has been used for comparison purpose to validate the friction model, and this model is not suitable in head and following waves because large pitch motions may occur.

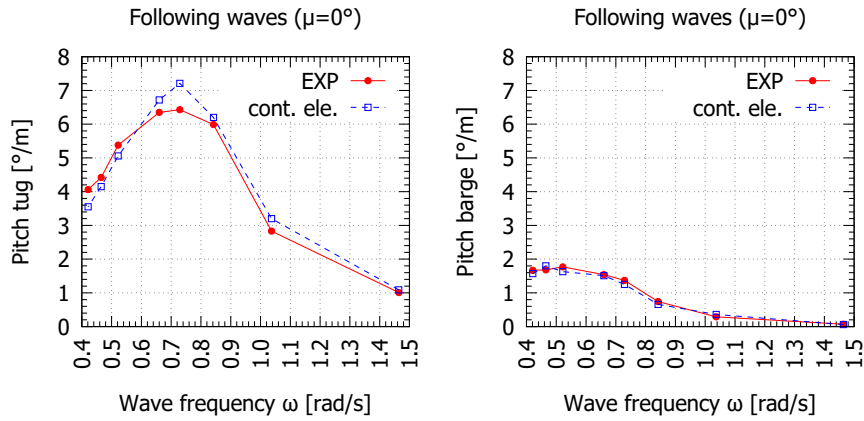


Figure 4.12: Pitch motions per wave amplitude of convoy bodies in following waves at zero forward speed.

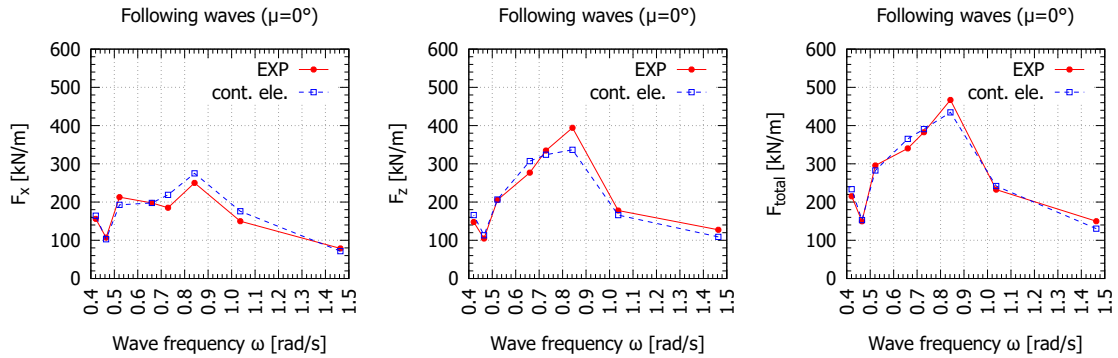


Figure 4.13: Coupling forces per wave amplitude in following waves at zero body forward speed.

Figure 4.12 shows the pitch motion responses per wave amplitude in following waves. Computed pitch motions have generally tracked the respective measurements fairly well, especially, pitch motions of the barge, which agree more favorably than those of the tug. This is possibly due to the relatively higher pitch responses

of the tug and its sensibility to external forces. The sudden rise of barge's simulated pitch at 0.47 rad/s is believed to be due to simulation inaccuracies because neither the force predictions, the measurements, nor the tug's motion predictions and measurements have registered such a rise. The wave frequency of 0.52 rad/s is among the wave frequencies for which the prismatic joints are believed to be wetter (less tensed) because measurements could not be reconstructed unless about 75% of contact element forces for $\omega < 0.52$ rad/s simulations are applied. Analogous to beam waves, the simulated vertical components of reaction forces are underpredicted, and the horizontal components are overpredicted; hence, the reaction forces comprised these two components have led to a satisfactory track of the resulting force measurements.

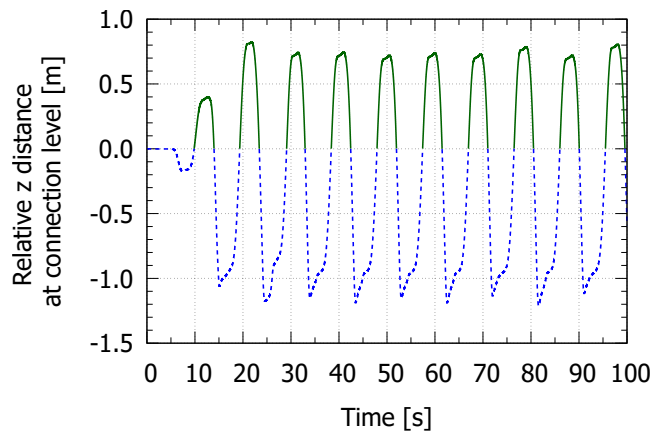


Figure 4.14: Time series of relative vertical distance between bodies at port and starboard connections. Positive relative distances indicate that the tug moves vertically above the barge with respect to the initial connection level. Tests are conducted in following waves of 9.5 s of period, ≈ 140 m of length and amplitude of 1.22 m at zero body forward speed.

In the following waves, an 80kN pretension force at each connection proves to be not sufficient enough to keep both vessels attached at and about the initial coupling points in following waves. Figure 4.14 shows high relative vertical motion of bodies. This figure reveals impacts at the top and bottom extremities of prismatic joints. The unequal maxima and minima (peaks of blue and green lines) reflect the unequal coverable distance below and above the initial coupling position (distance between blue and green surfaces in figure 4.3 are not equal). Comparing figure 4.14 with figure 4.17 reveals not only the exposure and shielding characteristics of the tug in following and head waves, but also the pretension effects. Indeed, these two figures emphasize the necessity to identify the significant wave directions in the sailing zone to optimize pretension force accordingly.

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Simulations in head waves are generally similar to those in following waves. These simulations are numerically less complicated than those in following waves, and relatively less effort is required to obtain numerically stable simulations. Again, instead of spring links, suppression of appropriate degrees of freedom of the barge is imposed. The wet surface problem issued in following waves has been of minor issue, and the concentration could again be set mainly on finding the appropriate friction coefficients, μ . Similar to runs in following waves, relative motions of bodies at both starboard and port couplings are the same.

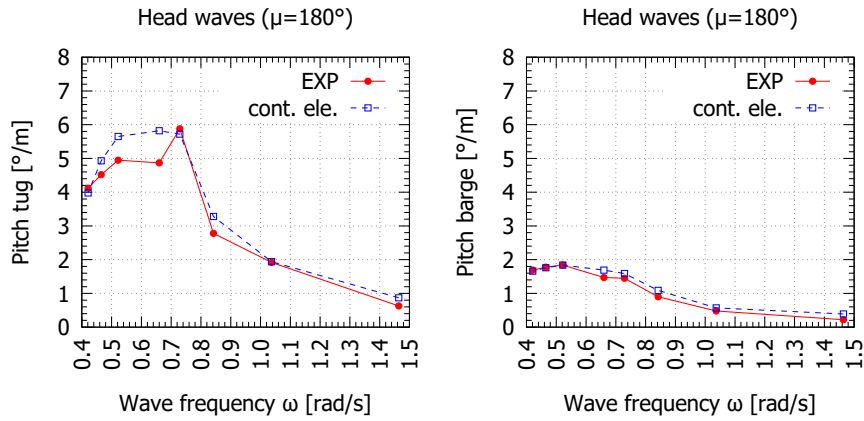


Figure 4.15: Pitch motions per wave amplitude of convoy bodies in head waves at zero forward speed.

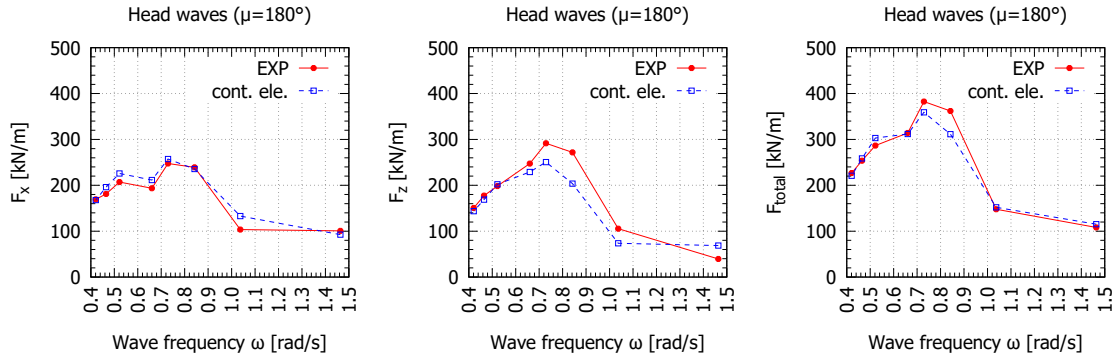


Figure 4.16: Coupling forces per wave amplitude in head waves at zero body forward speed.

Relative vertical distances between evaluation point pairs (figure 4.17) are negligible in comparison with their following wave counterparts. Figure 4.15 and figure 4.16 show the motion and force responses per wave amplitude for head wave

simulations. As already experienced, simulations provide smooth progression of motion response amplitude operator curves; however, the corresponding experiment curves show pitfalls or bumps. The abrupt pitfalls in measurement curves of pitch RAOs have not been tracked by the computations. Recurrently, overprediction of horizontal forces and underprediction of vertical forces are observed, which ultimately result in satisfactory agreement of total force responses to comparative measurements. Generally, contact element models have predicted the force response traces fairly well, except for $\omega > 1.04$ rad/s. The reason for this inaccuracy have been the increase of nonlinearities with decreasing wave lengths, and the behavior of the coupling in short waves. In short waves, the friction forces change more intensively and the contact element model is subjected to more frequent pretension changes.

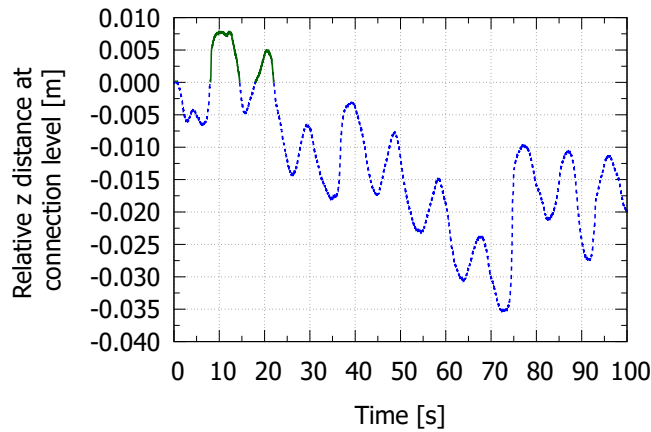


Figure 4.17: Time series of relative vertical distance between bodies at port and starboard connections. Positive relative distances indicate that the tug moves vertically above the barge with respect to the initial connection level. Tests are conducted in head waves of 9.5 s of period, ≈ 140 m of length and amplitude of 1.22 m at zero body forward speed.

4.2.2.3 Effect of Pretension

After having the eligibility of contact elements verified, the efficiency of pretension are examined. It is a time consuming and expensive task to do parametric model tests to study the effect of different pretension force magnitudes on the convoy behavior. The contact element model provides a qualified method to conduct parametric studies efficiently. Understanding the effectivity of pretension has a decisive role in choosing the proper machinery at the design stage. Choosing overdimensioned machinery decreases deadweight and increases lightweight.

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Additionally, overdimensioned machineries are usually bulkier and more resource-demanding. Underdimensioned machinery is also unacceptable because it may fail at the very moment.

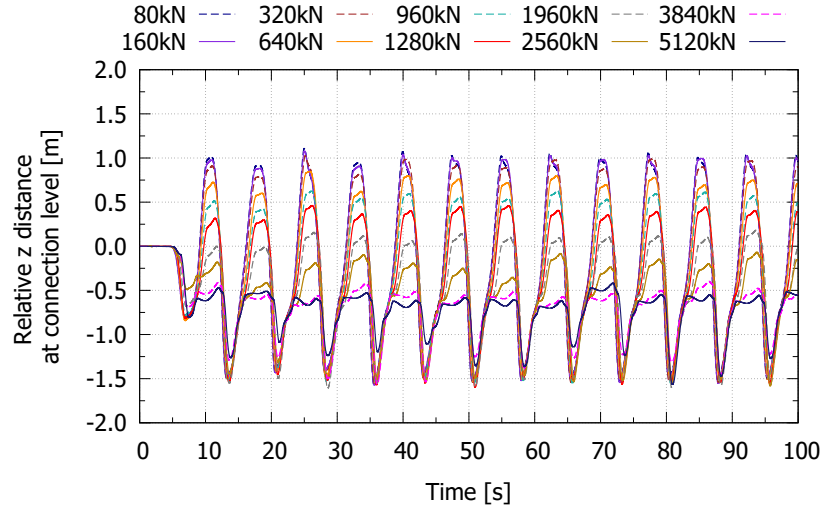


Figure 4.18: Time series of relative vertical distance between bodies at port and starboard connections for different pretensions. Positive relative distances indicate that the tug moves vertically above the barge with respect to the initial connection level. Tests are conducted in following waves of 7.5 s of period, ≈ 87 m of length and amplitude of 1.22 m at zero body forward speed.

In order to assess the effect of pretension, relative vertical motion between bodies is assessed because this motion is most significant for the coupling mechanism. The assessment of relative vertical motion by means of the defined evaluation point pairs has already been demonstrated. As relative vertical motion is generally more prominent in following waves, following waves with the amplitude of 1.22 m and frequency of 0.74 rad/s, which correspond to waves of 7.45 s of period and of ≈ 87 m of length, are chosen. Exposed to these waves, the convoy undergoes severe relative motions with hard impacts on prismatic joint ends. Figure 4.18 shows the time series of relative vertical motion at couplings. Due to the suppression of roll motion both starboard and port point pairs register identical time series. Therefore, only the times series of one side are shown. The influence of pretension is evident as the peaks mitigate with increasing pretension magnitude. How effective the pretension decreases the relative vertical motion, i.e. how tense the pretension keeps both bodies attached, is quantified by measuring the distance between maximum and minimum peak values, i.e. relative distance. Figure 4.18 shows that pretension increment does not decrease the relative distance linearly. Regres-

sion studies based on linear, exponential and second order polynomial functions show that the two latter functions predict satisfactorily the relationship between pretension and relative distance.

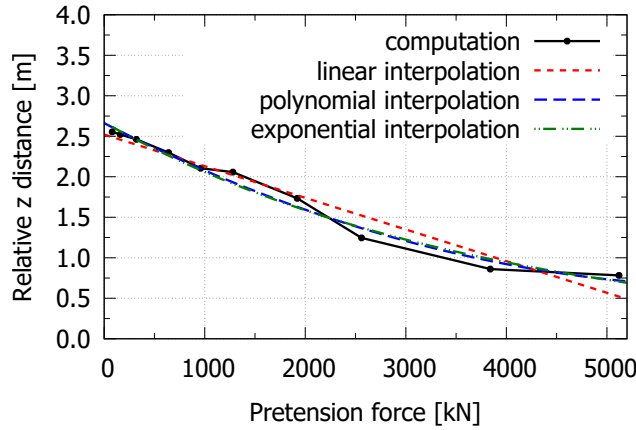


Figure 4.19: Relative vertical distance measured from pretension time series in figure 4.18 vs. pretension. Tests are conducted in beam waves of 7.5 s, period and ≈ 87 m of length and amplitude of 1.22 m at zero body forward speed.

4.2.2.4 Nonlinear Effects

To examine the influence of nonlinear Froude-Krylov forces on articulation forces and body motions, wave amplitudes are multiplied by the factors 1.3, 1.5 and 2.0. Theoretically, more nonlinear affects are expected in steeper waves. For the wave frequency of 1.46 rad/s, the highest amplification factor is 1.8 to avoid wave breaking. Table 4.3 shows the considered wave angular frequencies ω , wave amplitudes h and steepnesses $\frac{H}{\lambda}$.

Figure 4.20 shows the resulting pitch motion of the tug and vertical force on couplings obtained from linear and nonlinear solutions in head waves. Experiment measurements, denoted **EXP**, are shown for comparison. **Amp** in figure 4.20 indicates the applied amplitude amplification factor. Nonlinear affects can be identified clearly for almost all frequencies. Generally, nonlinear effects have improved response predictions. However, nonlinear effects have increased inaccuracies of the contact element model to predict vertical coupling forces at high wave frequencies.

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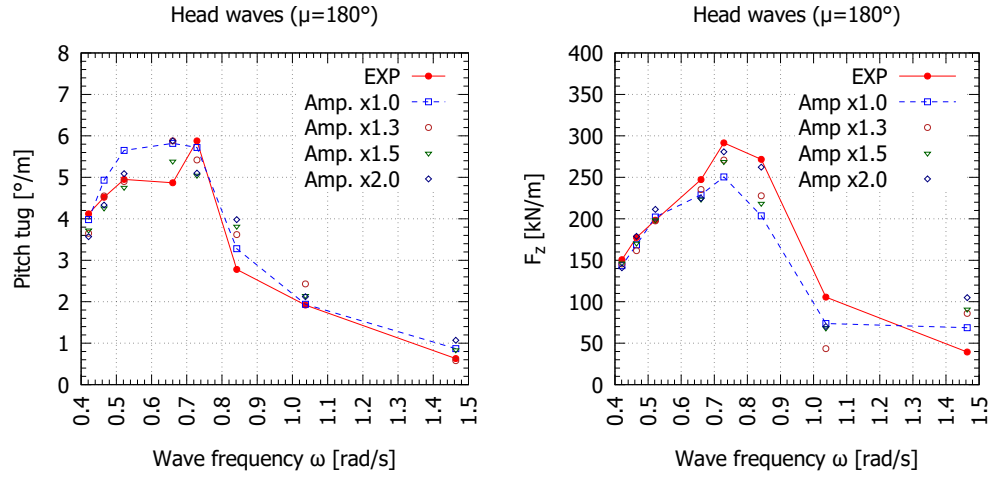


Figure 4.20: Effect of nonlinear Froude-Krylov forces on tug's pitch and coupling vertical force responses per wave amplitude in head waves of different amplitudes. Amp denotes amplitude amplification factor.

Table 4.3: Wave angular frequencies, amplitudes, and steepnesses.

ω [rad/s]	λ [m]	h [m]	$100\frac{H}{\lambda}$ [-]	$1.3h$ [m]	$100(1.3\frac{H}{\lambda})$ [-]	$1.5h$ [m]	$100(1.5\frac{H}{\lambda})$ [-]	$2.0h$ [m]	$100(2.0\frac{H}{\lambda})$ [-]
1.46	28.76	1.12	7.79	1.46	10.13	1.68	11.68	2.02 [†]	14.02 [†]
1.04	57.32	1.05	3.66	1.37	4.76	1.58	5.50	2.10	7.33
0.84	86.94	1.22	2.81	1.59	3.65	1.83	4.21	2.44	5.61
0.73	115.98	1.15	1.98	1.50	2.58	1.73	2.97	2.30	3.97
0.66	141.50	1.22	1.72	1.59	2.24	1.83	2.59	2.44	3.45
0.52	226.21	1.09	0.96	1.42	1.25	1.64	1.45	2.18	1.93
0.47	285.06	1.14	0.80	1.48	1.04	1.71	1.20	2.28	1.60
0.42	347.76	1.20	0.69	1.56	0.90	1.80	1.04	2.40	1.38
1.04	57.32	1.05	3.66	1.37	4.76	1.58	5.50	2.10	7.33
0.84	86.94	1.22	2.81	1.59	3.65	1.83	4.21	2.44	5.61

[†]Amplification factor 1.8 used instead.

5 Summary and Conclusions

This thesis presented a method to articulate the bodies of multibody systems by means of contact elements. The contact element model was validated and its ability to model complex mechanical couplings was demonstrated. Reliable prediction of motion behavior of articulated floating bodies and joint forces depends on the reliability of coupling models, hydrodynamic methods, as well as equation of motion solvers. Therefore, different hydrodynamic methods in adherence with linear and nonlinear equations of motion, and coupling models were also studied in this work.

Boundary element and RANS methods were applied. Two boundary element methods of zero-speed Green function type were chosen. Both methods were capable of handling non-zero forward speeds by means of encounter frequency approach, but one of them corrected additionally the added mass and radiation damping coefficients to account for forward speed effects. Moreover, an unsteady RANS, a frequency domain Rankine source, as well as a time domain boundary element methods were used. In frequency domain methods, the forces on the wetted surface of the bodies up to mean water level were computed using linearized Bernoulli equation, and applied in linearized equation of motion to compute the motion responses of the bodies. For time domain simulations, instantaneous water level was taken in each time step to compute nonlinear restoring and Froude-Krylov forces. The time domain boundary element method solver used the linearized equation of motion to compute the motion responses. Unsteady RANS used also the instantaneous water level to compute hydrodynamic loads on the bodies nonlinearly, but was coupled to a nonlinear equation of motion solver. Volume-of-Fluid method was utilized in RANS simulations to capture free surface. Depending on the simulation, mesh morphing or Chimera methods were applied to couple fluid and equation of motion solvers.

In conjunction with different hydrodynamic solvers, two different methods to treat mechanical couplings were chosen, i.e. contact element method - the to be validated approach, and kinematic constraints - the comparison approach. Hinge joints were applied for validation studies. However, rigid joints were also used auxiliarily.

5 Summary and Conclusions

Kinematic constraints were formulated to represent joints in boundary element methods. It was also demonstrated how constraints were applied in time domain boundary element method and RANS solvers. Kinematic constraints augmented the motion equations. In order to close the augmented motion equations mathematically, Lagrange multipliers were introduced, from which the induced forces on the joints were computed. Rigid and hinge joints were simulated in the frequency and time domain boundary element method models, and hinge joints were simulated in the RANS model. Contact elements were applied in conjunction with time domain boundary element method. The validation studies were conducted in a successive manner. At first, stand-alone bodies were studied. Then, multibody arrangements without mechanical couplings were analyzed. Finally, articulated multibody simulations were performed. The Rankine source method, called **RSM** herein, was not nominated for articulated multibody simulations because unphysical oscillations in the motion responses of hydrodynamically coupled bodies occurred. The oscillations were attributed to unoptimized free surface mesh for a wave frequency range. A river-sea convoy consisting of a pushing tug and a barge was taken in this work. Predicted motion responses of the bodies and articulation forces were taken for validation studies. Simulation results were compared with the model test measurements. The barge heave and pitch responses were very well predicted in stand-alone, hydrodynamically coupled, and articulated conditions. Being a small vessel, the tug proved to be very sensitive to external forces. Its sensitivity made its motion predictions difficult. The tug heave and pitch responses for zero speed were predicted well in stand-alone, unarticulated, and articulated multibody conditions. In case of forward speeds, the heave responses of frequency domain solvers for the tug showed incorrect humps in response amplitude operator curves, if hydrodynamic coefficients were not corrected for forward speeds for stand-alone and unarticulated cases. For these cases, time domain boundary element solvers were apt to underpredict the tug heave responses. The RANS solver predicted the responses well.

For the stand-alone and unarticulated cases for forward speeds, relatively high tug's responses were predicted. Articulation of the tug harnessed high motion pitch amplitudes significantly, and submitted another evidence on vulnerability of the tug to external forces. Connection of the tug to the barge impacted the tug's motion for forward speeds so considerably, that the respective pitch response amplitude operators resembled those of the barge's. The boundary element methods tended to overestimate the measured pitch responses of the tug for forward speeds. Pitch responses predicted by RANS agreed better with measurements, but with slightly underestimation tendency. As a whole, fairly well agreement of pitch responses of the tug with measurements in stand-alone and unarticulated cases for forward speeds was obtained.

Small deviations between tug's computed and measured responses in hingedly articulated simulations at zero forward speed were observed. Similarly, barge motion responses in zero and non-zero speeds were satisfactory. Comparison of motion responses provided the evidence that the contact elements could represent hinge joints successfully because the motion responses predicted by kinematically constrained and contact element hinges are qualitatively at the same level. For the barge, no hydrodynamic solvers showed any superiority to the others. Same could be concluded for the tug at various wave attack angles and at zero forward speed.

The longitudinal and vertical hinge forces were estimated well by all solvers for zero forward speed at various wave attack angles. However, the boundary element methods in association with kinematic constraints had overestimation tendencies. RANS showed better agreements of longitudinal forces with the measurements, but not necessarily for the vertical hinge forces. The boundary element method in association with contact elements provided the best agreement of predicted longitudinal and vertical hinge forces at zero forward speed. In case of forward speeds, still fairly well agreement between simulation predictions of hinge forces and measurements was achieved. Considering the total hinge forces consisting of longitudinal and vertical force components, the hinges represented by constraining the body kinematics at acceleration level in RANS, as well as the hinges constructed by means of contact elements submitted the most agreeable results. These forces were apt to overestimation for the constraint hinges in frequency and time domain solvers. It could be concluded that neither hydrodynamic nor articulation methods could be judged alone for accurate prediction of motion behavior of articulated bodies and articulation forces. The interaction of hydrodynamic and coupling models is also decisive in reliable prediction of articulated multibody behaviors. Additionally, the results proved that contact elements are undoubtedly eligible to represent hinge joints.

Rigid joints were also modeled by kinematic constraints in the aforementioned frequency and time domain solvers. Similar to hinge joint cases, motion behaviors were estimated well. Again, similar to hinge cases, the joint forces were overestimated by boundary element methods.

Following the validation studies to characterize contact element qualification in representation of articulations, a case study was presented. The convoy bodies were connected by a mechanical coupling consisting of prismatic joints. The prismatic joints were modeled by contact elements. The case study proved that contact elements could be utilized successfully to represent prismatic joints. The mechanical coupling mechanism had the capability to exercise pretension at the coupling locations. Two different pretension models were conceived. One model represented

the pretension as a force exerted in a constant direction. The other model referenced to the effect of the pretension on the system, i.e. increment of friction along prismatic joint axis. Since friction force is direction-dependent, an algorithm was implemented to determine in which direction, and how the friction forces were to be applied. The algorithm intervened at each time step. The equivalence of both pretension models were checked in beam waves. Motion responses were predicted well at various wave attack angles, but longitudinal joint forces were slightly overestimated. On the contrary, transverse joint forces in beam waves, and vertical forces in head and following waves were slightly underpredicted. Therefore, the total reaction forces agreed well with measured forces. Furthermore, it was demonstrated that the pretension efficiency could be assessed by the presented contact element model.

5.1 Major Findings of the Work

The most considerable outcomes of the presented work are the followings:

- Contact elements can be used to represent articulations efficiently, and to assess articulation forces reliably.
- With reference to the predicted coupling force responses, the contact element model is as reliable as, or even superior to, kinematic constraint models in representing articulations.
- The contact element model has a general shortcoming that finding the appropriate contact element coefficients is difficult.
- Hinge joints can be represented conveniently by contact elements. However, representation of prismatic joints is comparatively intricate because of difficulties in finding the appropriate contact element coefficients.
- For prismatic joints, the contact element model does not deliver reliable results in short waves.
- Hydrodynamic, kinematic, and articulation methods should not be judged alone for accurate prediction of motion behavior of articulated bodies and articulation forces. The interaction of these models is also decisive in reliable prediction of the responses.
- Effects of different hydrodynamic, kinematic and coupling methods are more evident in coupling force responses than in motion responses.

- In absence of nonlinear hydrodynamic effects, coupling forces are overpredicted.
- Nonlinear hydrodynamic effects, such as Froude-Krylov forces, improve prediction of motion and coupling force responses. However, nonlinear effects also intensify the inaccuracies of the contact element model at high wave frequencies.
- Considering pre-, postprocessing, and computation time, frequency domain boundary element methods coupled with kinematic constraint models are the most practical methods to assess motion and coupling force responses of articulated multibody systems.

5.2 Suggestions for Future Works

Application of Reynolds-averaged Navier–Stokes equation solvers in dealing with marine and offshore problems is expected to draw increasing attention from practical point of view. Therefore, it is suggested to extend the investigations of this work to couple the contact element model with nonlinear equation of motion and RANSE solvers. Such investigations provide the opportunity to scrutinize further the maneuverability of ships articulated by means of contact elements with rotating propellers and moving rudders.

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